

AQUALOOP

PROJECT CODE: B1092301

PCB NUMBER: 23W005

PCB VERSION: 1

PCB SIZE: 419.48 mm*190.6 mm

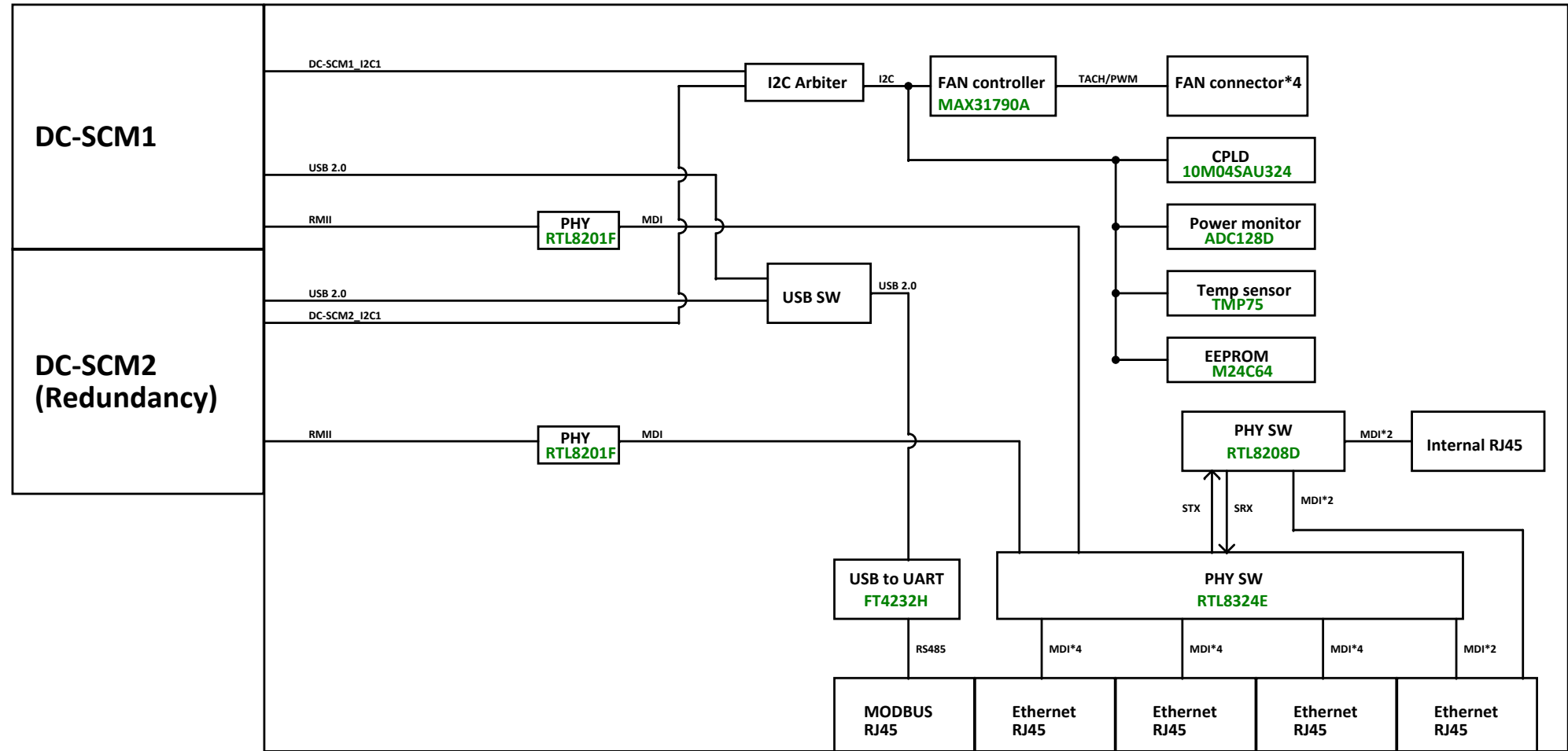
PCB Layer: 8 Layer

PCB THICKNESS: 1.57 $\pm$ 8% mm

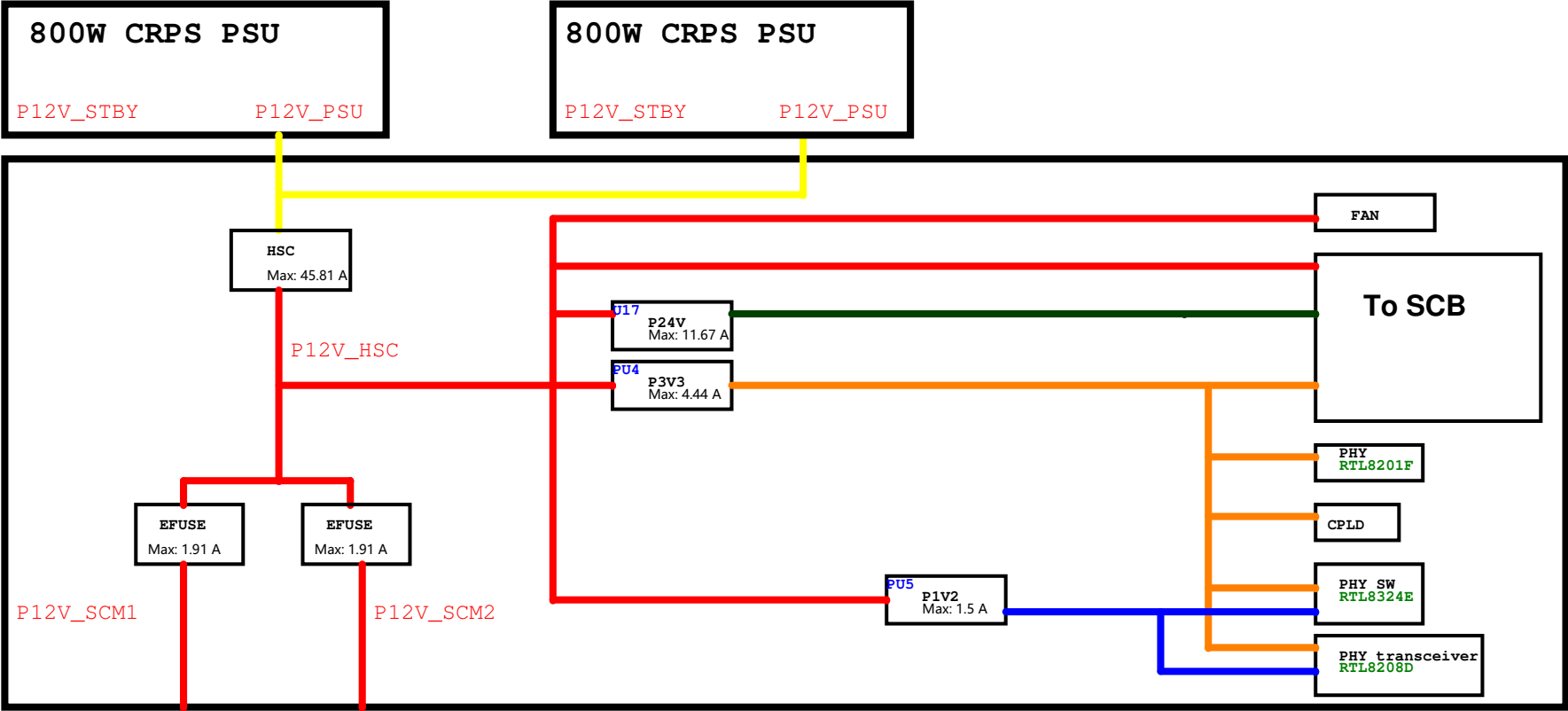
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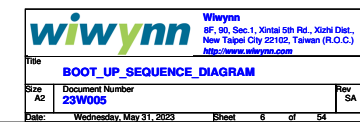
BLOCK DIAGRAM (IMMERSION TANK)



POWER DELIVER DIAGRAM



DC-SCI

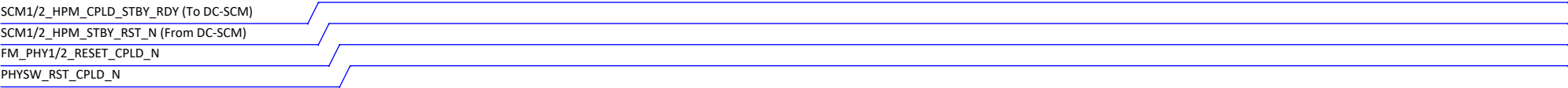


POWER_ON_OFF_SEQUENCE

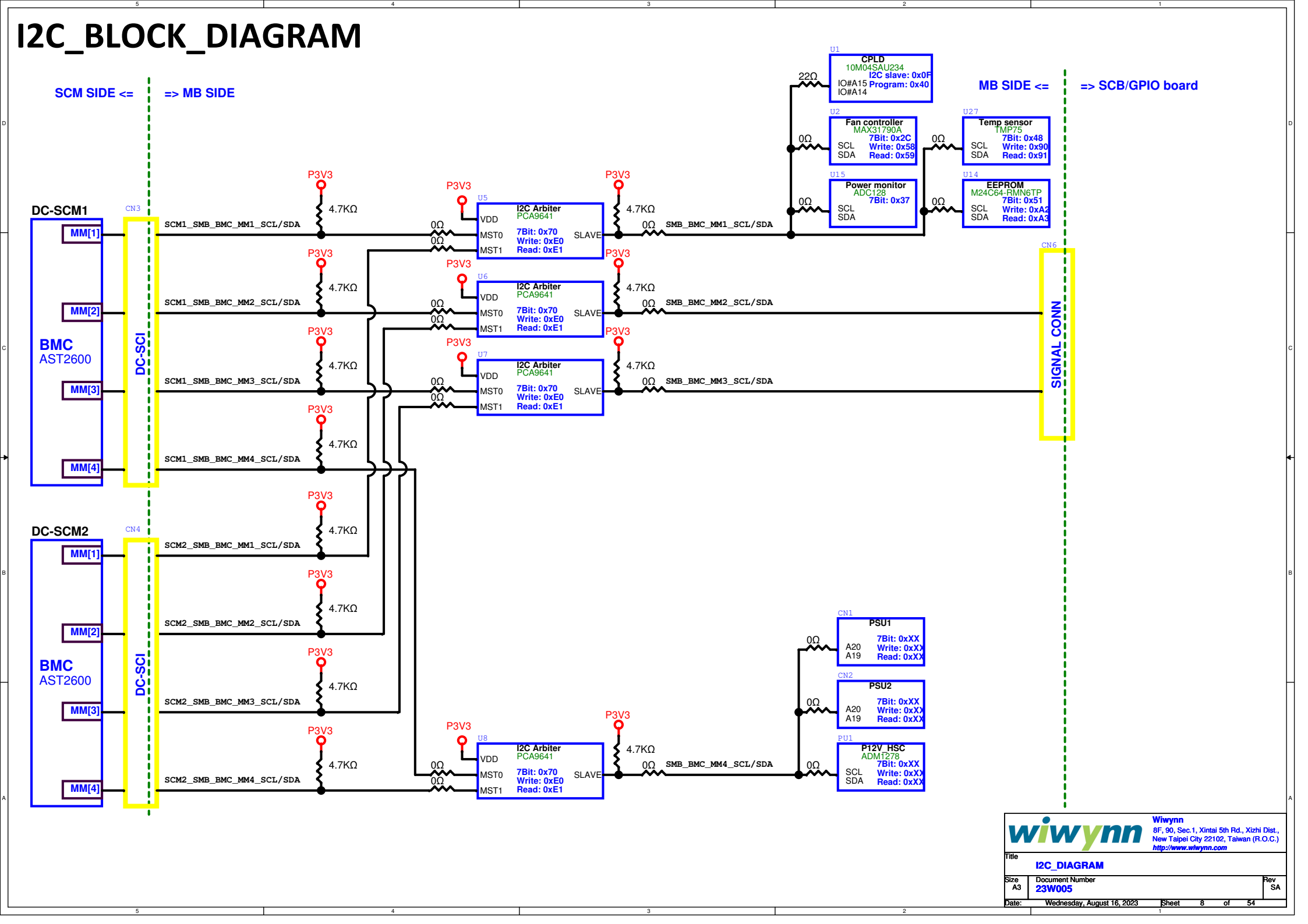
Power



Reset

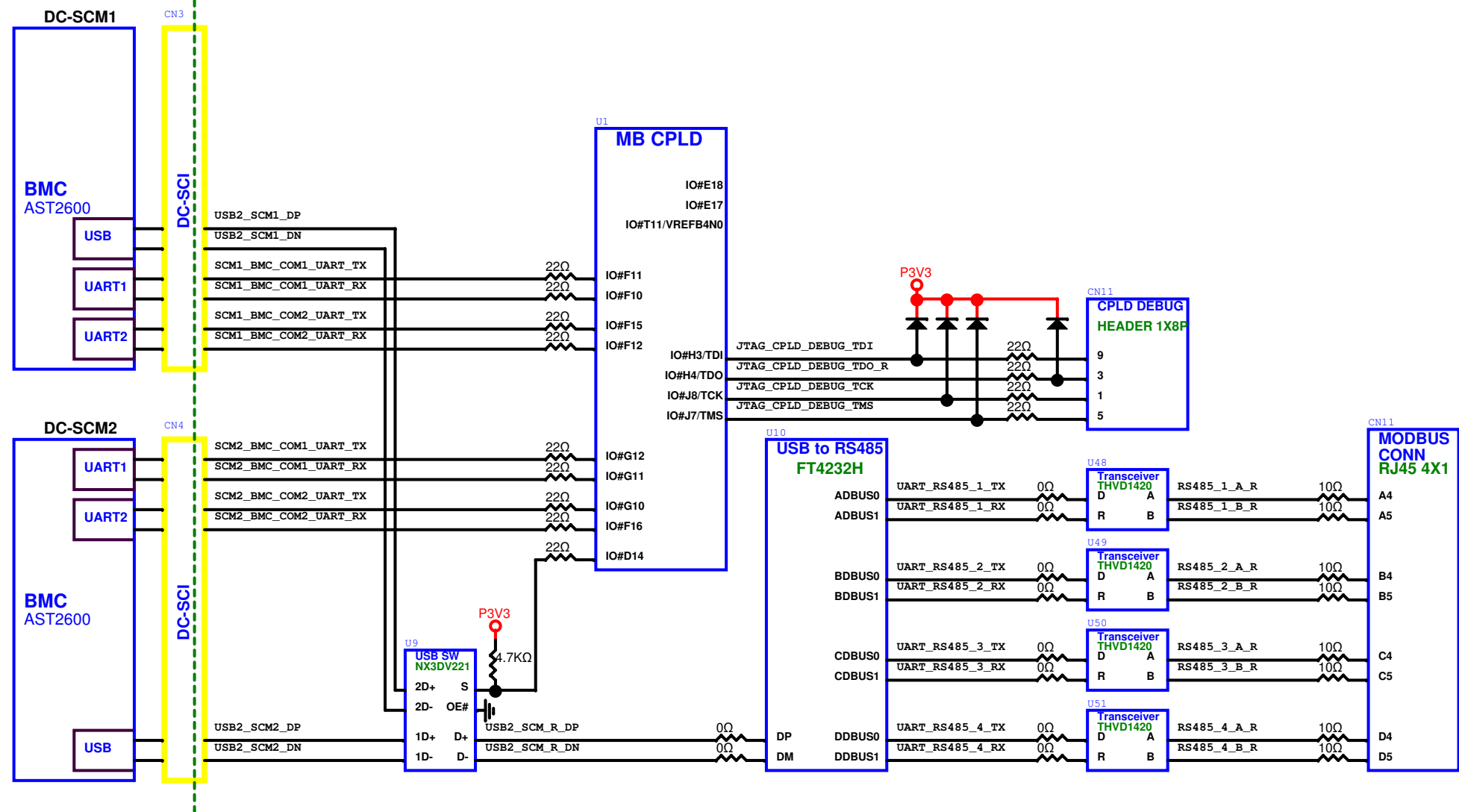


I2C_BLOCK_DIAGRAM

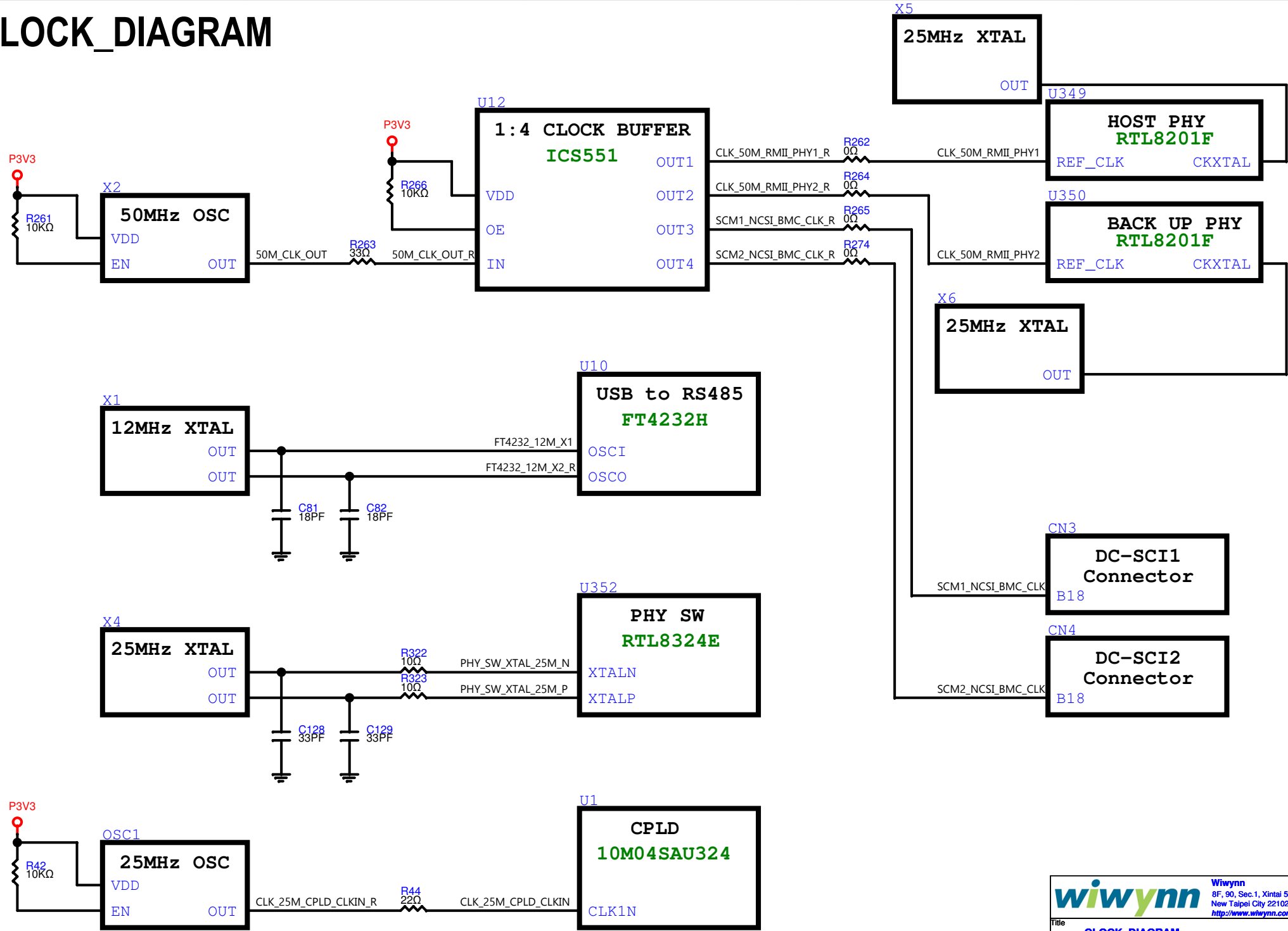


MANAGEMENT_DIAGRAM

SCM SIDE <== ==> MB SIDE



CLOCK_DIAGRAM



LAN_DIAGRAM

SCM SIDE <=> MB SIDE

DC-SCM1

BMC
AST2600

MAC

CN3

DC-SCI

SCM1_NCSI_BMC_CRS_DV
SCM1_NCSI_BMC_TX_EN
SCM1_NCSI_BMC_TXD0
SCM1_NCSI_BMC_TXD1
SCM1_NCSI_BMC_RXER
SCM1_NCSI_BMC_RXD0
SCM1_NCSI_BMC_RXD1

33Ω
33Ω
33Ω
33Ω
33Ω
33Ω
33Ω

U349

HOST PHY
RTL8201F

CRS_DV
TXEN
TXD0
TXD1
RXER
RXD0
RXD1

RD+ RTL8201_1_TX_C_DP
RD- RTL8201_1_TX_C_DN
TD+ RTL8201_1_RX_C_DP
TD- RTL8201_1_RX_C_DN

PHY SW
RTL8324E/RTL8208D

LAN_P00_RX_DP/N
LAN_P00_TX_DP/N
LAN_P01_RX_DP/N
LAN_P01_TX_DP/N
LAN_P02_RX_DP/N
LAN_P02_TX_DP/N
LAN_P03_RX_DP/N
LAN_P03_TX_DP/N

XF1
XFORM

23/24 26/25
20/21 29/28
17/18 32/31
14/15 35/34
11/12 38/37
8/9 41/40
5/6 44/43
2/3 47/46

LAN_P00_XF_RX_DP/N
LAN_P00_XF_TX_DP/N
LAN_P01_XF_RX_DP/N
LAN_P01_XF_TX_DP/N
LAN_P02_XF_RX_DP/N
LAN_P02_XF_TX_DP/N
LAN_P03_XF_RX_DP/N
LAN_P03_XF_TX_DP/N

CN12
ETHERNET
CONN
RJ45 4X1

A3/A6
A1/A2
C3/C6
C1/C2
B3/B6
B1/B2
D3/D6
D1/D2

DC-SCM2

BMC
AST2600

MAC

CN4

DC-SCI

SCM2_NCSI_BMC_CRS_DV
SCM2_NCSI_BMC_TX_EN
SCM2_NCSI_BMC_TXD0
SCM2_NCSI_BMC_TXD1
SCM2_NCSI_BMC_RXER
SCM2_NCSI_BMC_RXD0
SCM2_NCSI_BMC_RXD1

33Ω
33Ω
33Ω
33Ω
33Ω
33Ω
33Ω

U350

BACKUP PHY
RTL8201F

CRS_DV
TXEN
TXD0
TXD1
RXER
RXD0
RXD1

RD+ RTL8201_2_TX_C_DP
RD- RTL8201_2_TX_C_DN
TD+ RTL8201_2_RX_C_DP
TD- RTL8201_2_RX_C_DN

LAN_P08_RX_DP/N
LAN_P08_TX_DP/N
LAN_P09_RX_DP/N
LAN_P09_TX_DP/N
LAN_P10_RX_DP/N
LAN_P10_TX_DP/N
LAN_P11_RX_DP/N
LAN_P11_TX_DP/N

XF3
XFORM

23/24 26/25
20/21 29/28
17/18 32/31
14/15 35/34
11/12 38/37
8/9 41/40
5/6 44/43
2/3 47/46

LAN_P08_XF_RX_DP/N
LAN_P08_XF_TX_DP/N
LAN_P09_XF_RX_DP/N
LAN_P09_XF_TX_DP/N
LAN_P10_XF_RX_DP/N
LAN_P10_XF_TX_DP/N
LAN_P11_XF_RX_DP/N
LAN_P11_XF_TX_DP/N

CN14
ETHERNET
CONN
RJ45 4X1

A3/A6
A1/A2
C3/C6
C1/C2
B3/B6
B1/B2
D3/D6
D1/D2

LAN_P12_RX_DP/N
LAN_P12_TX_DP/N
LAN_P13_RX_DP/N
LAN_P13_TX_DP/N
LAN_P14_RX_DP/N
LAN_P14_TX_DP/N
LAN_P15_RX_DP/N
LAN_P15_TX_DP/N

XF4
XFORM

23/24 26/25
20/21 29/28
17/18 32/31
14/15 35/34
11/12 38/37
8/9 41/40
5/6 44/43
2/3 47/46

LAN_P12_XF_RX_DP/N
LAN_P12_XF_TX_DP/N
LAN_P13_XF_RX_DP/N
LAN_P13_XF_TX_DP/N
LAN_P14_XF_RX_DP/N
LAN_P14_XF_TX_DP/N
LAN_P15_XF_RX_DP/N
LAN_P15_XF_TX_DP/N

CN15
ETHERNET
CONN
RJ45 4X1

A3/A6
A1/A2
C3/C6
C1/C2
B3/B6
B1/B2
D3/D6
D1/D2

LAN_P16_RX_DP/N
LAN_P16_TX_DP/N

XF5
XFORM

6/5 19/20
3/2 22/23

LAN_P16_XF_RX_DP/N
LAN_P16_XF_TX_DP/N

RJ1
INTERNAL
CONN
RJ45

3/6
1/2

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Title
LAN_DIAGRAM

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	Sheet	13 of 54

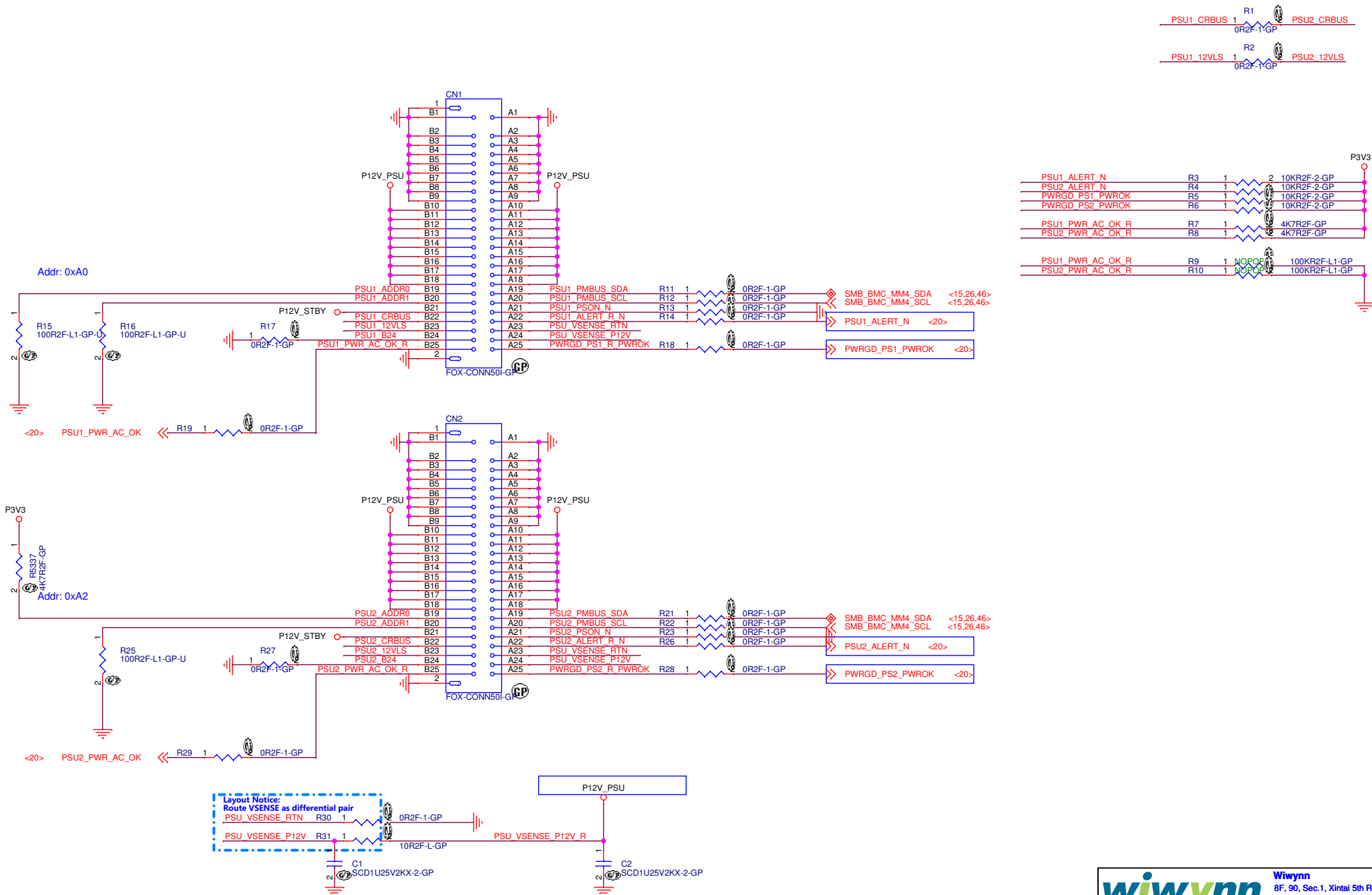
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Title		
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PSU CONNECTOR

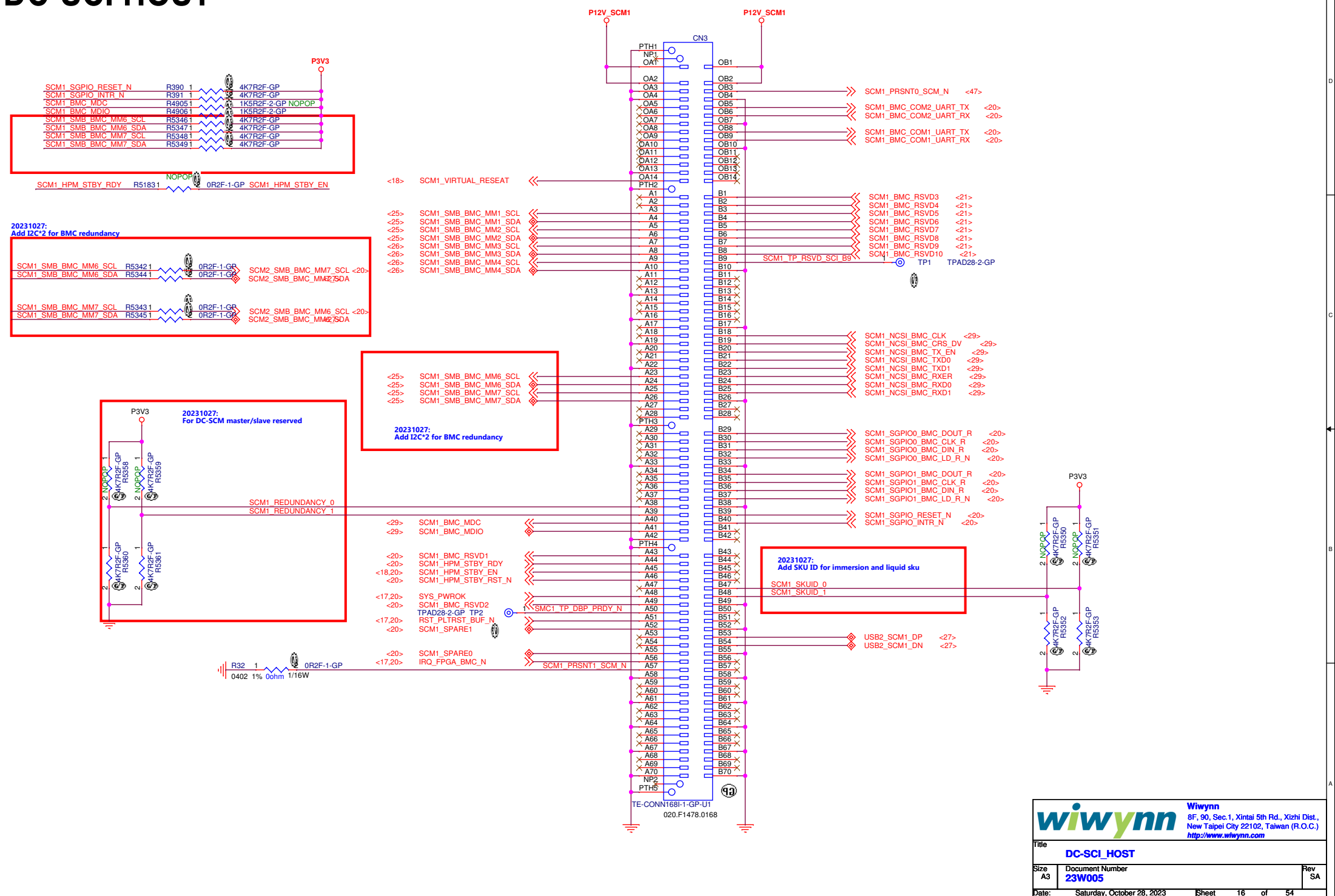


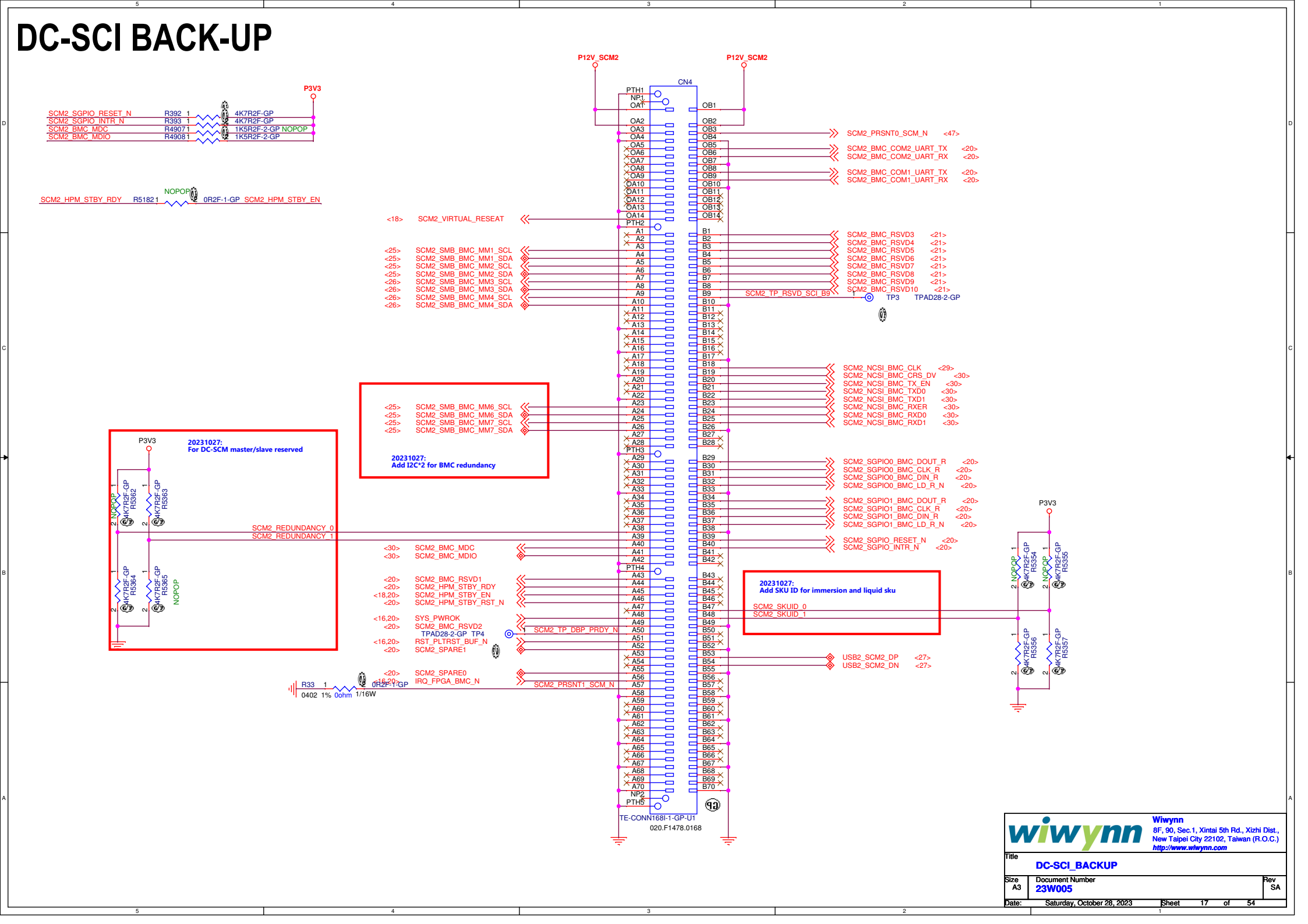
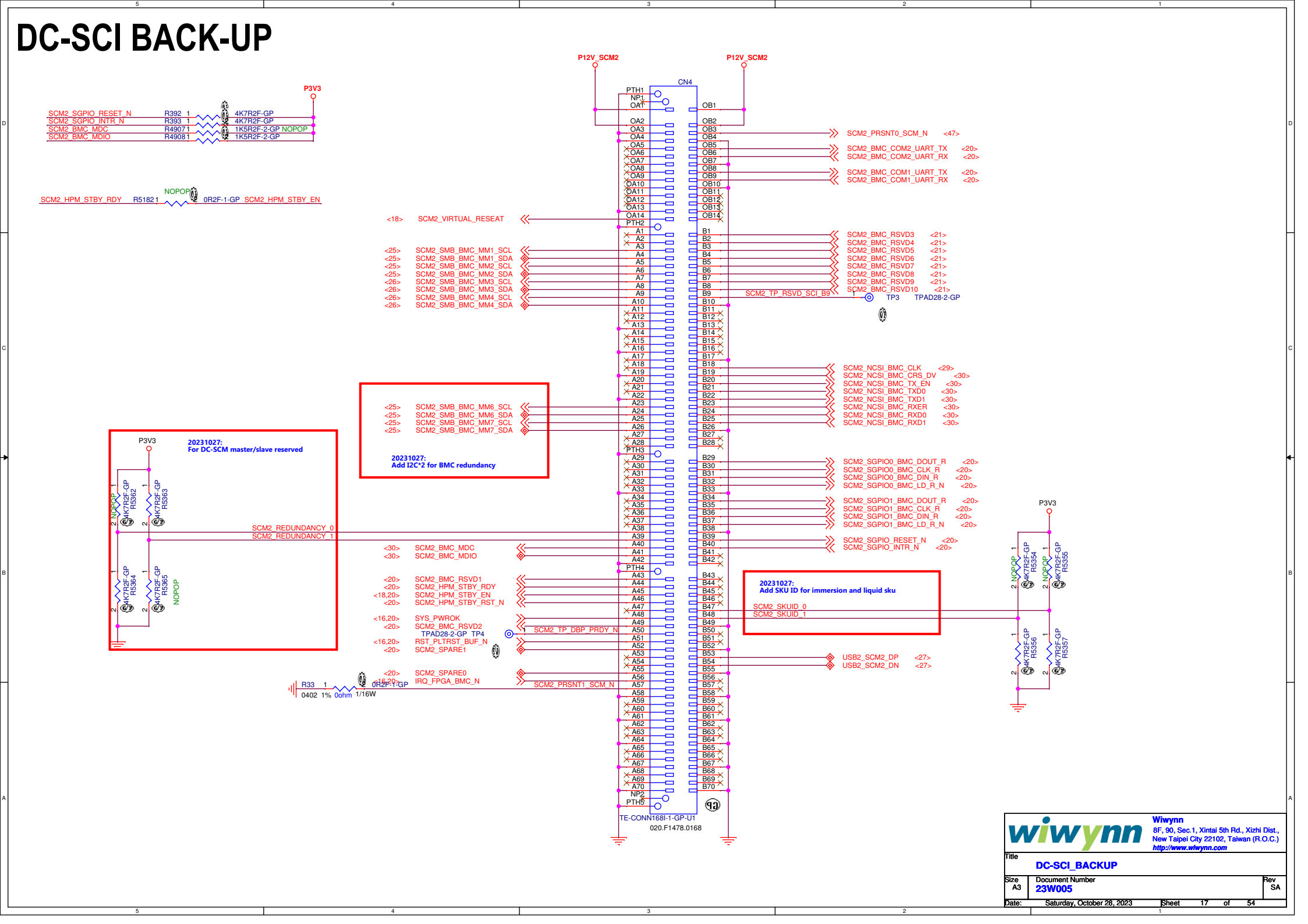
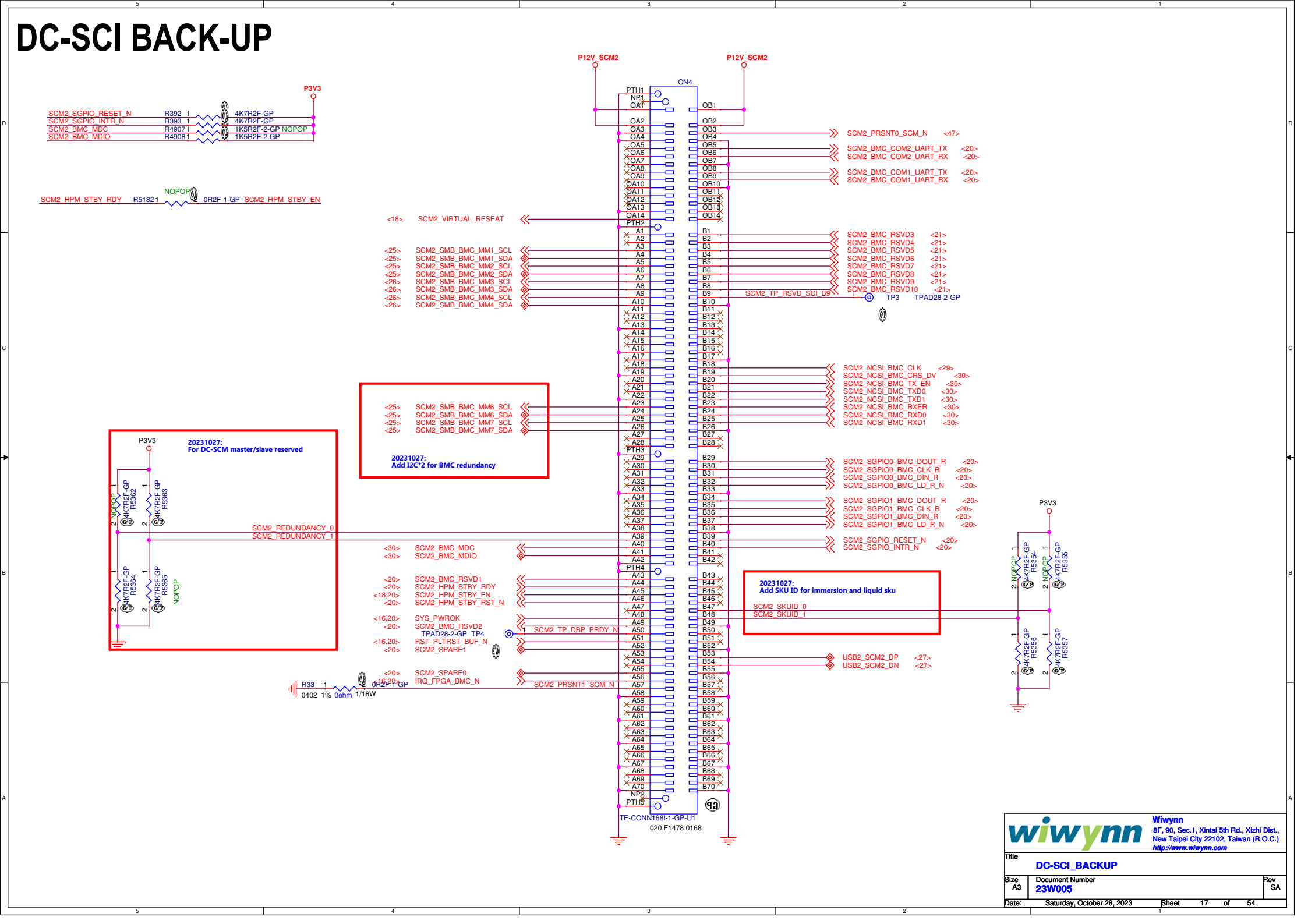
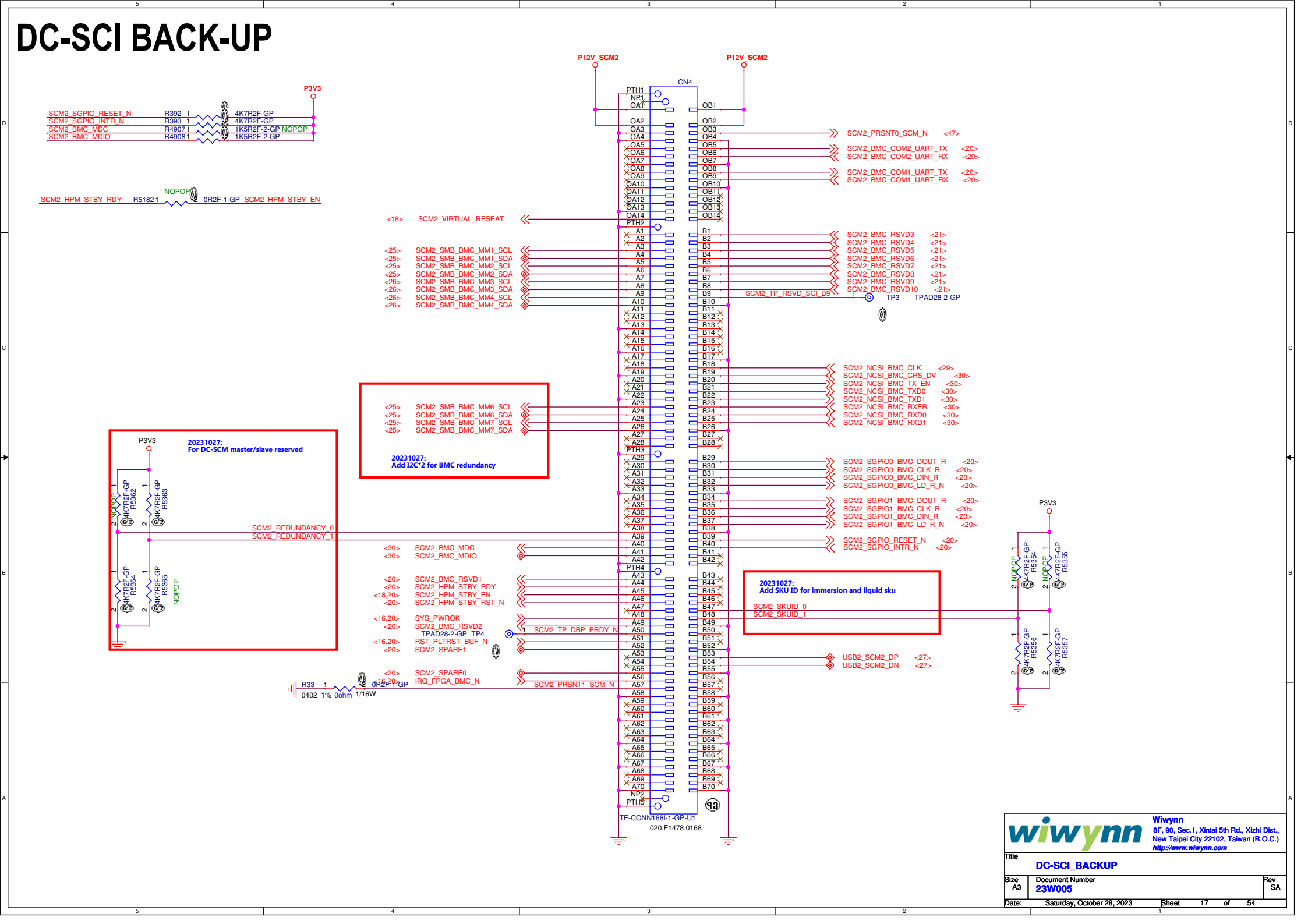
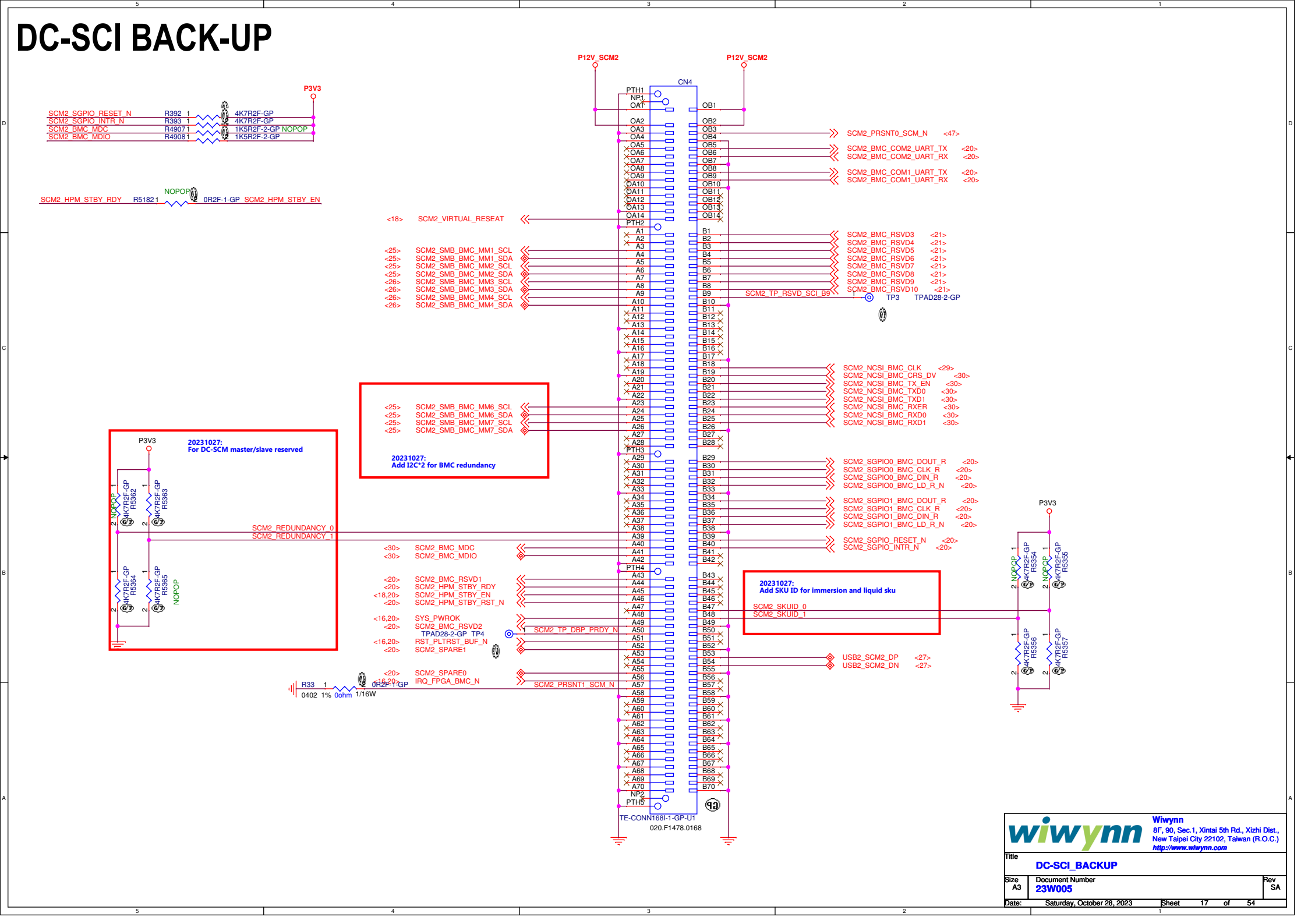
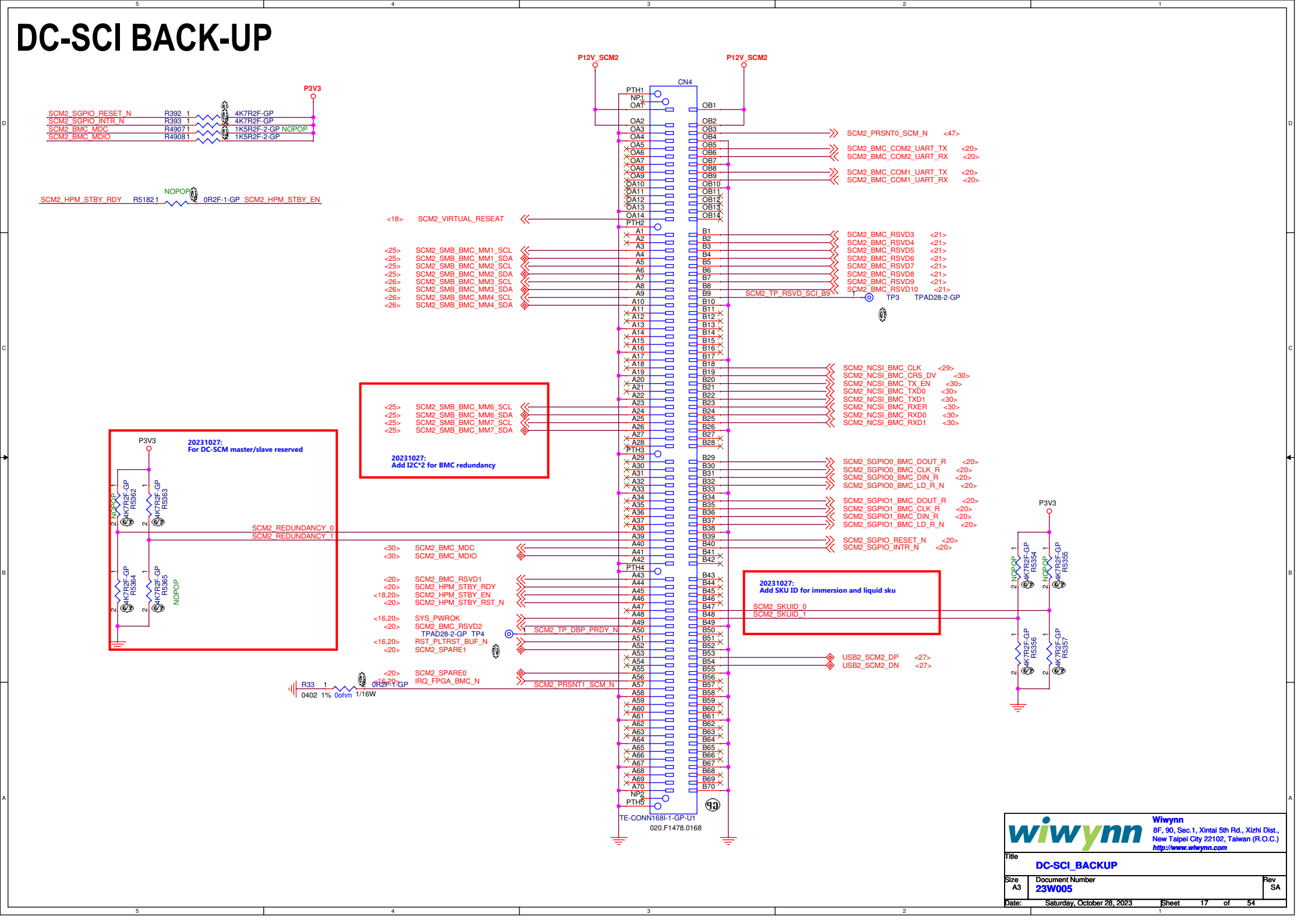
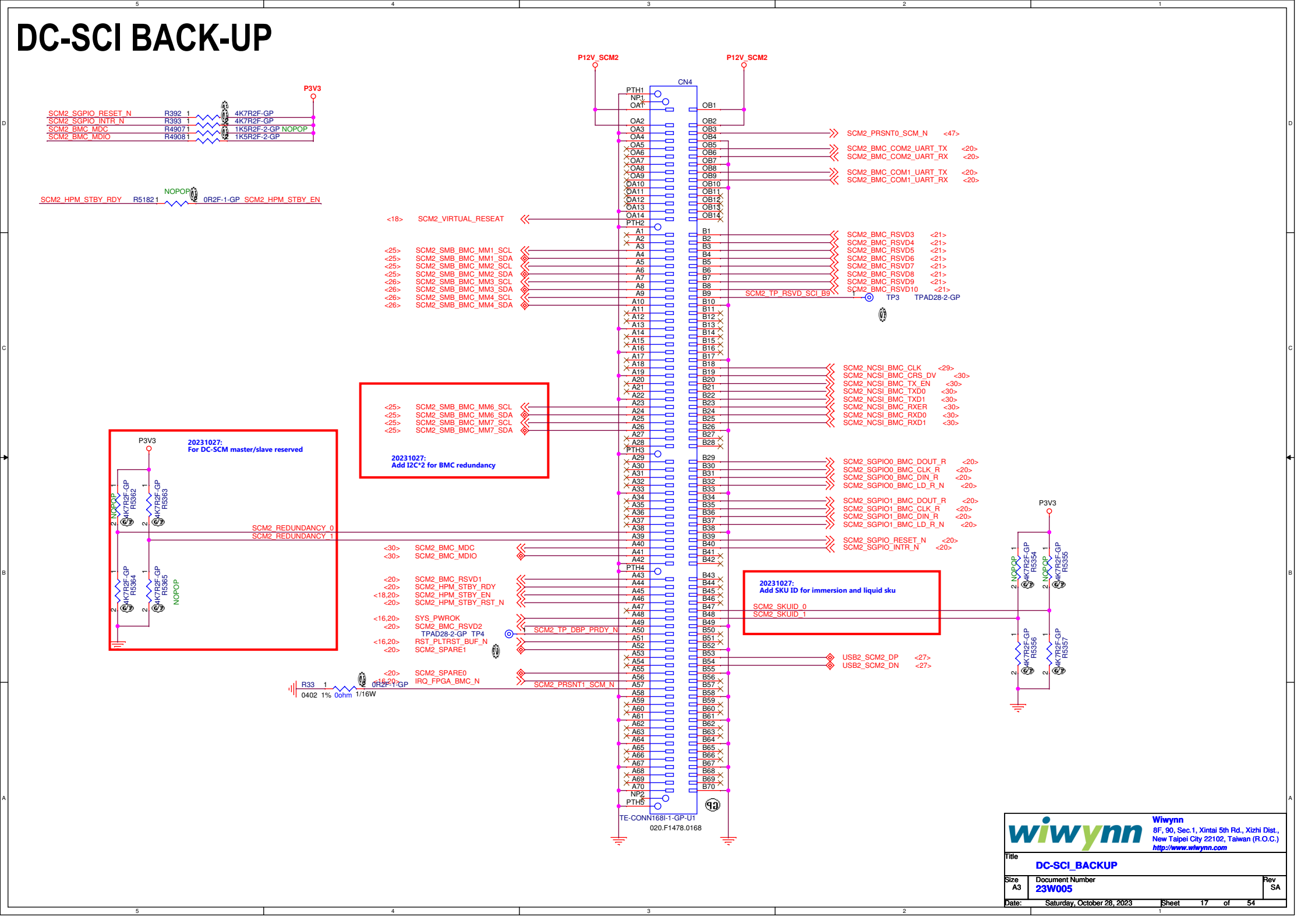
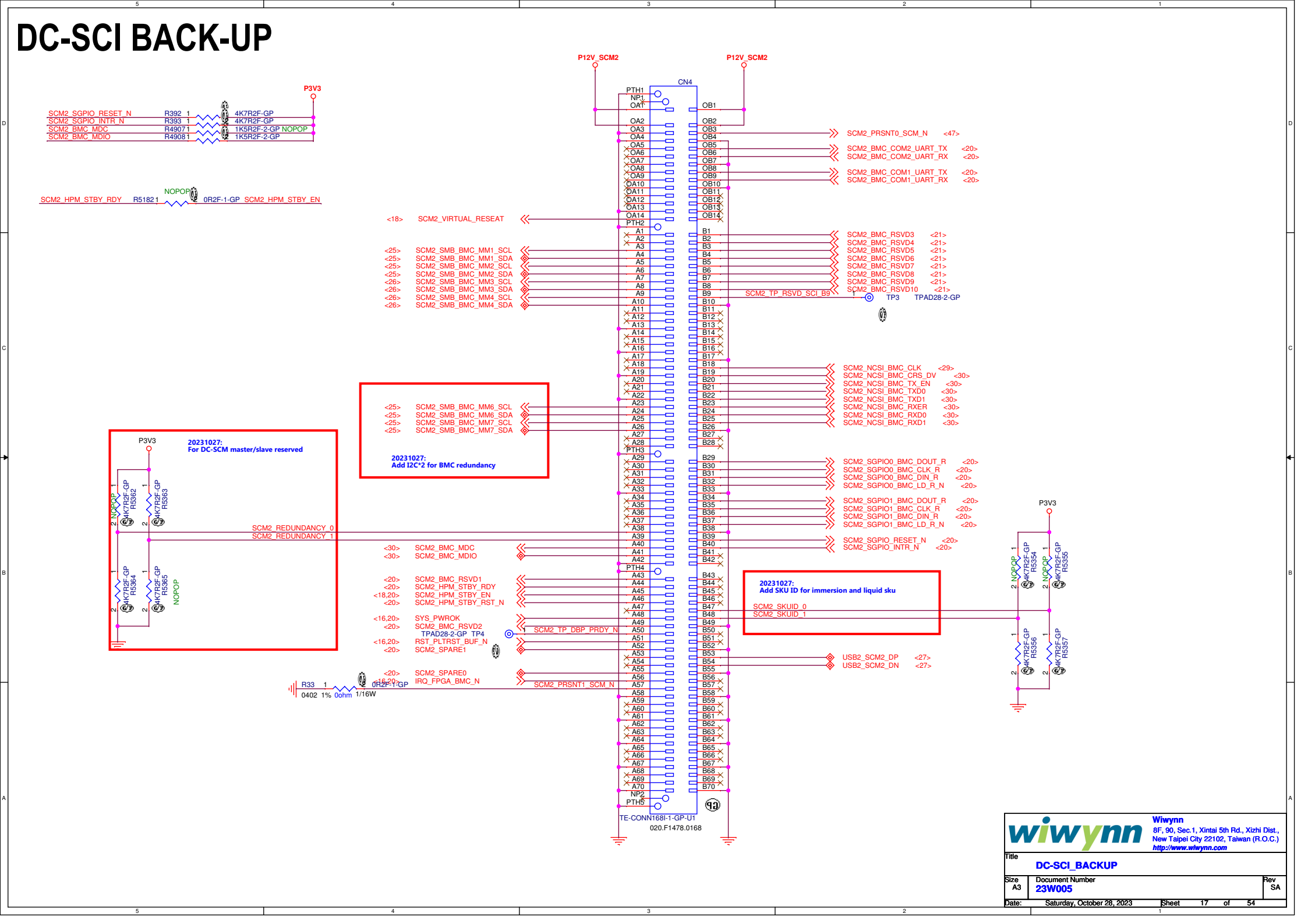
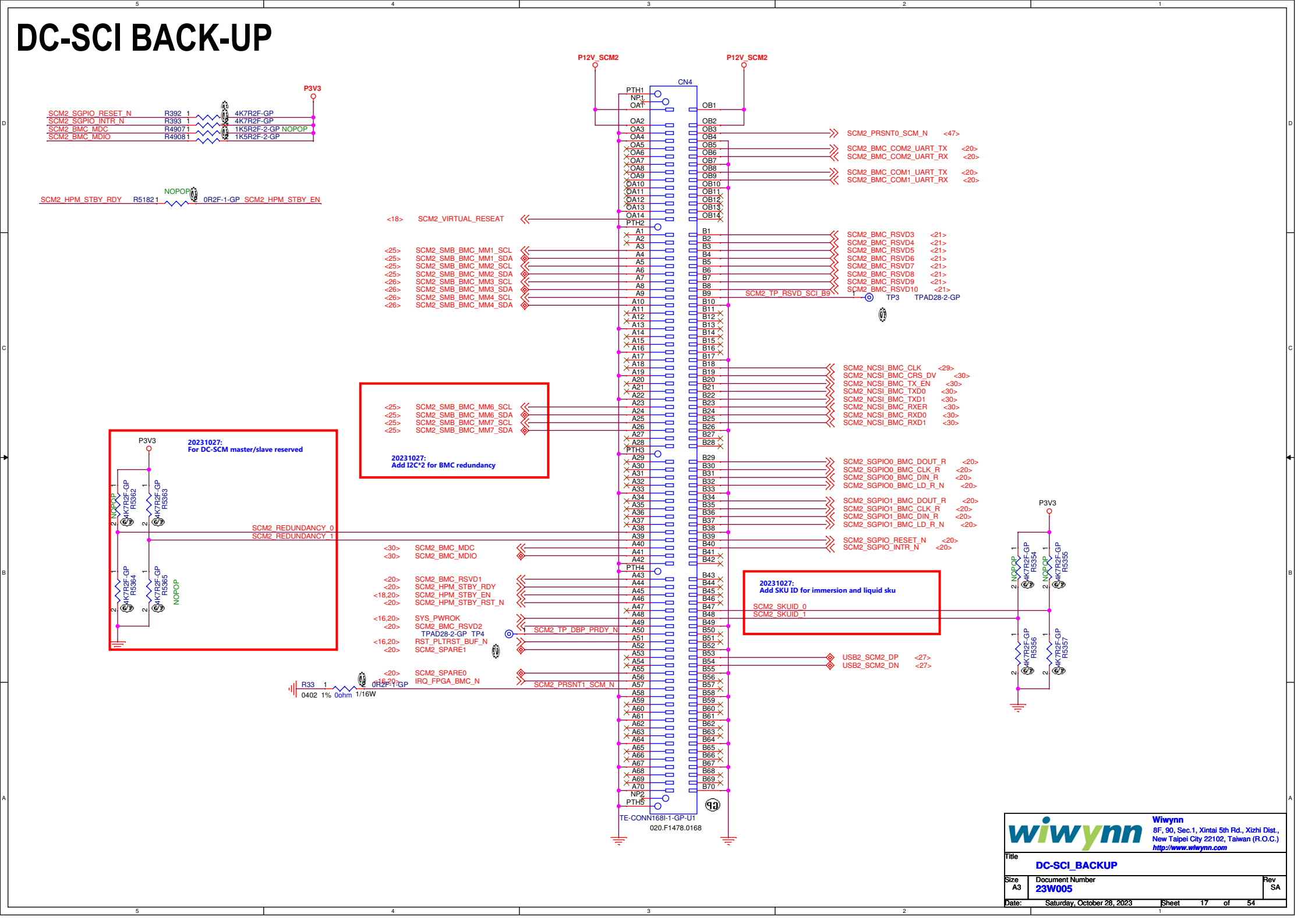
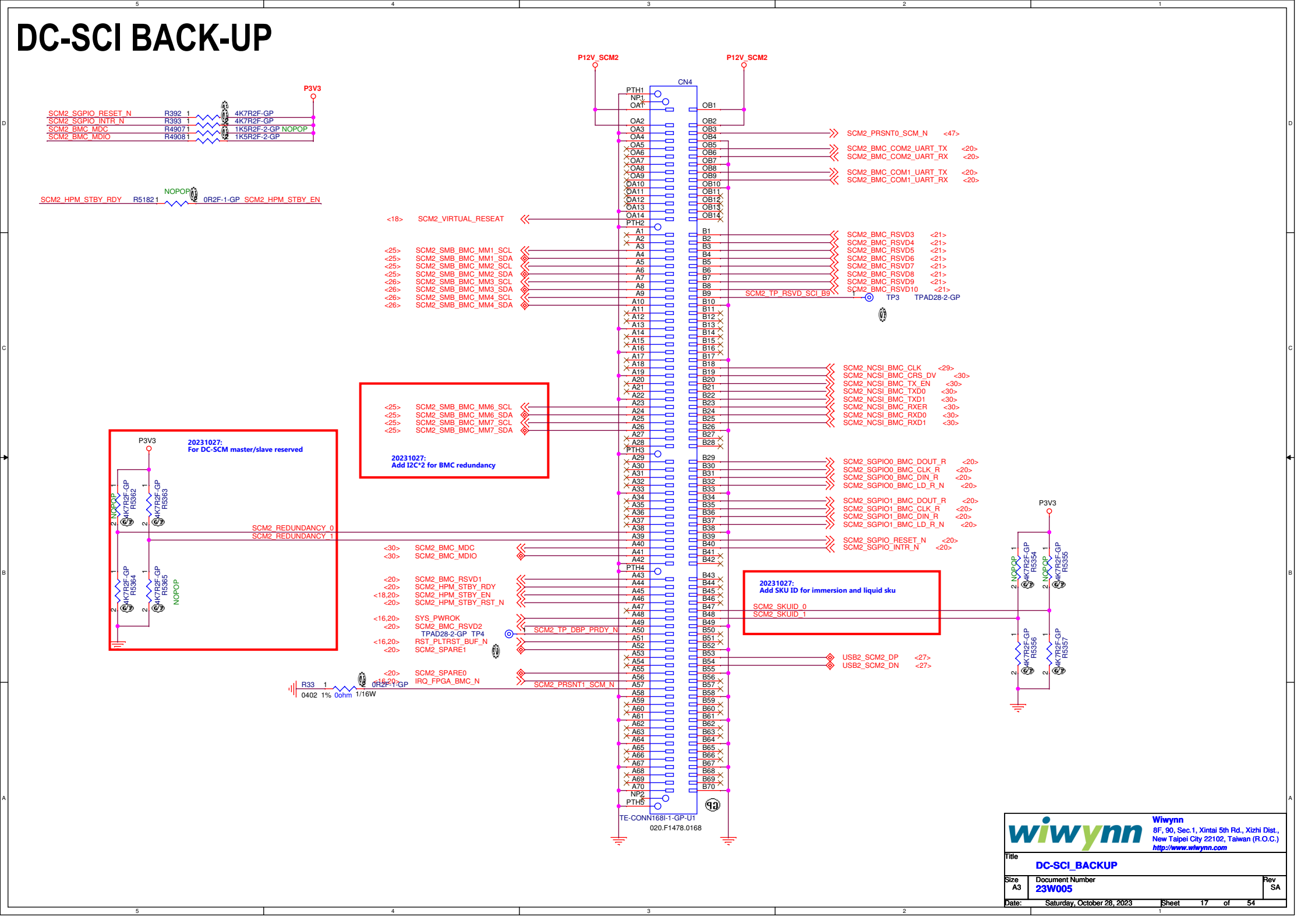
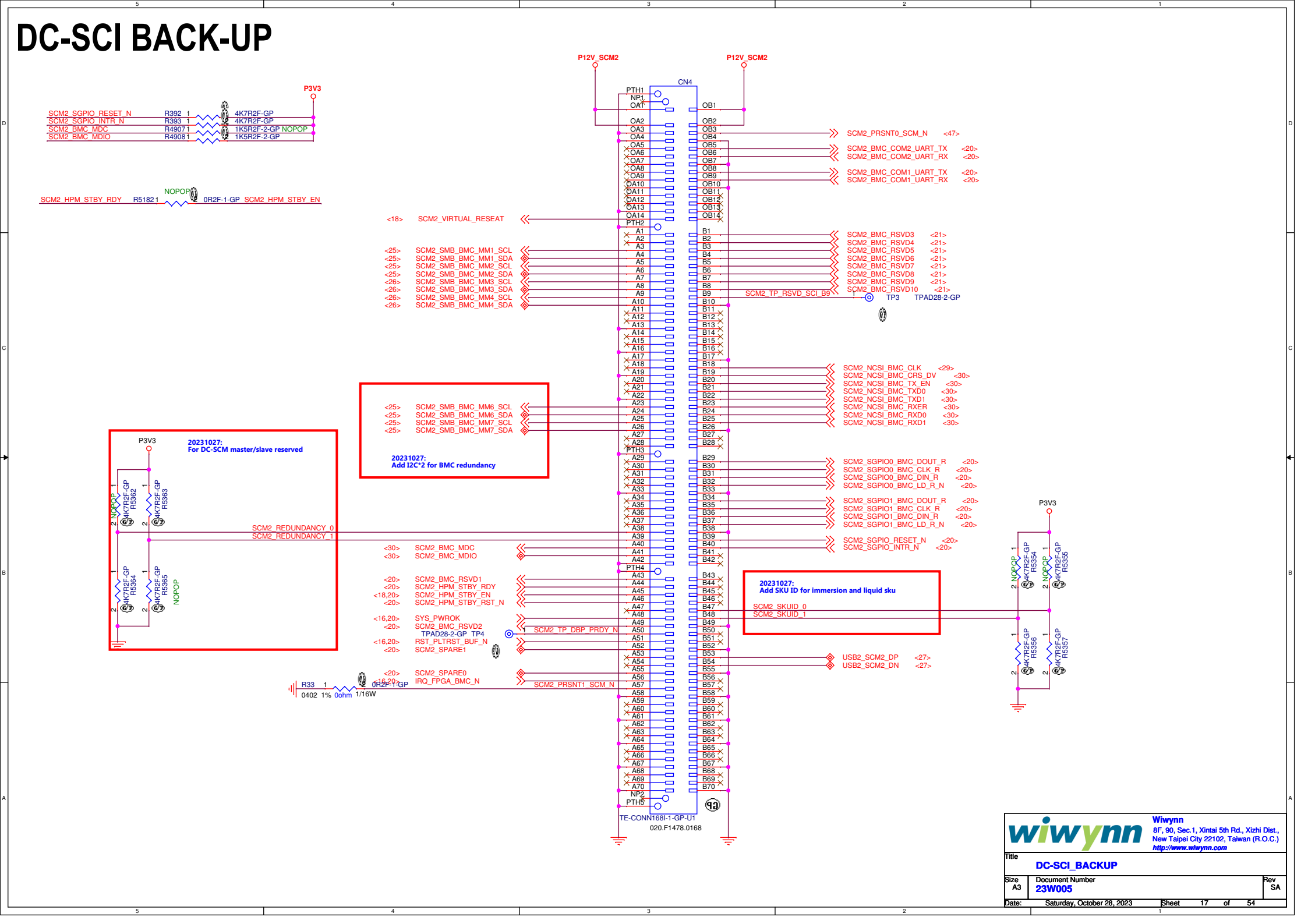
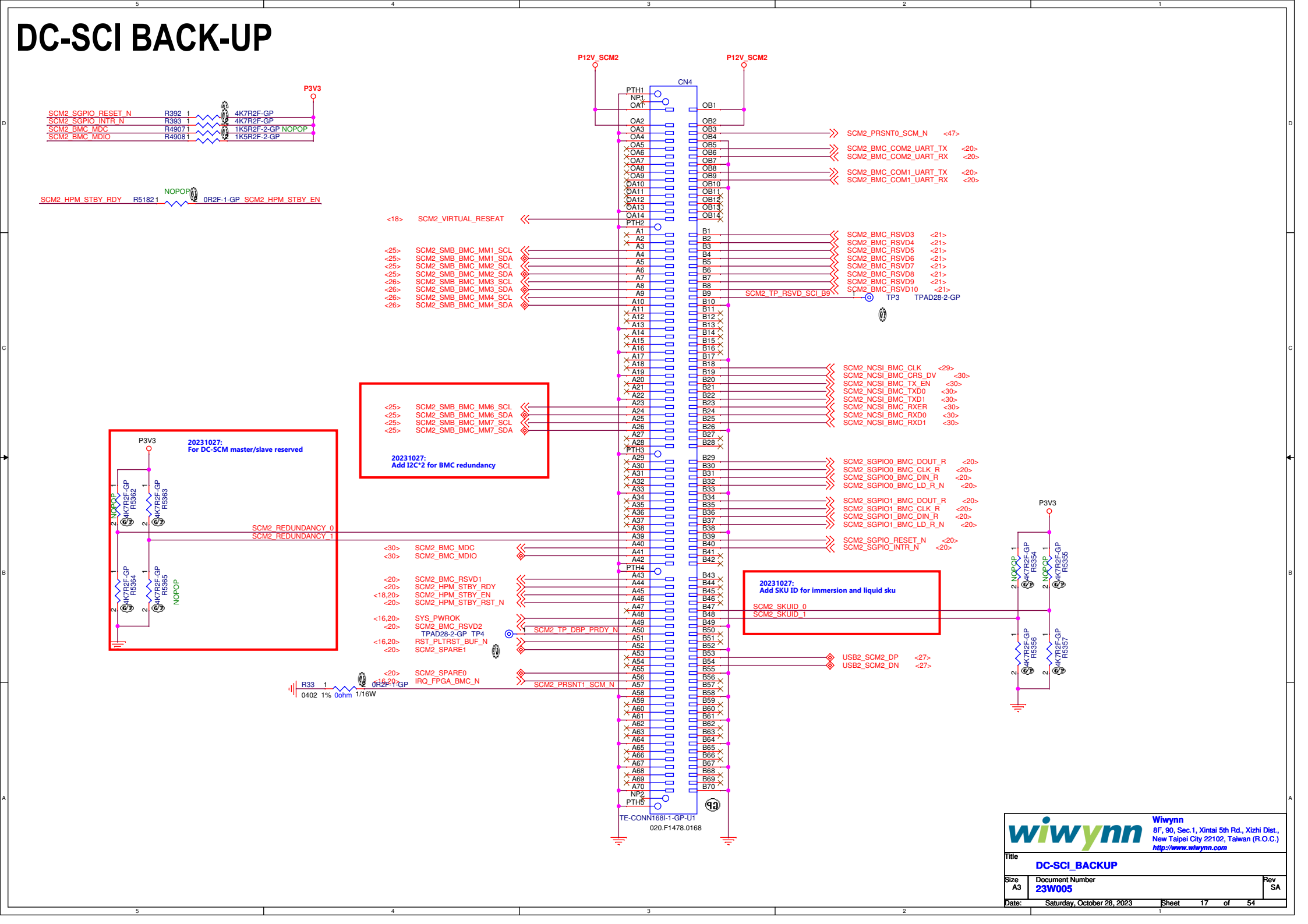
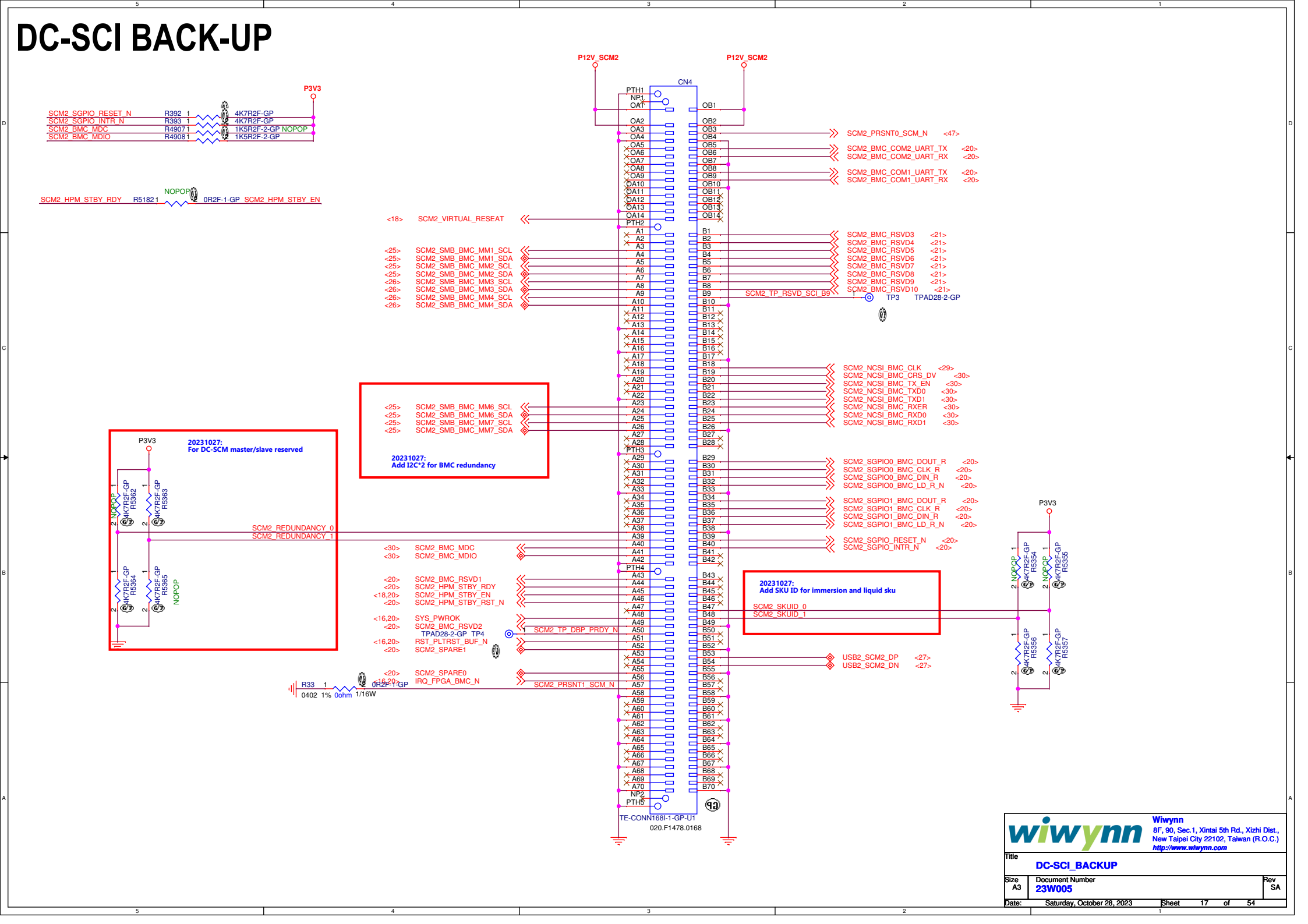
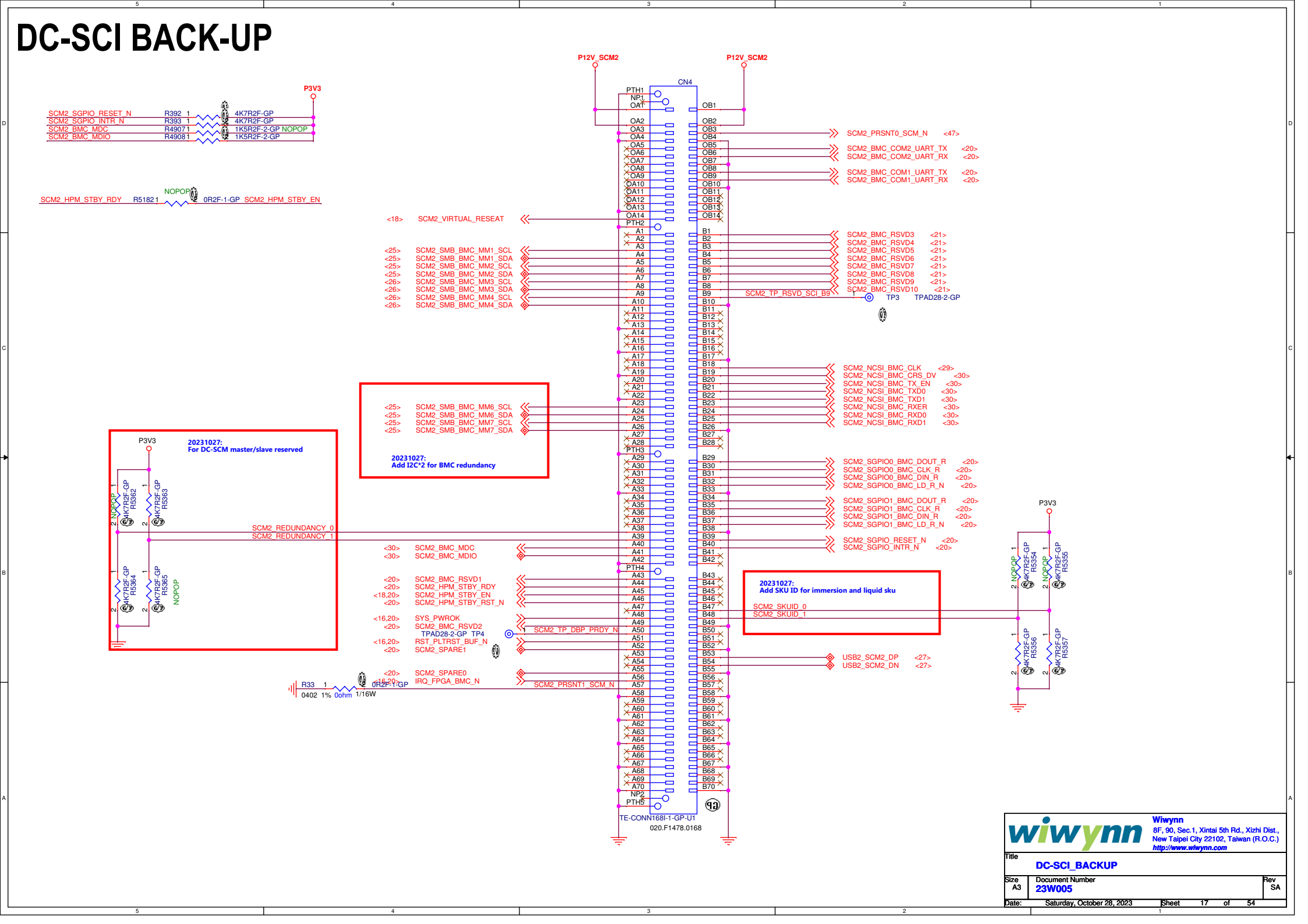
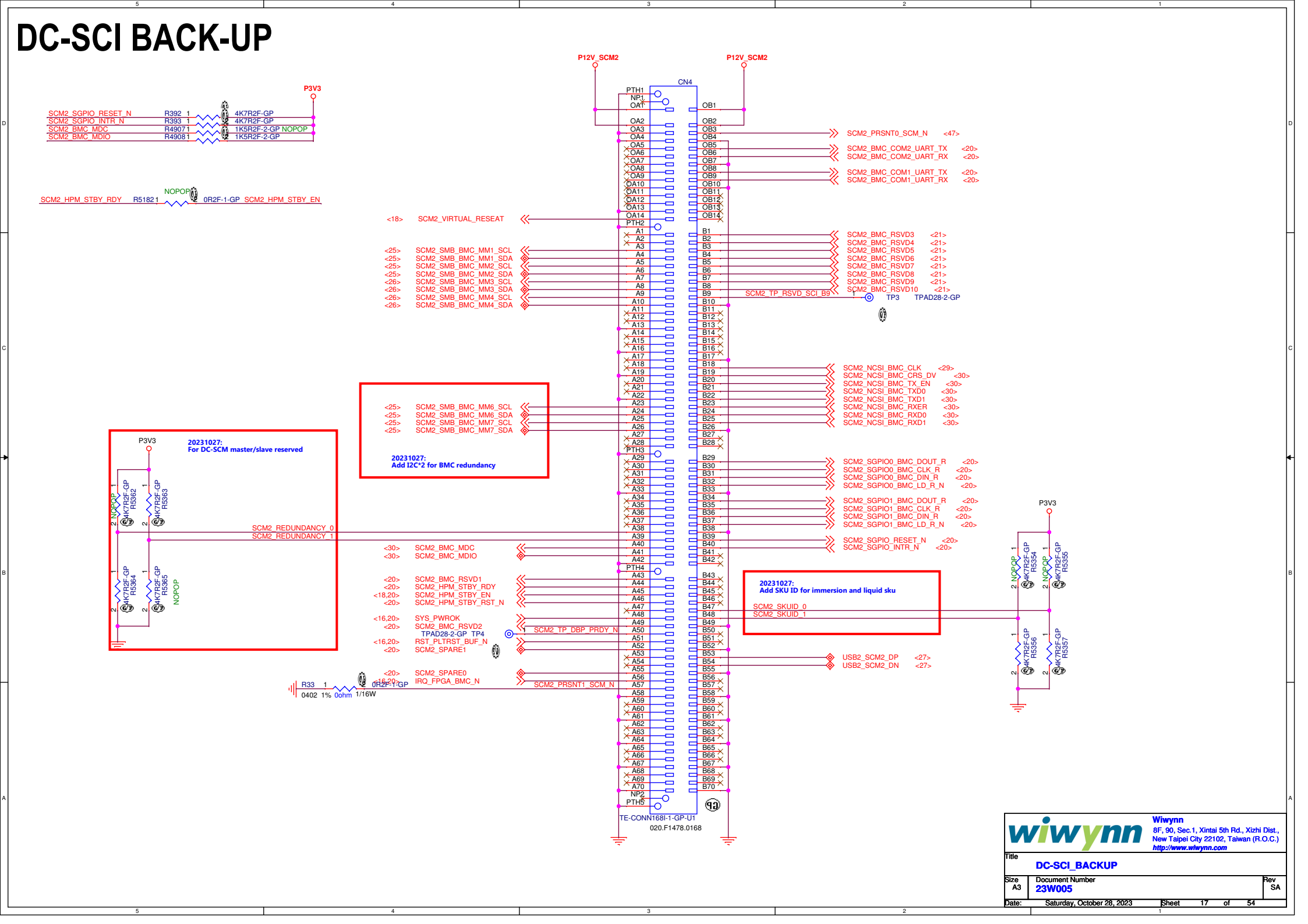
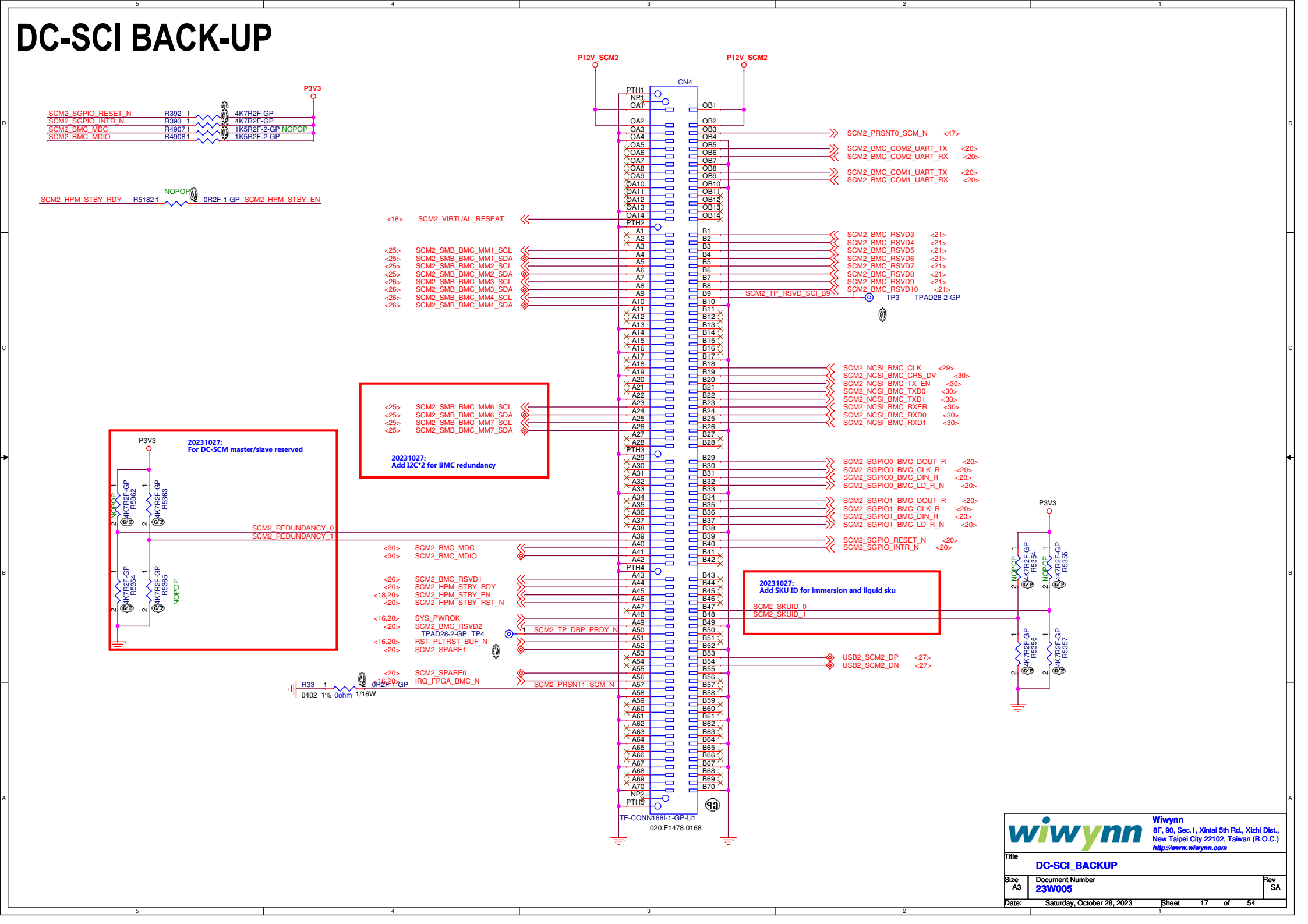
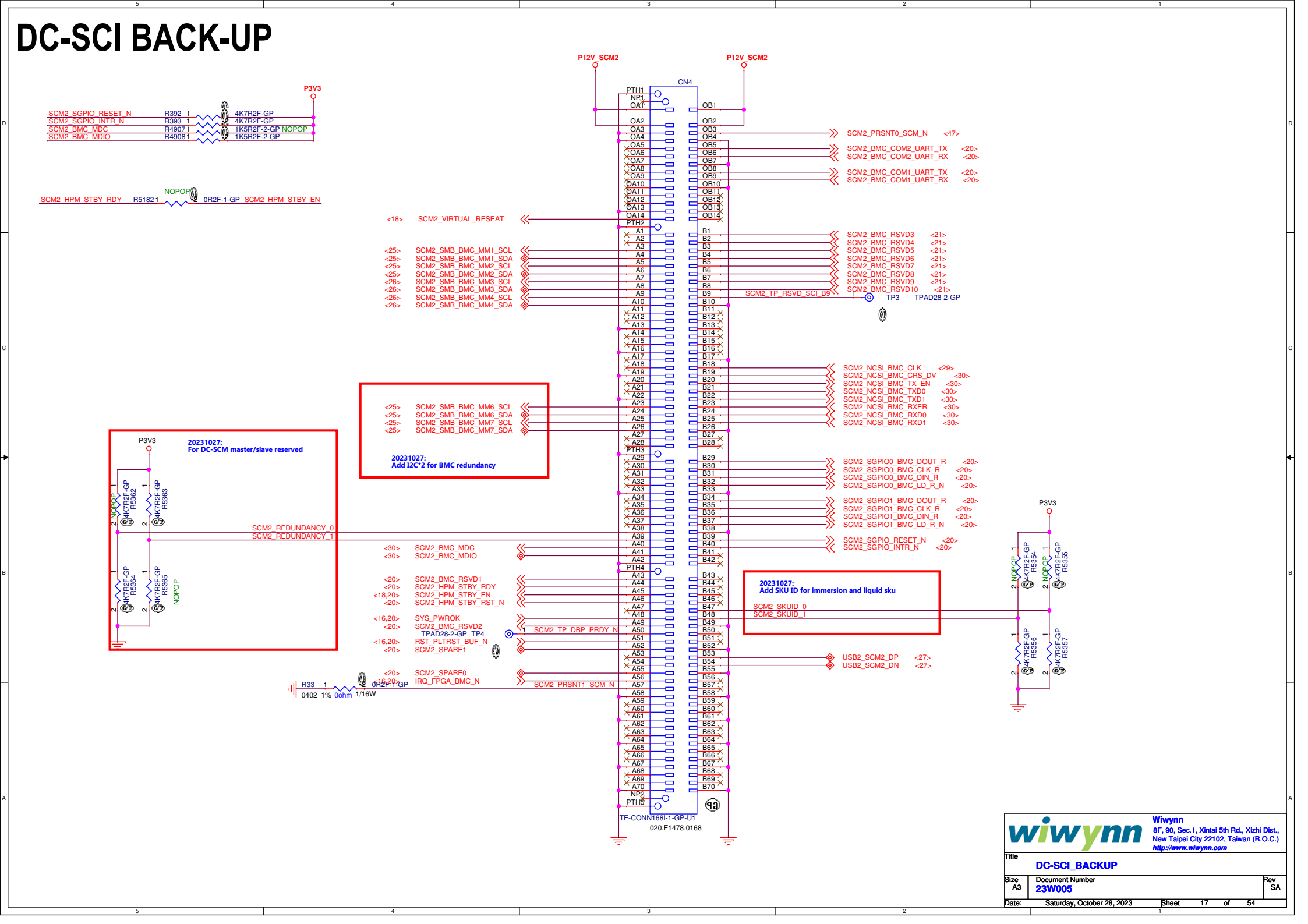
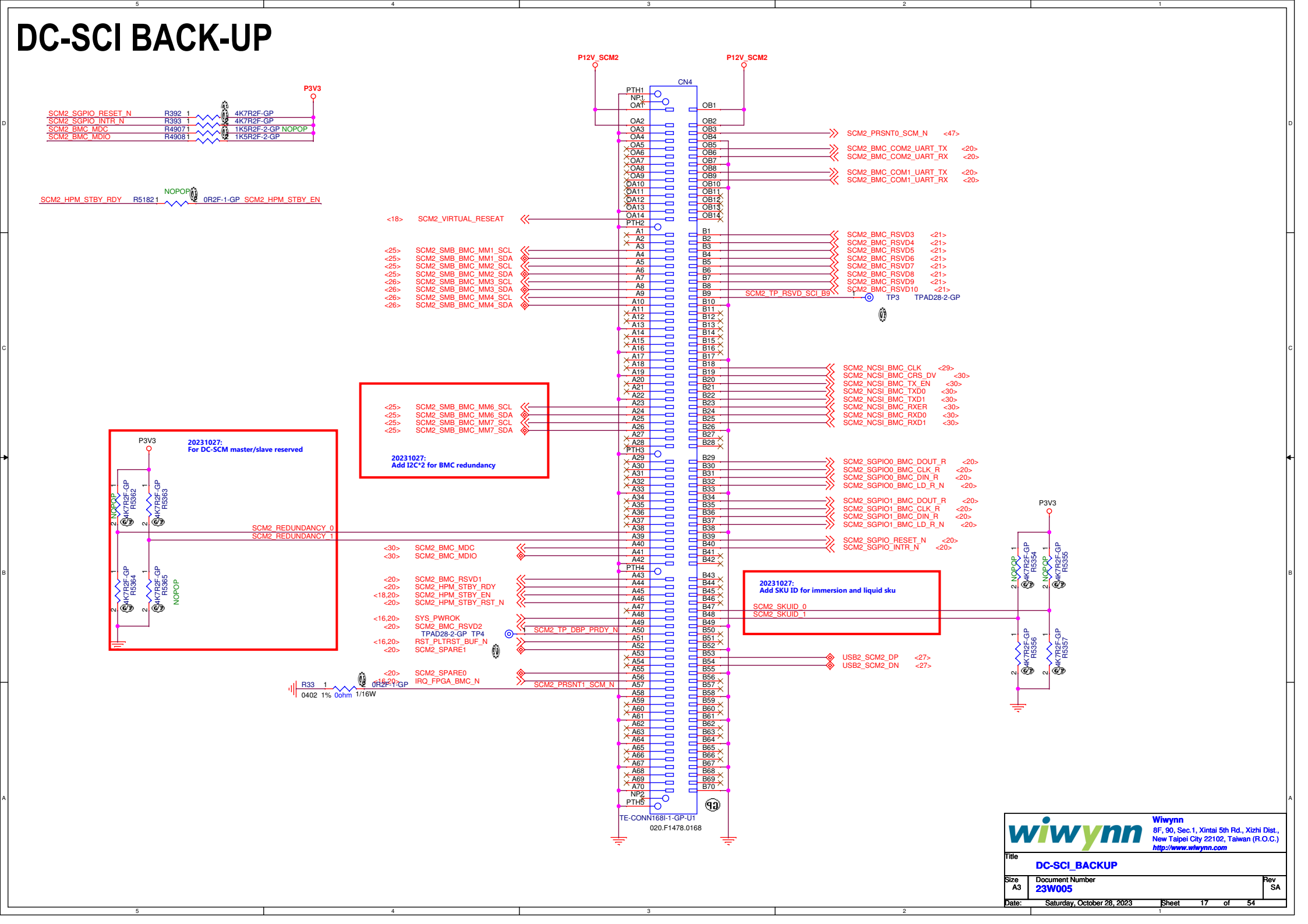
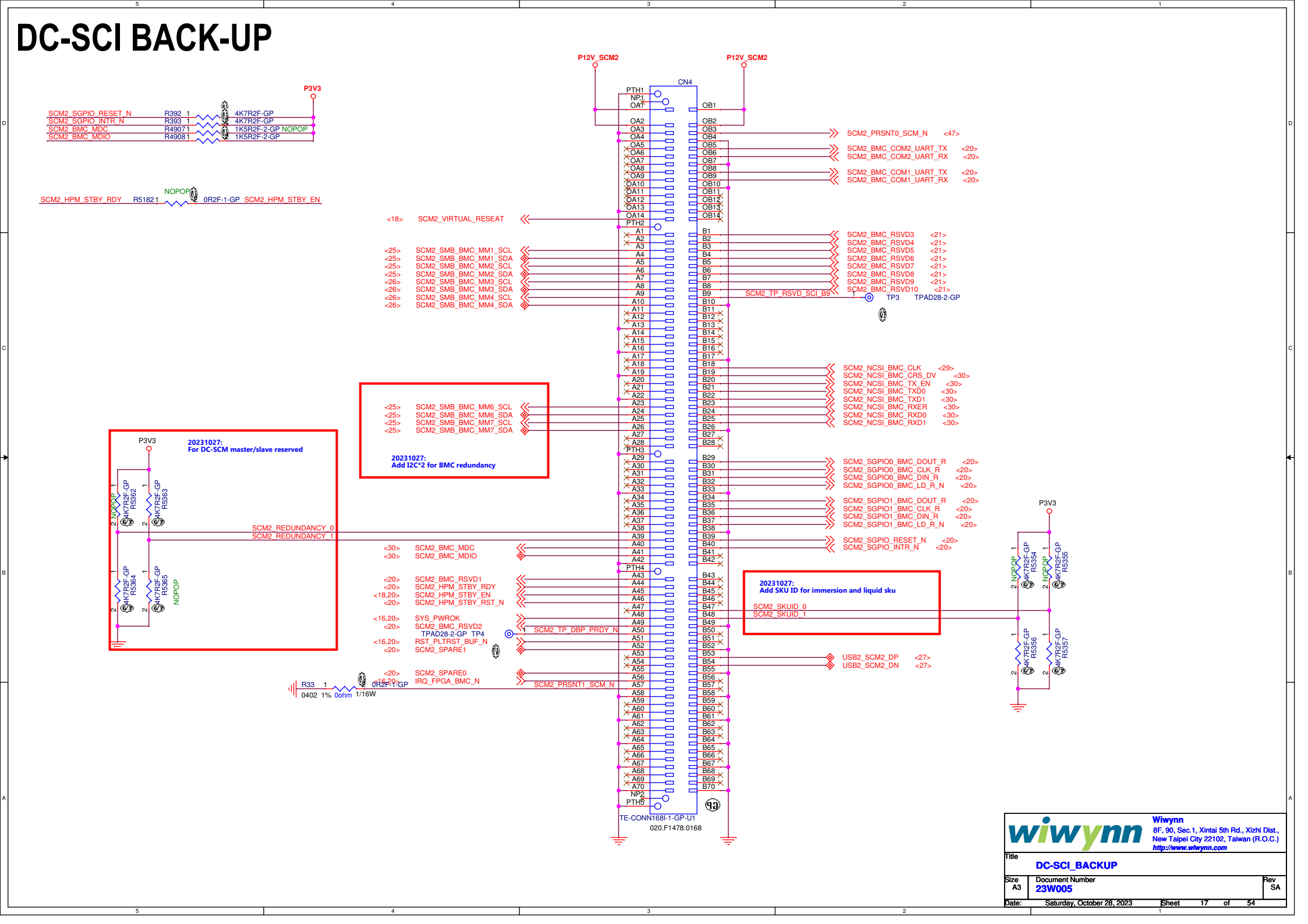
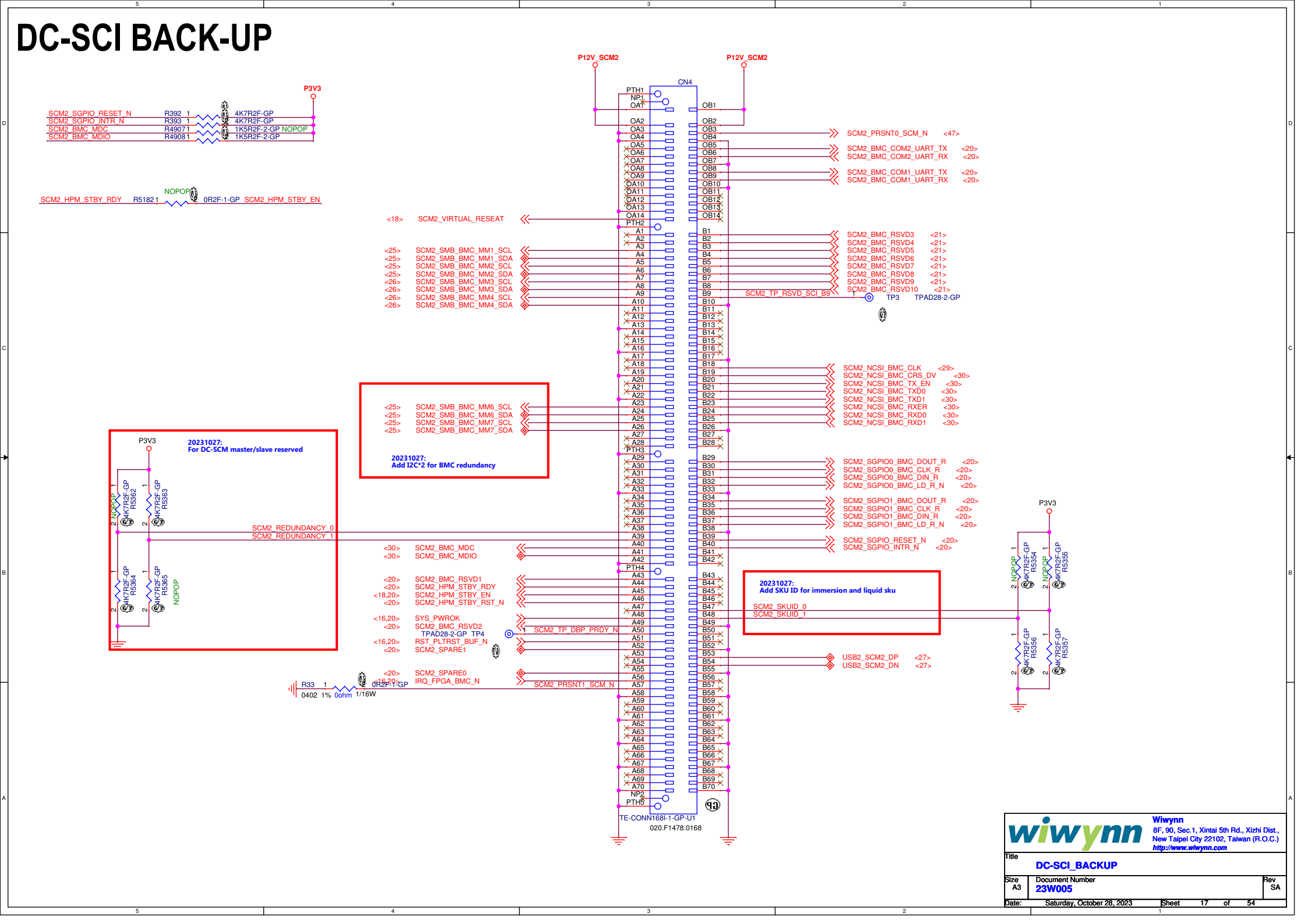
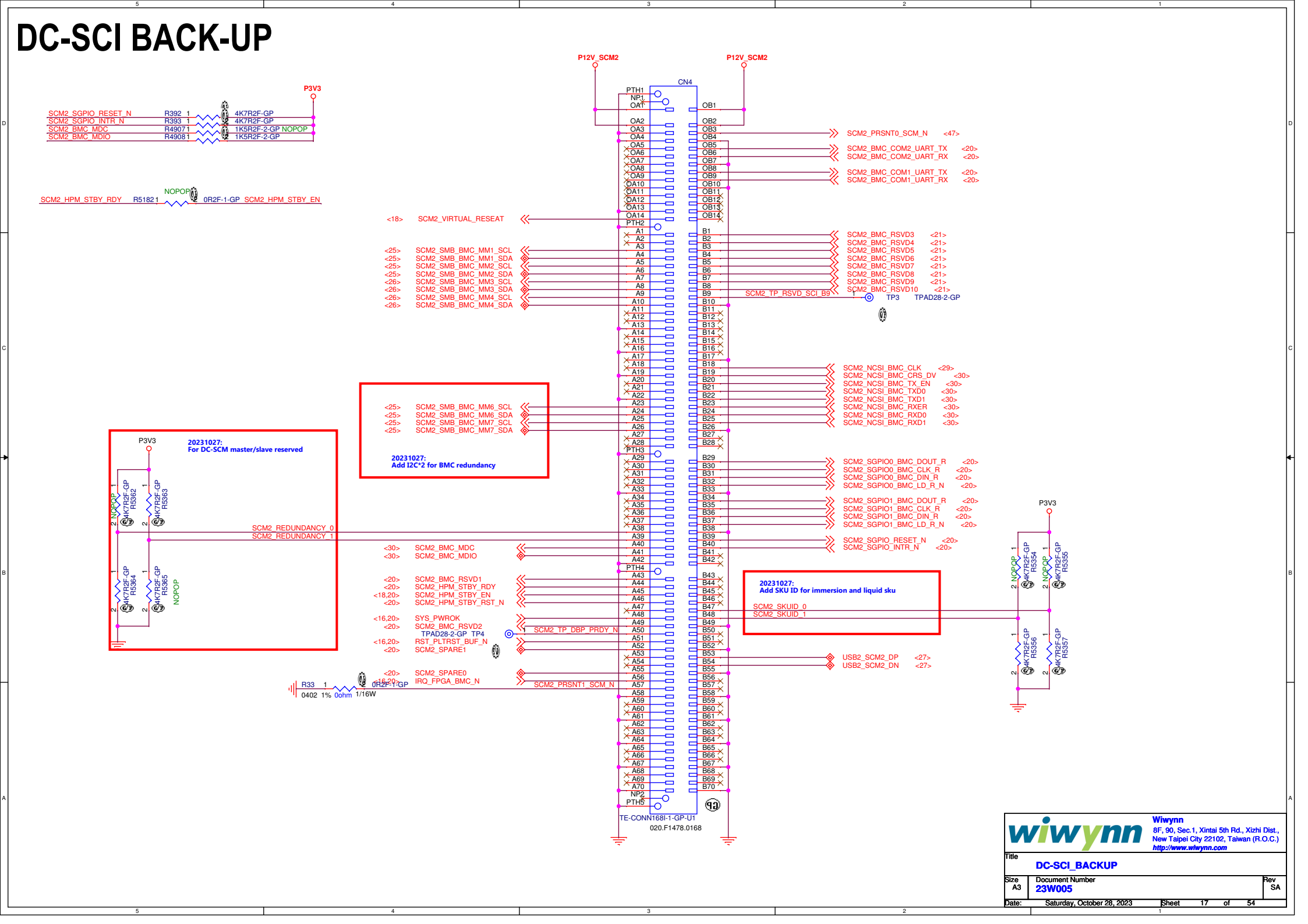
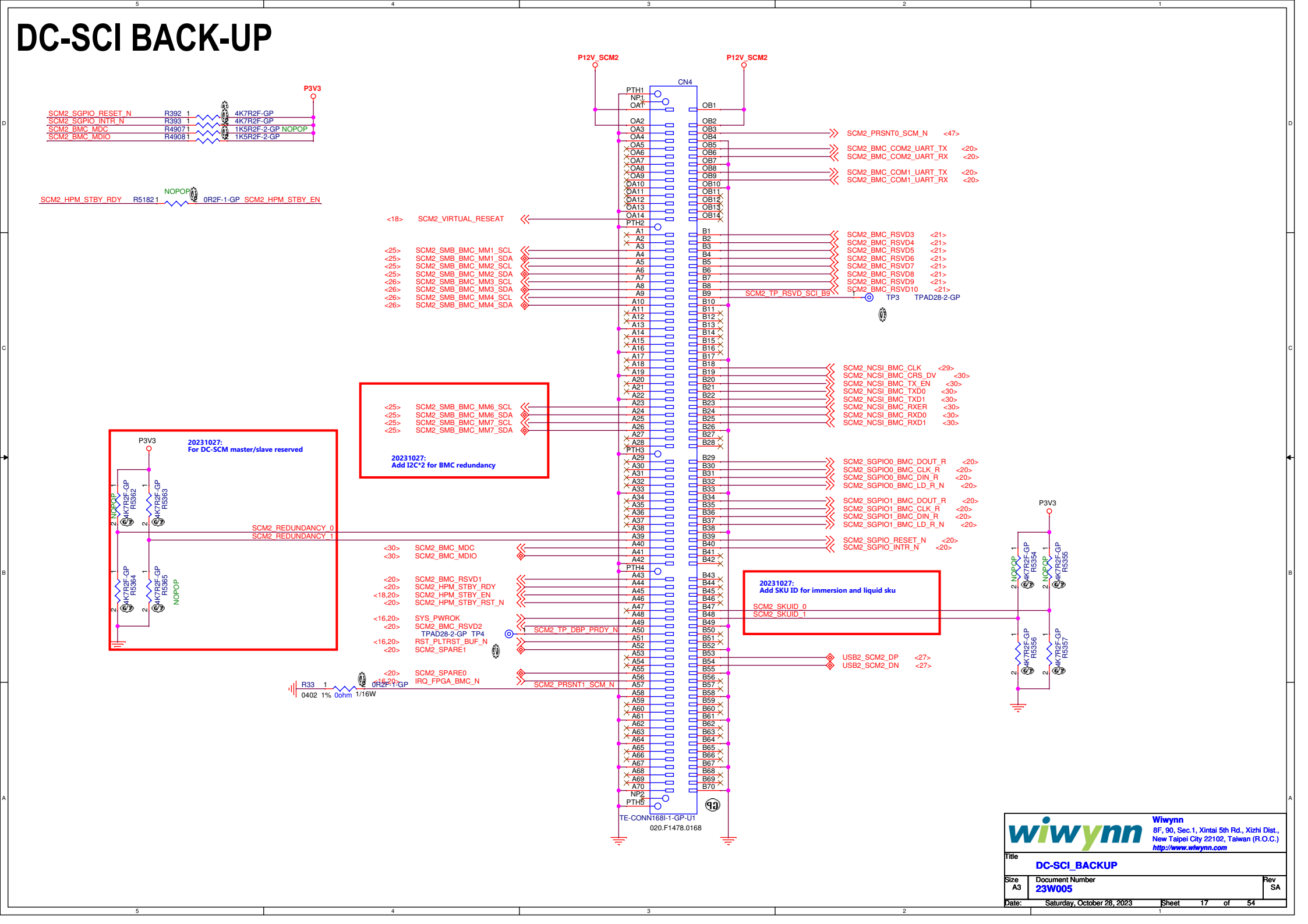
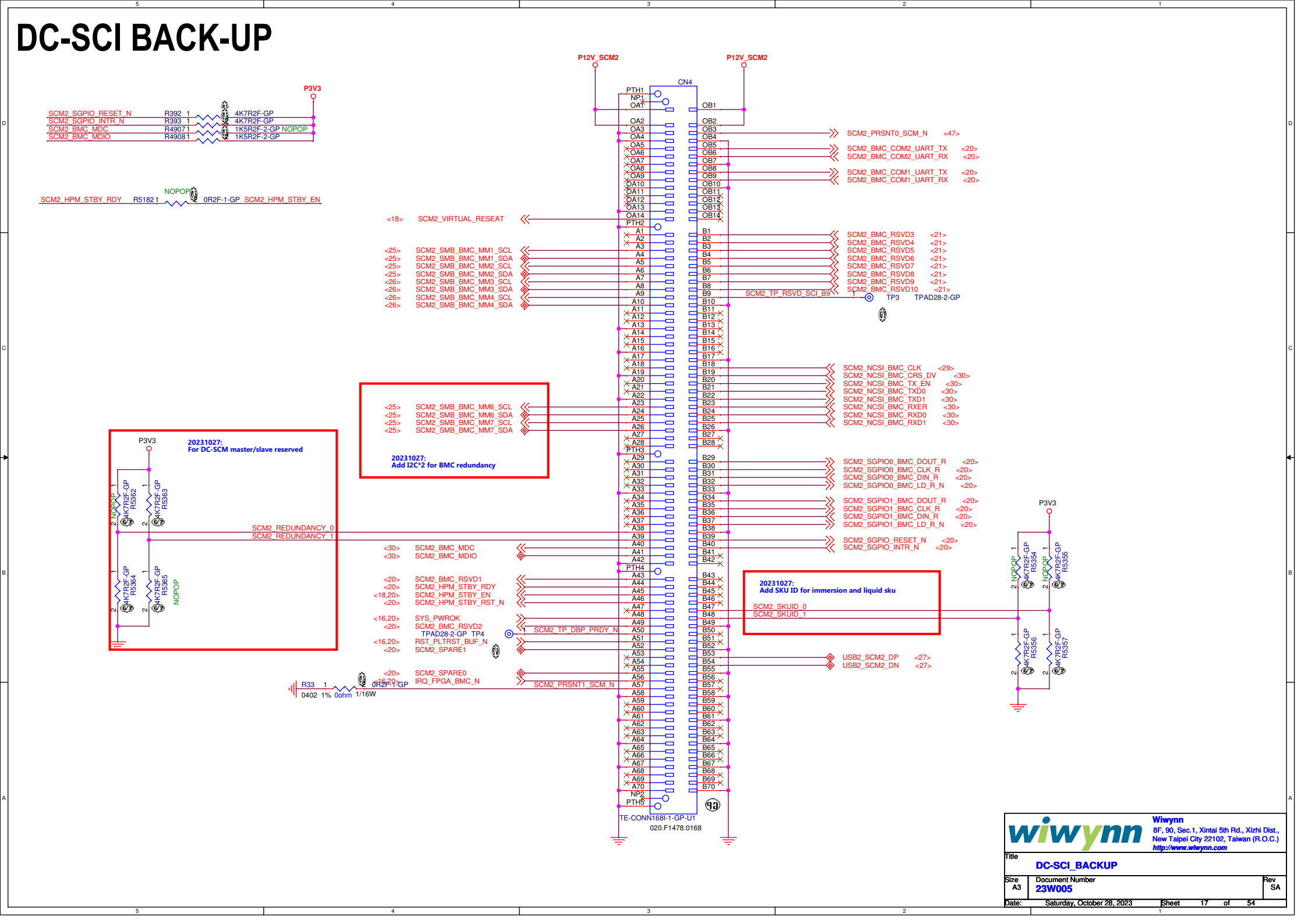
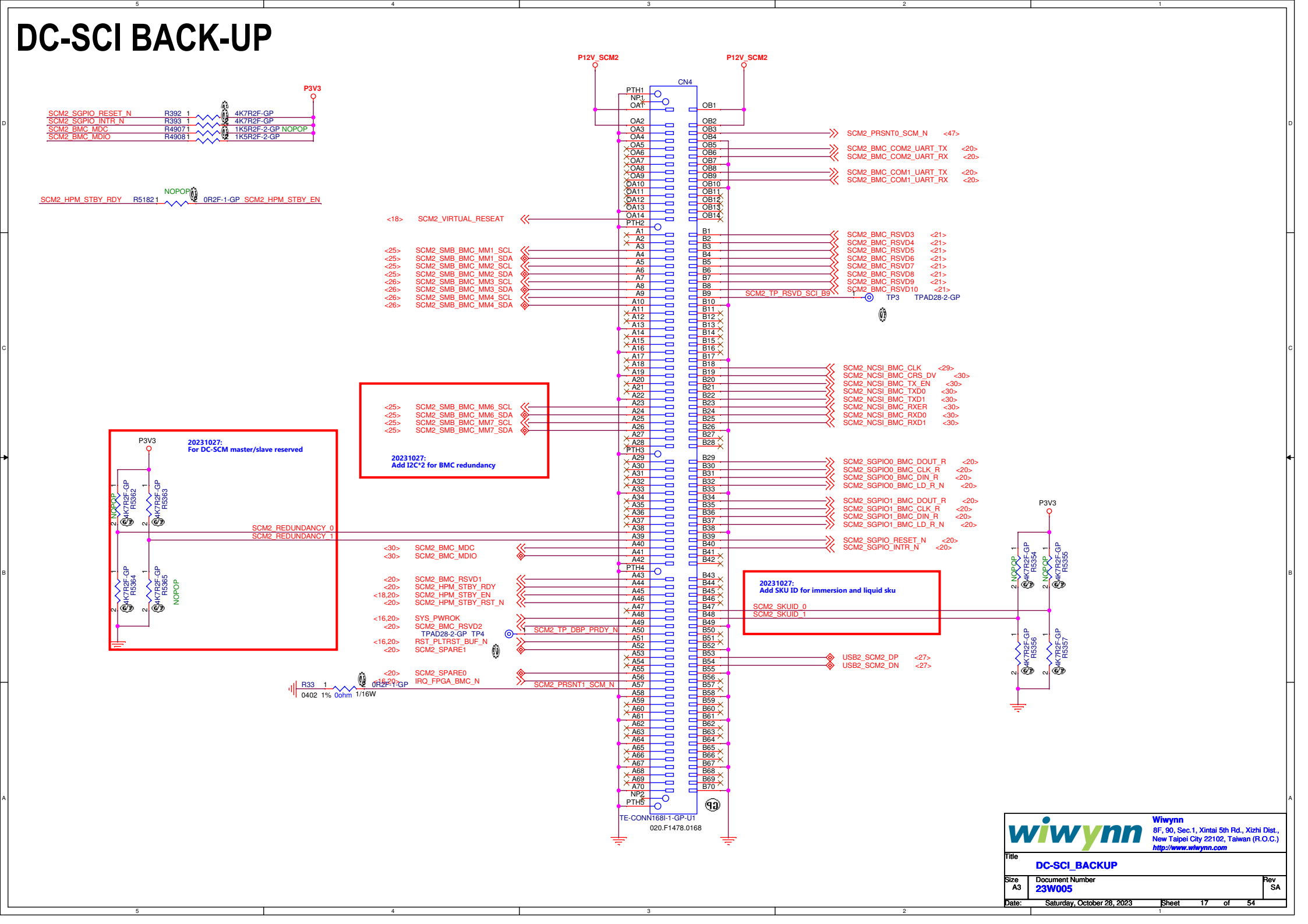
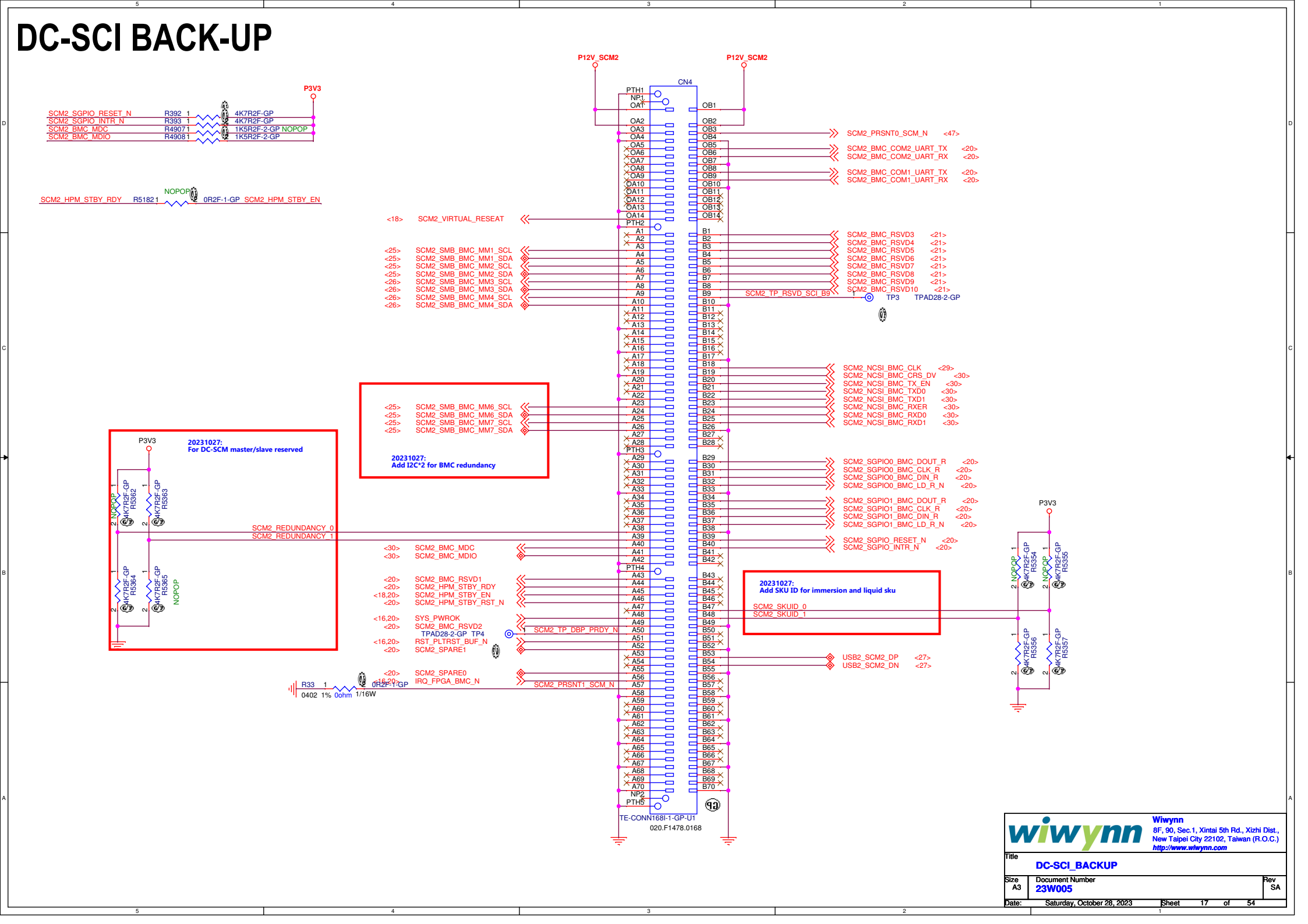
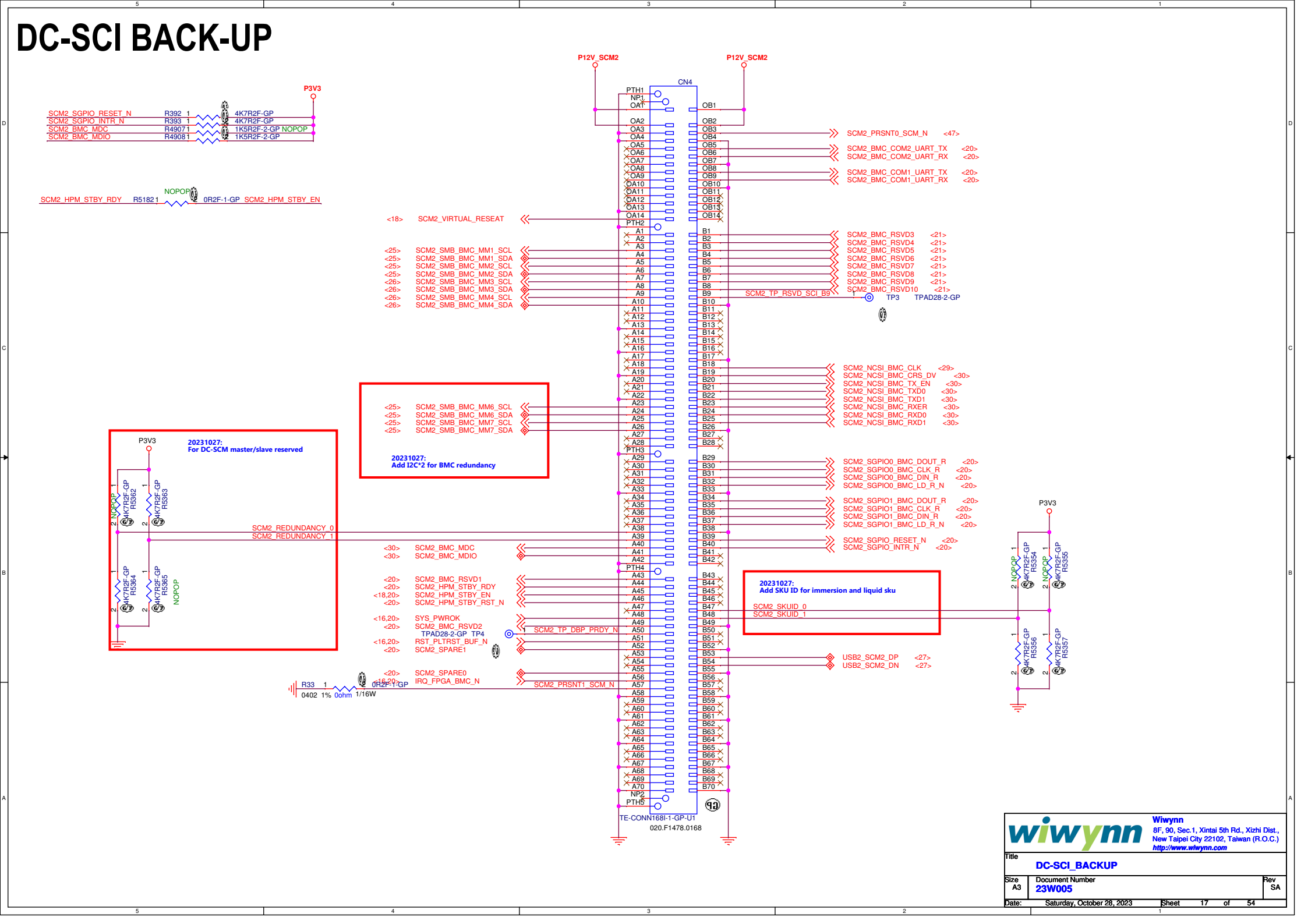
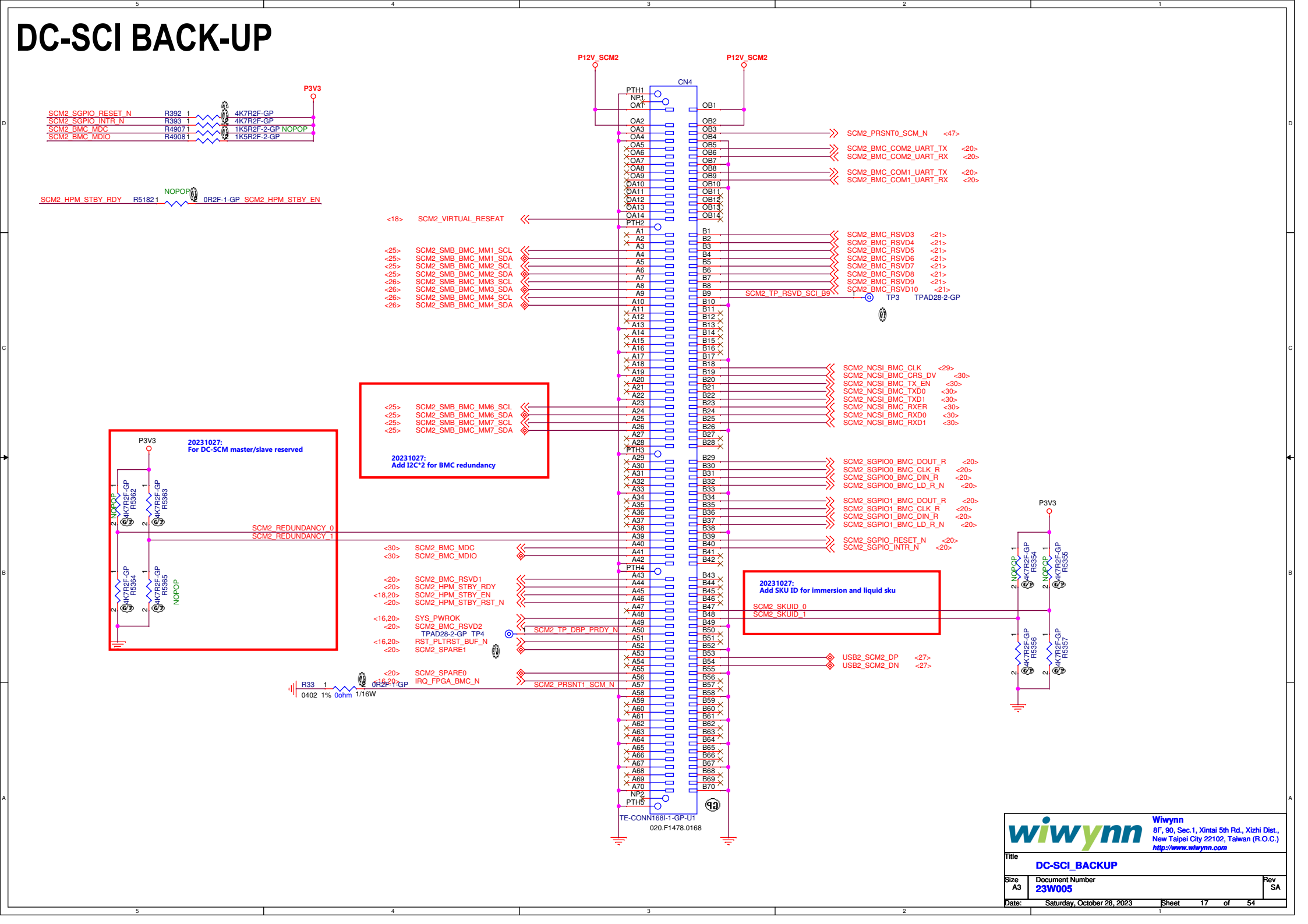
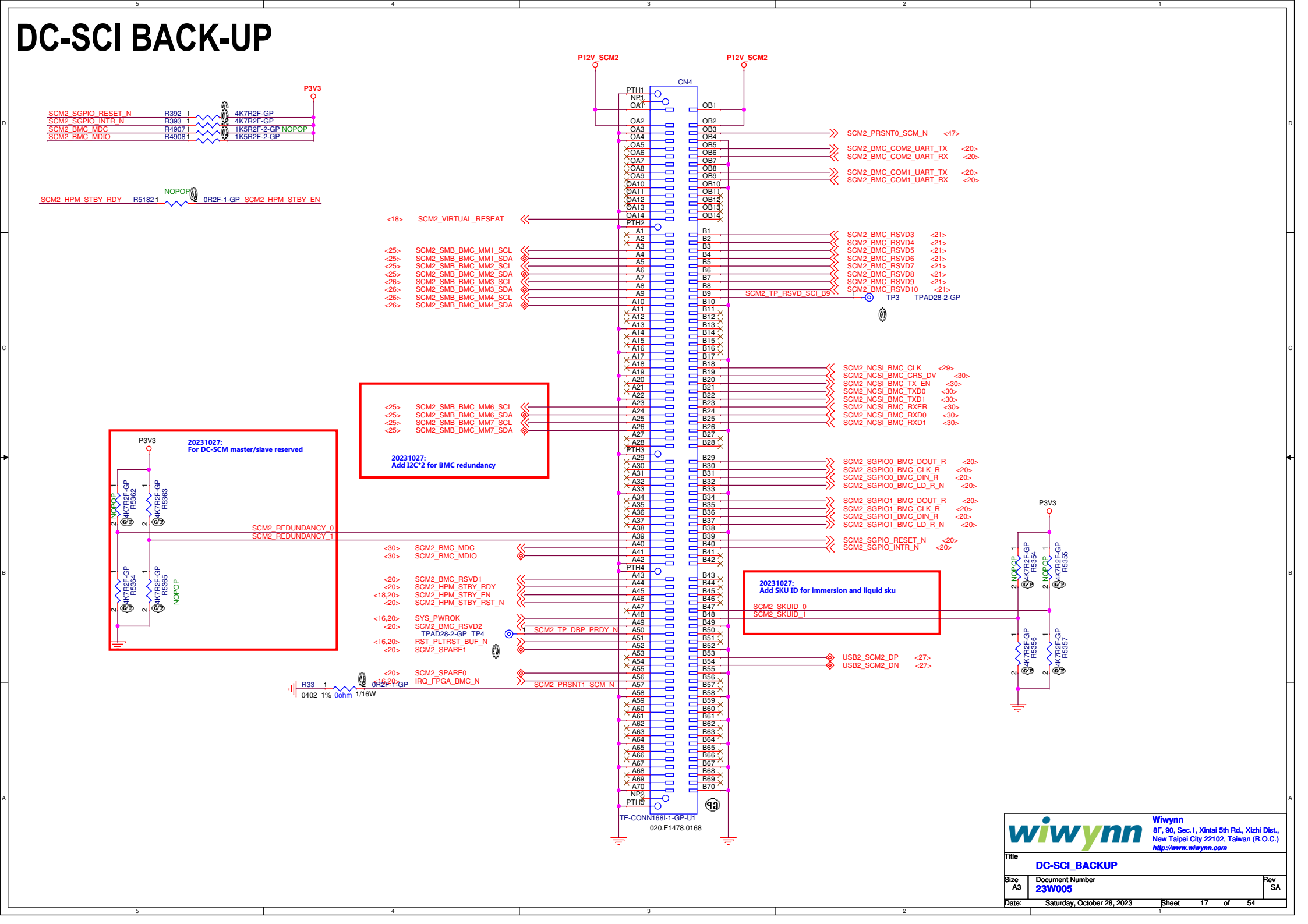
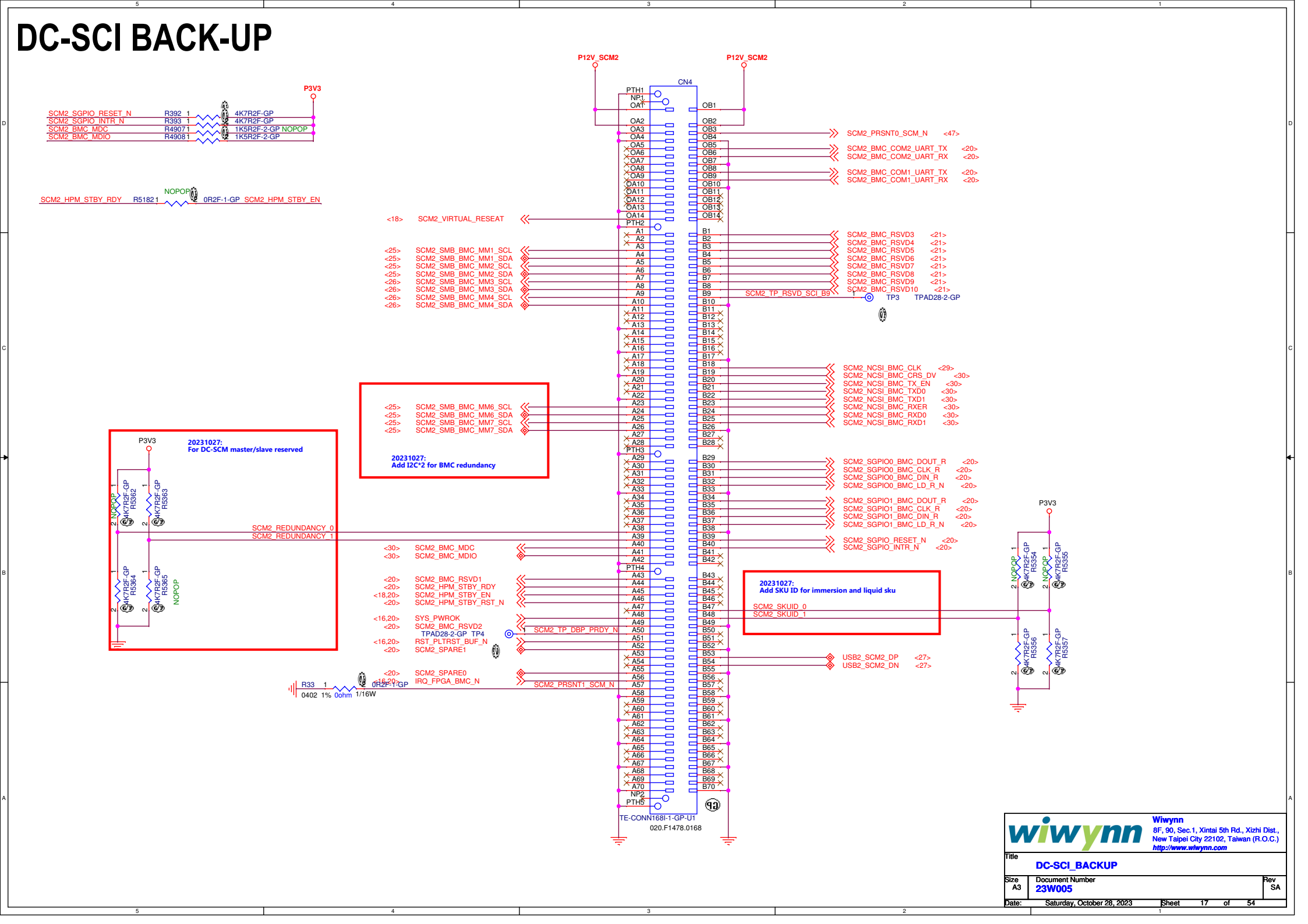
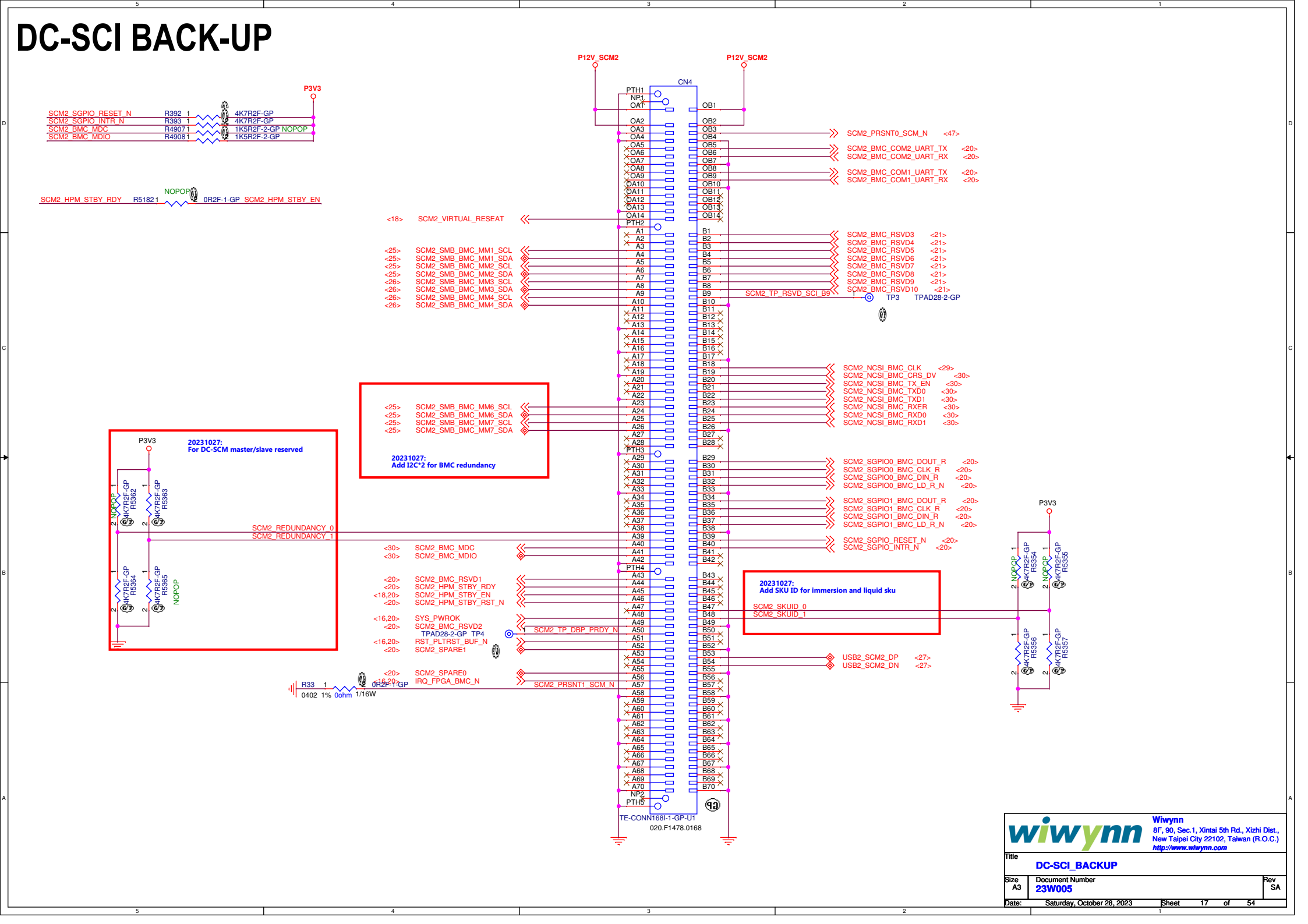
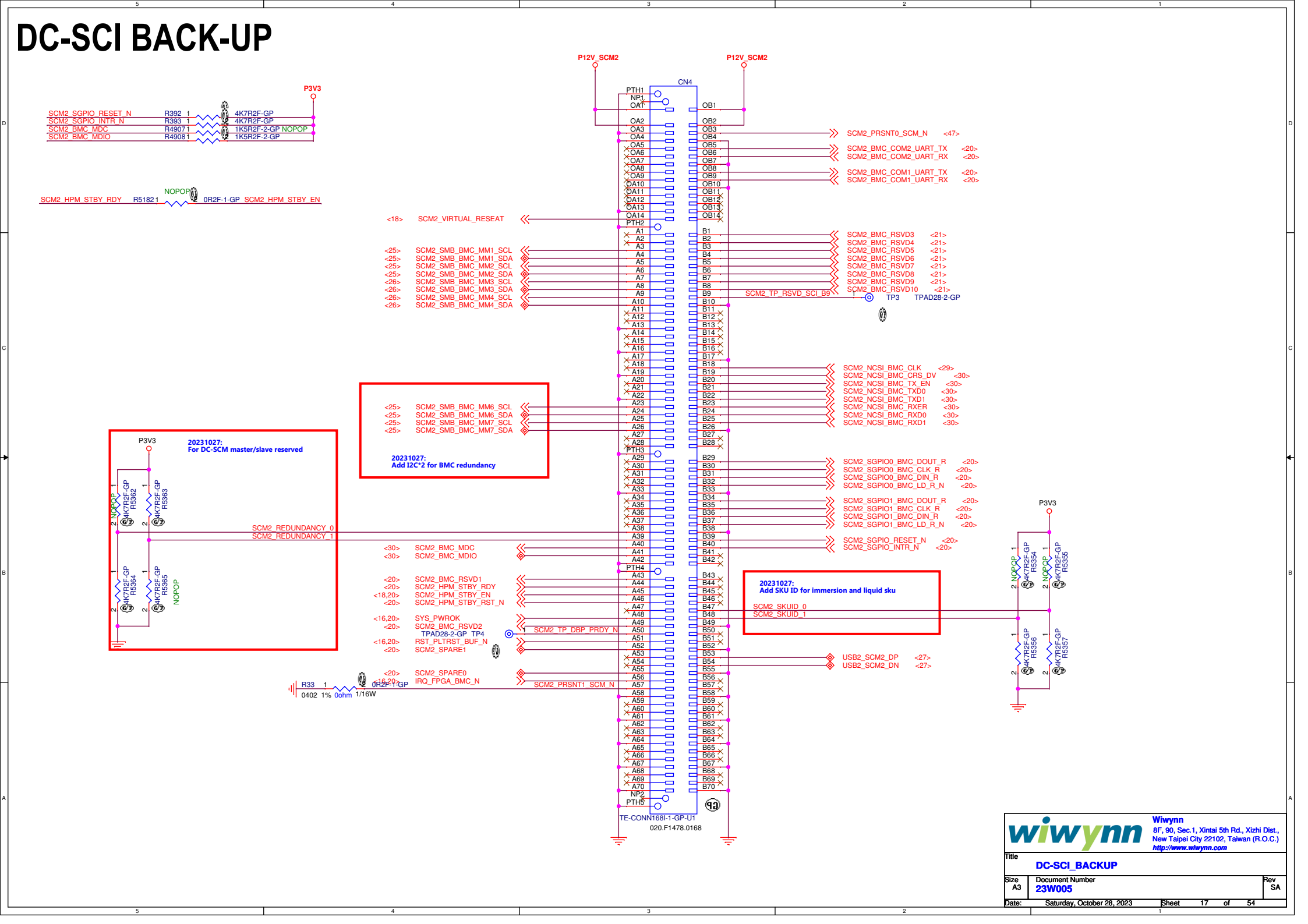
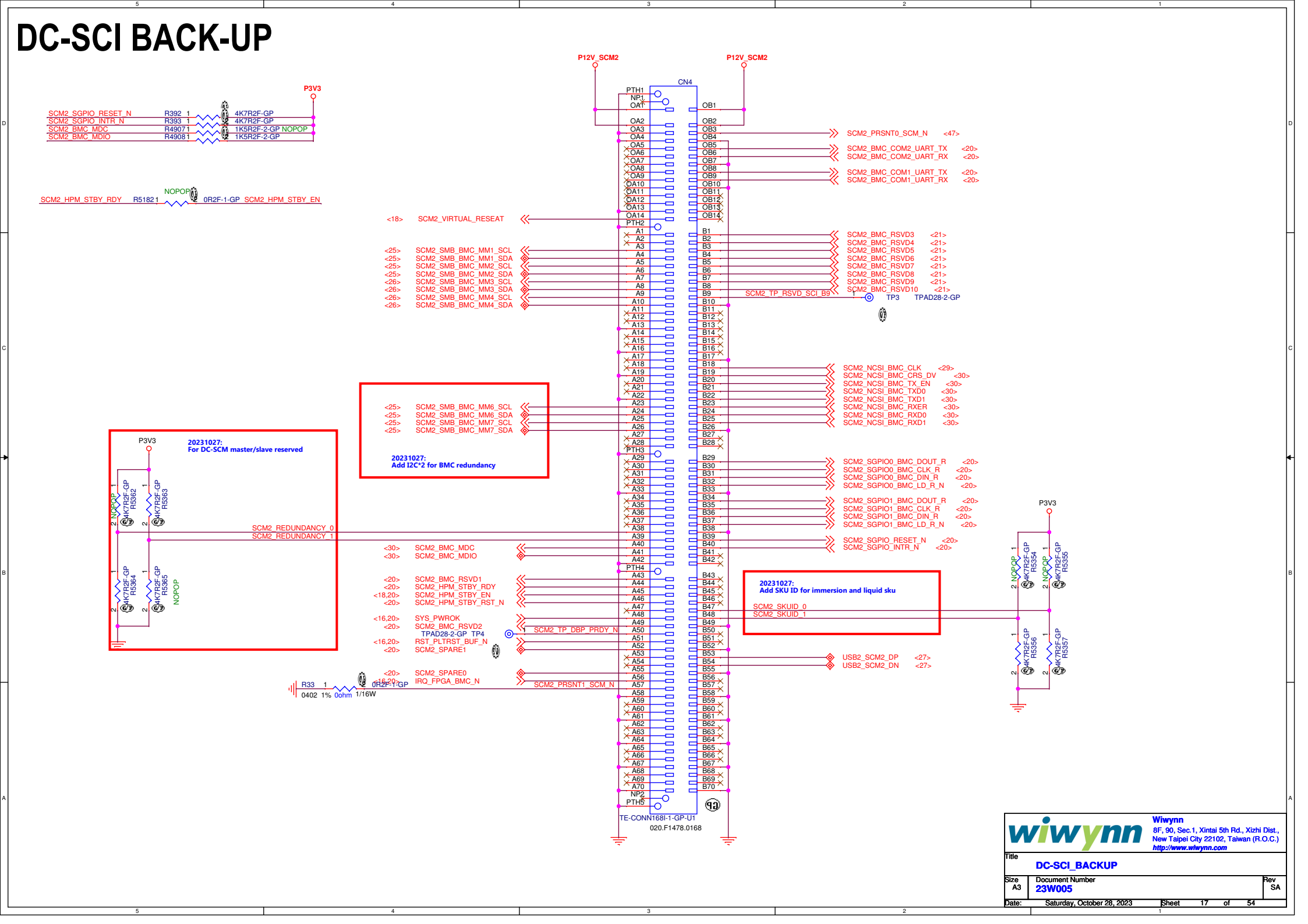
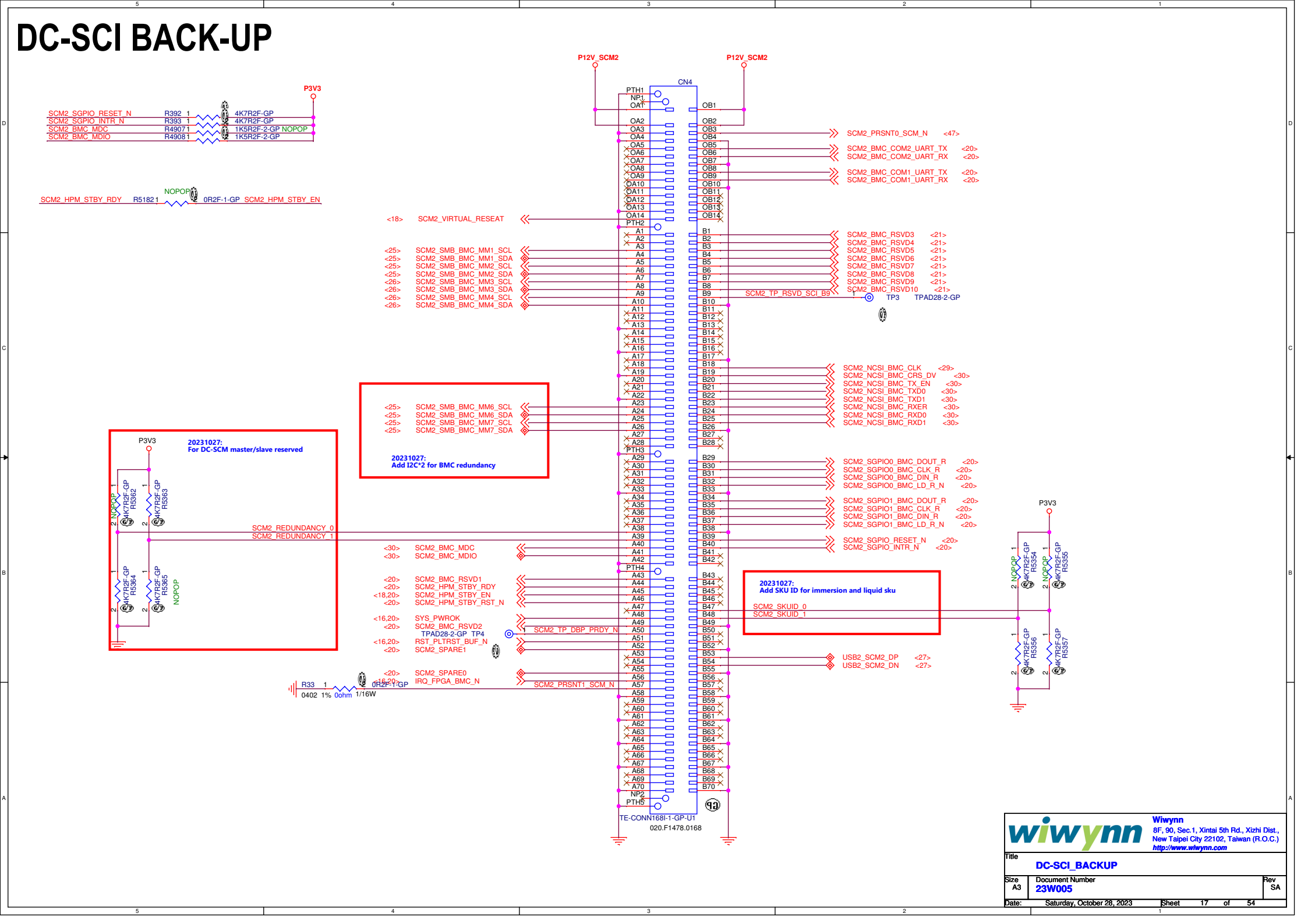
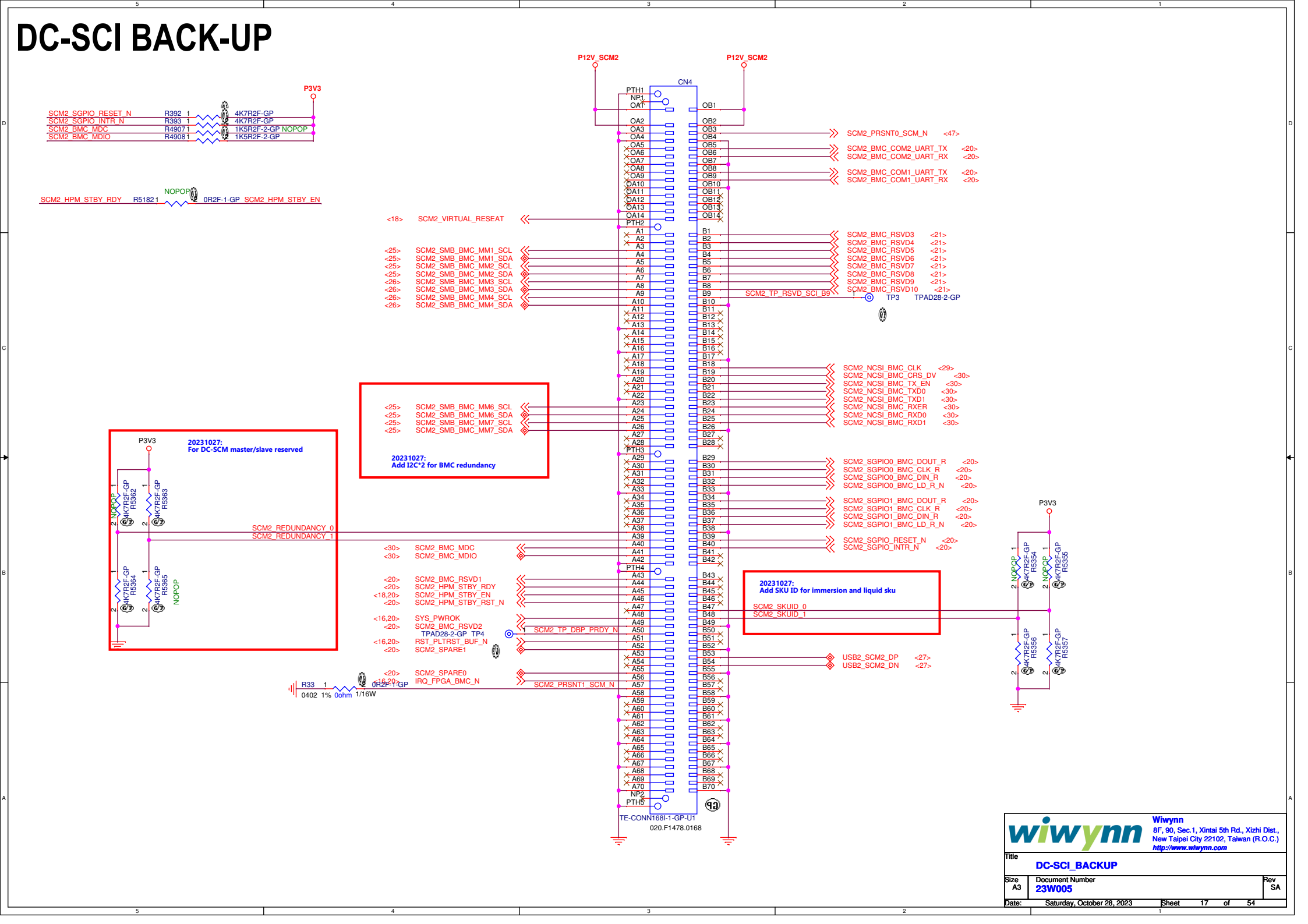
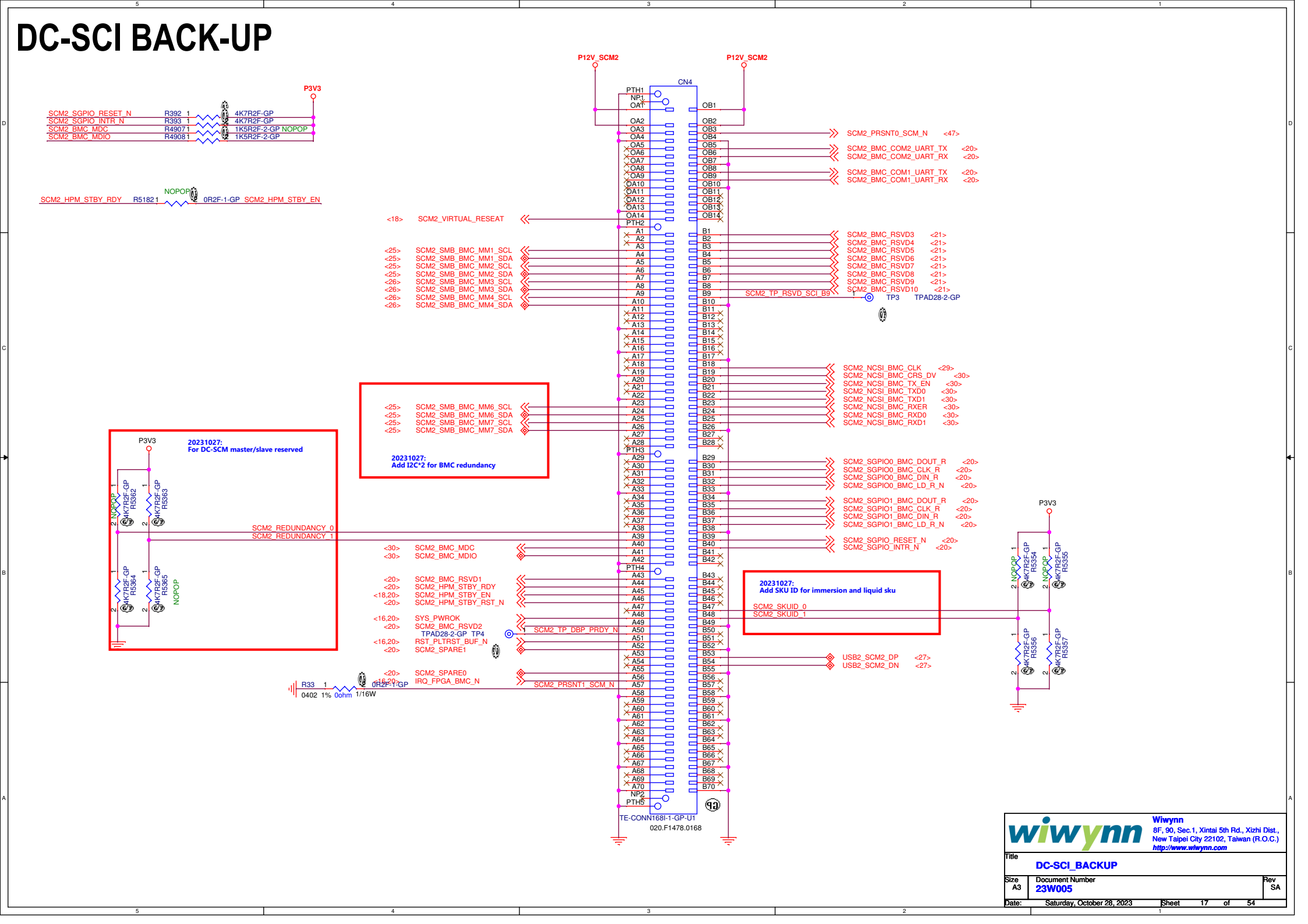
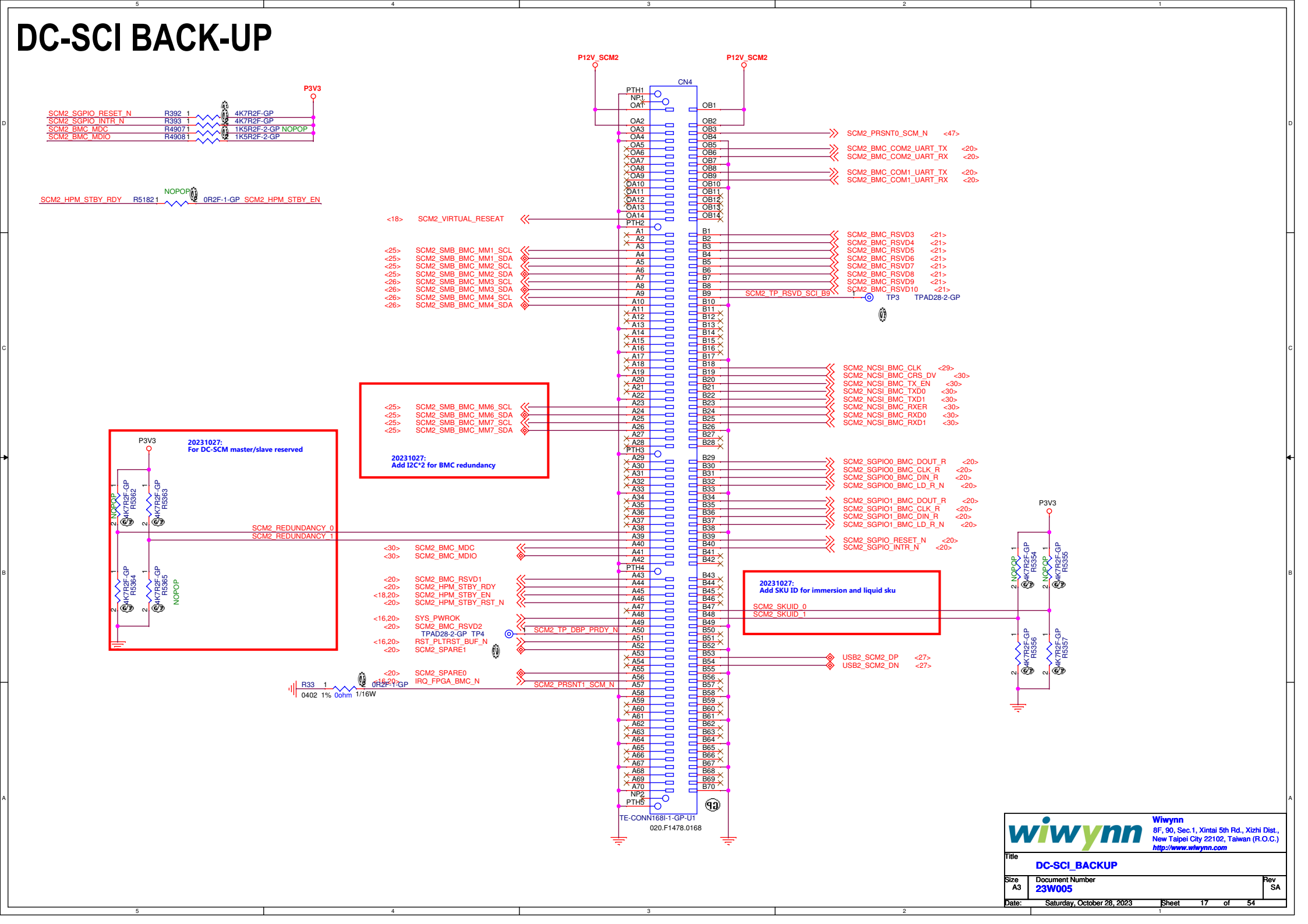
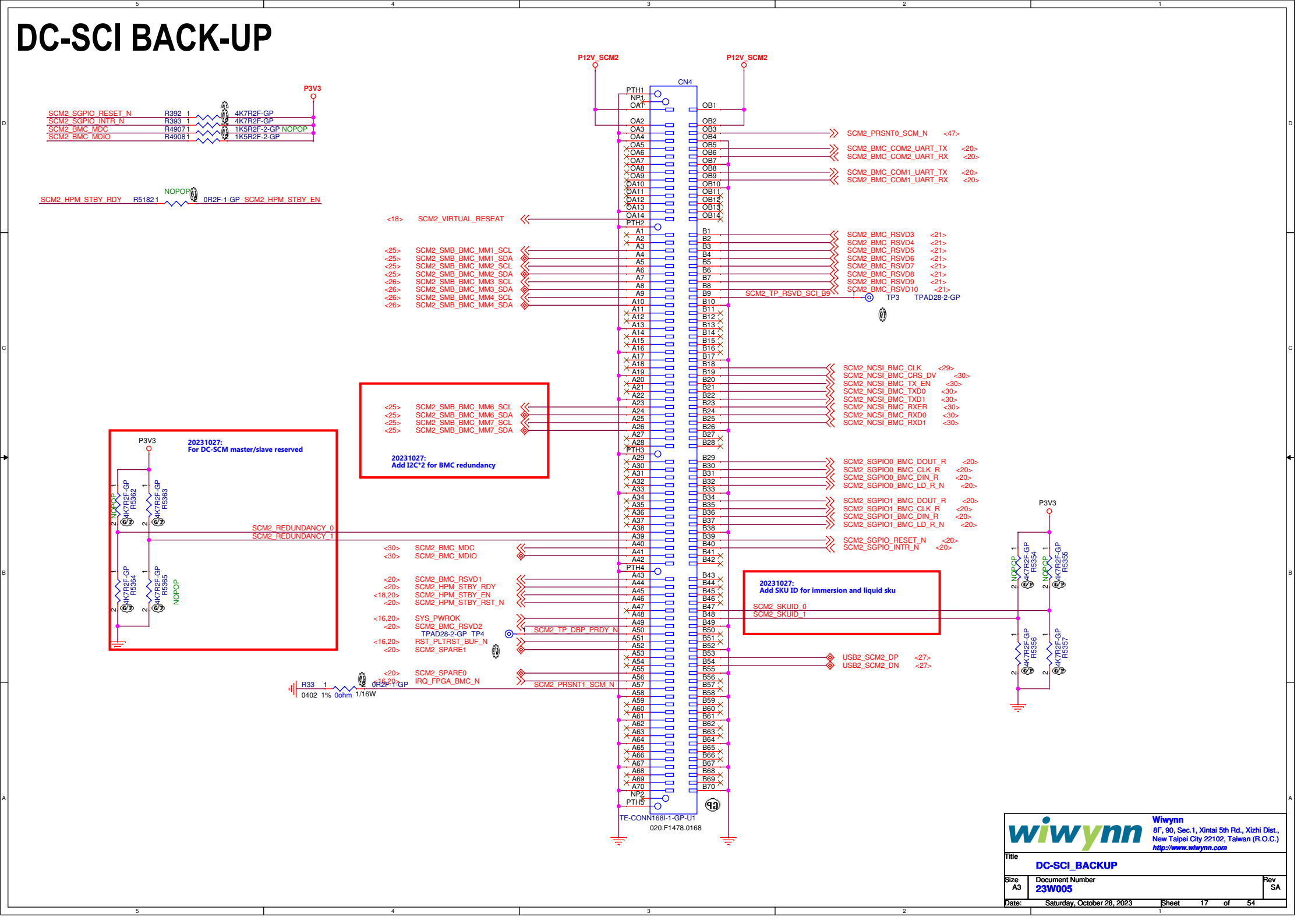
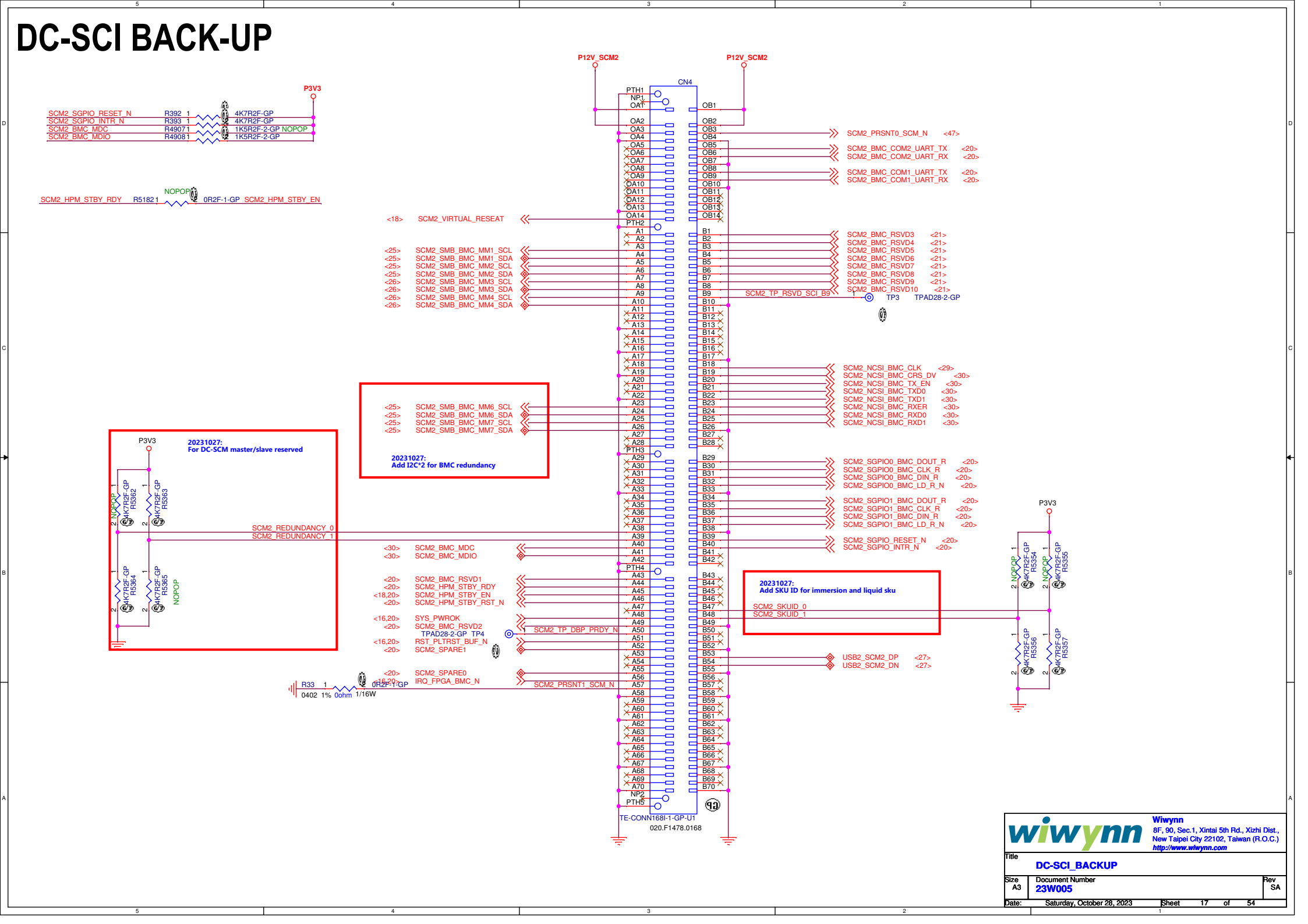
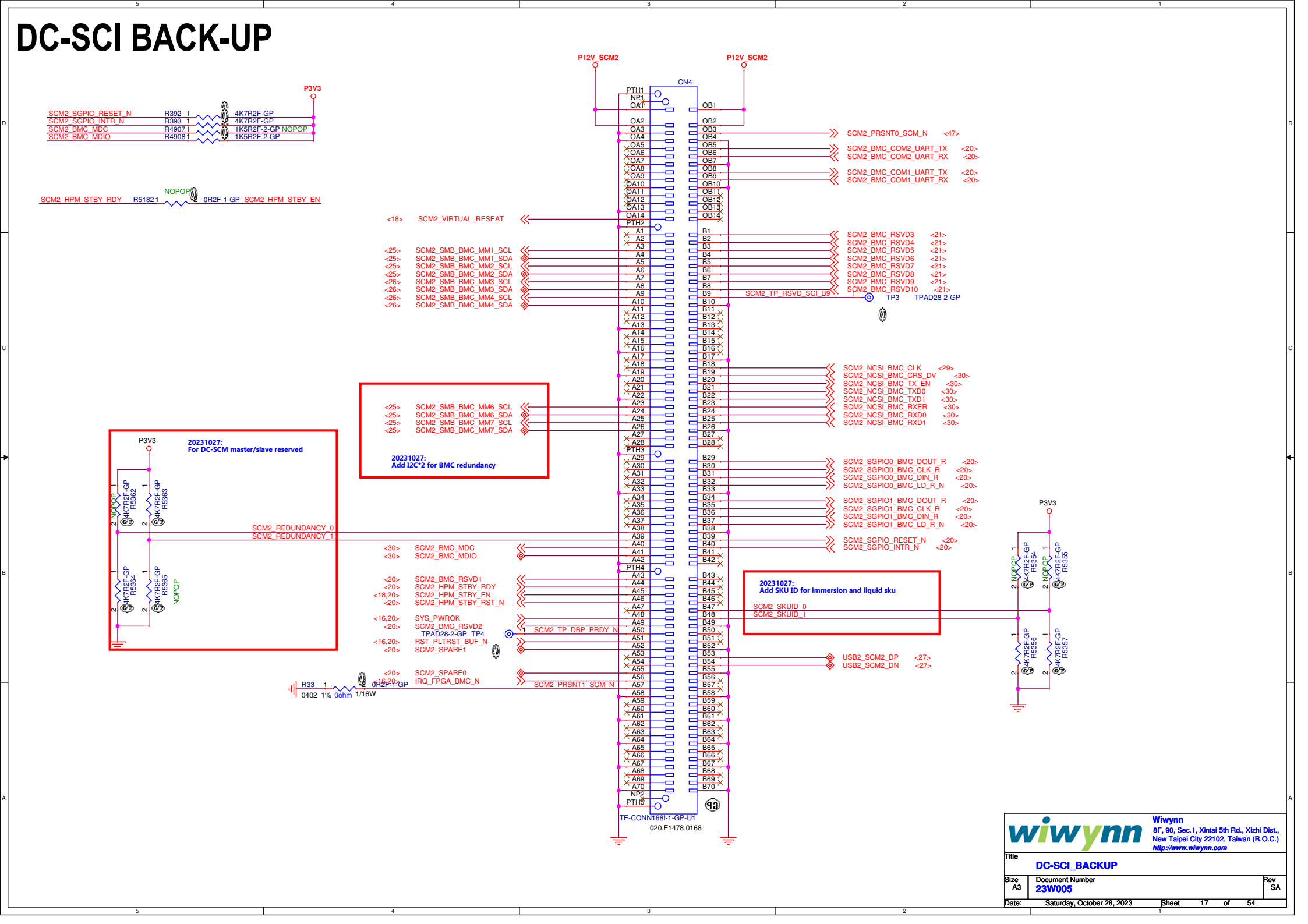
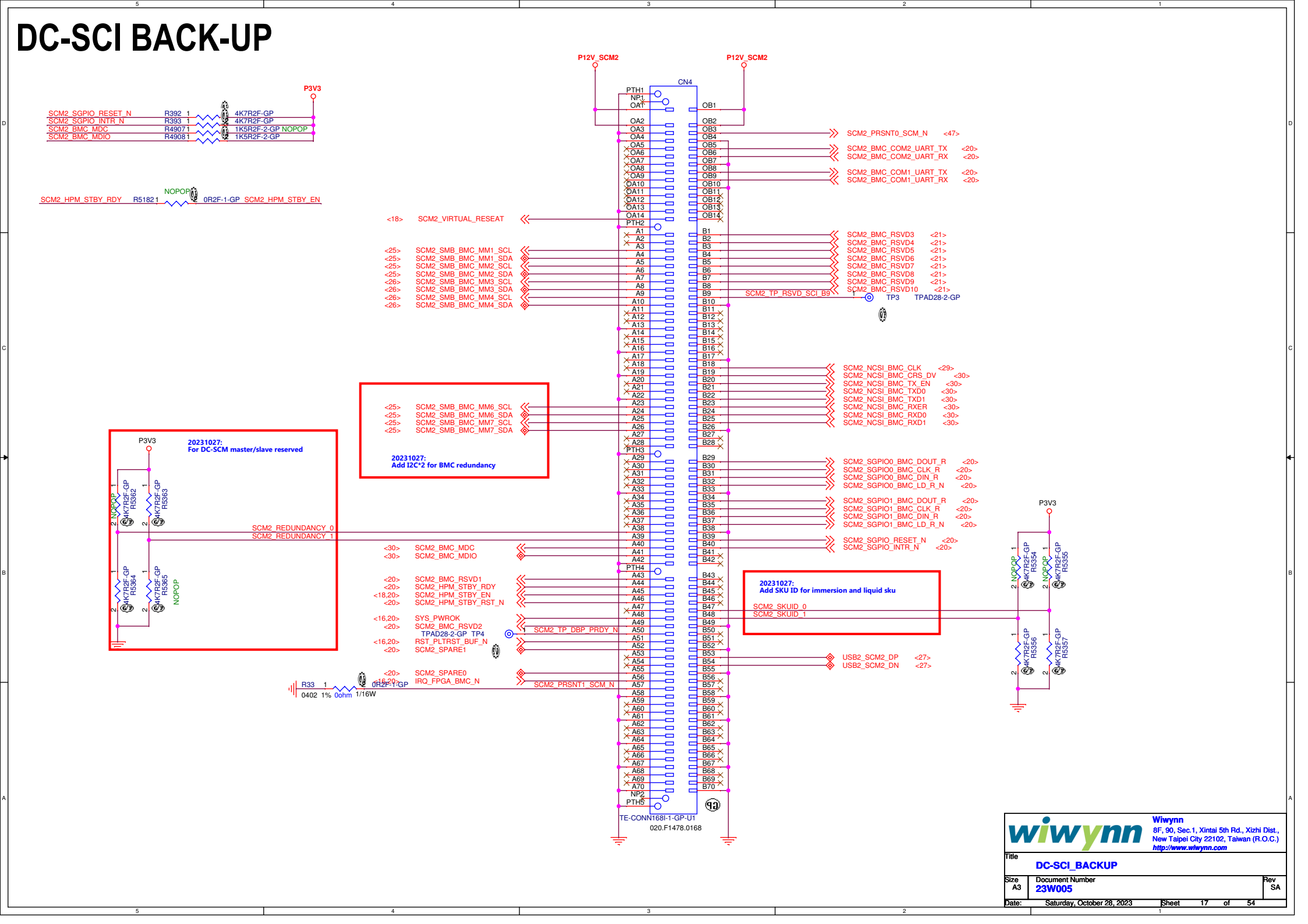
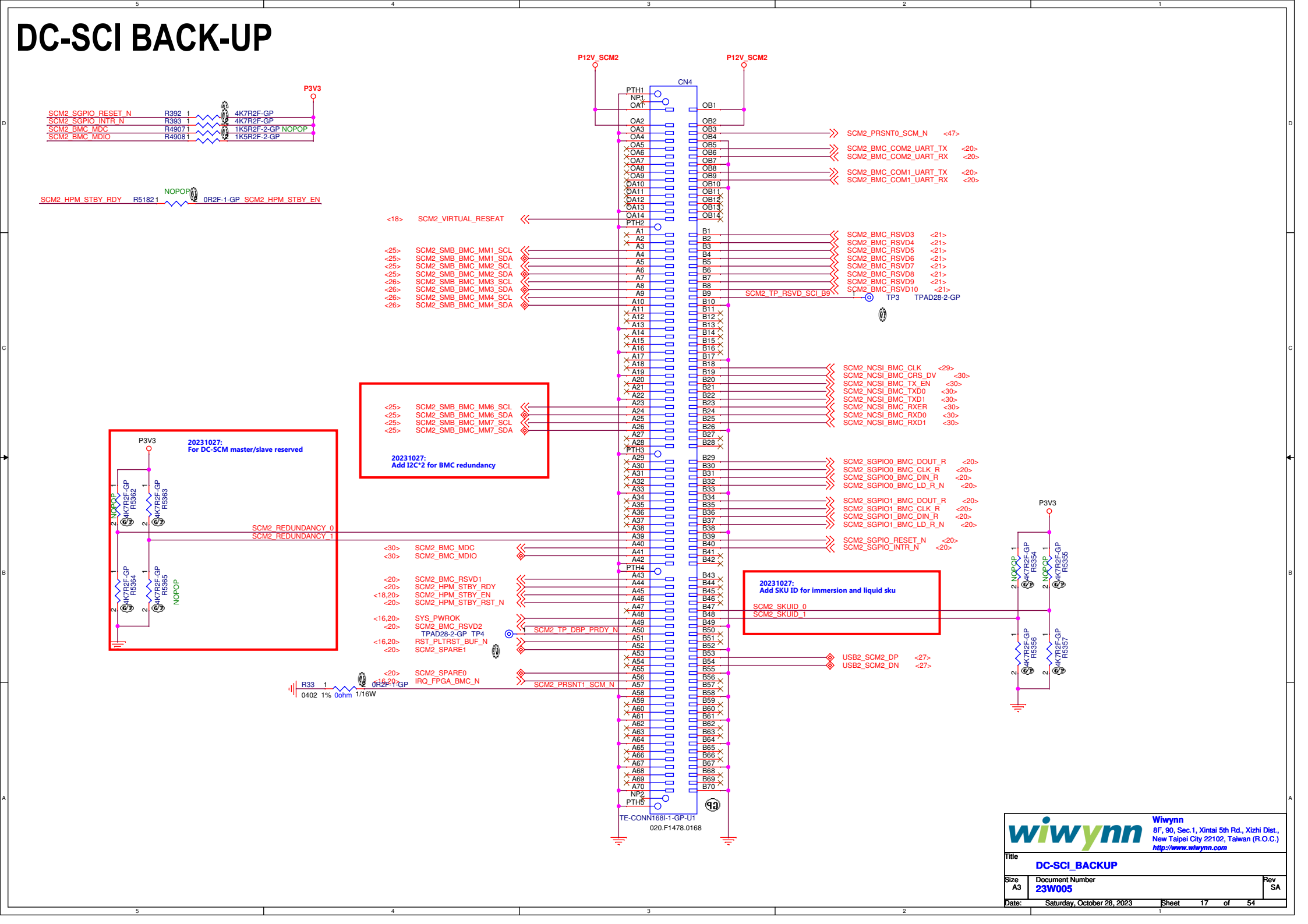
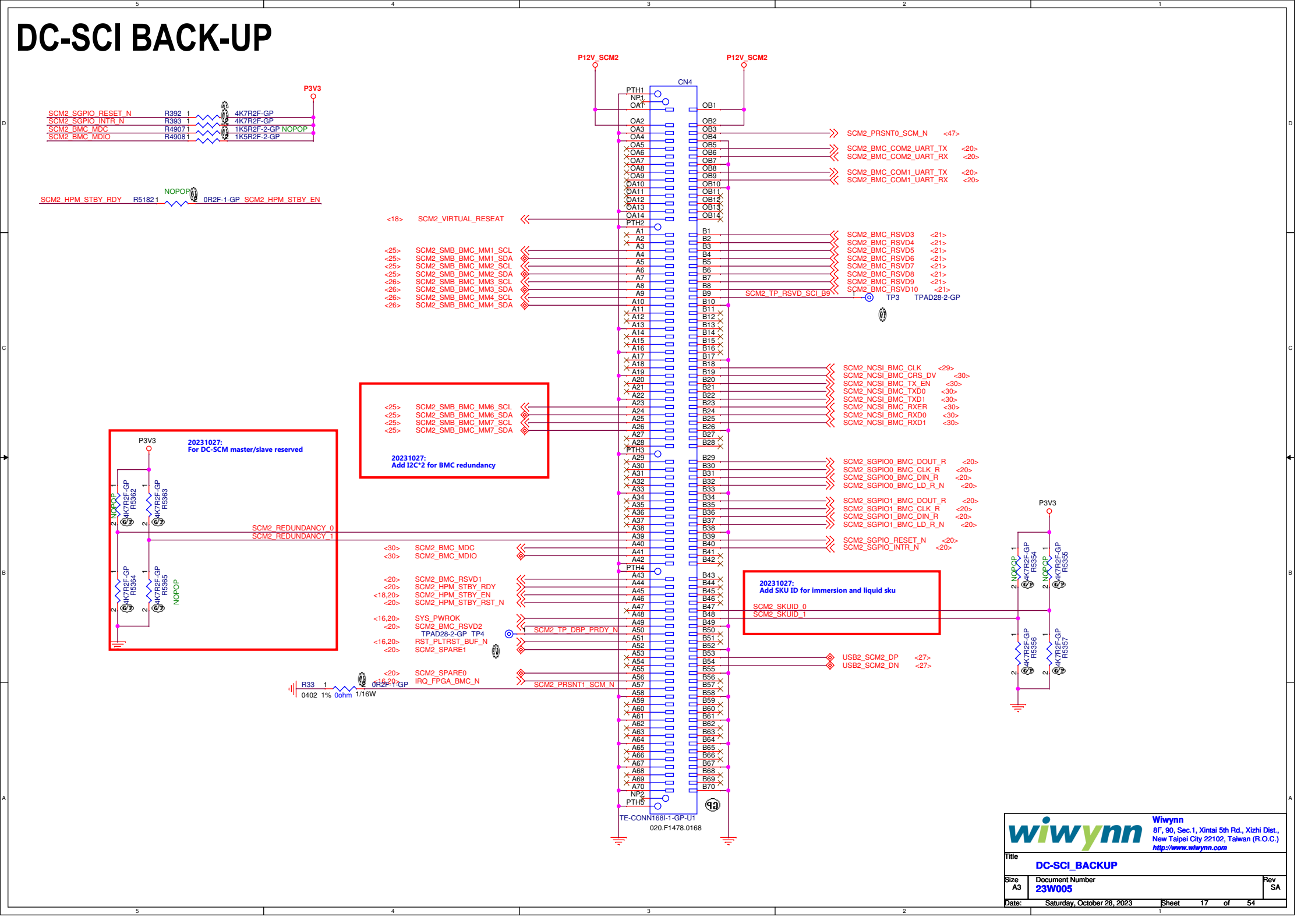
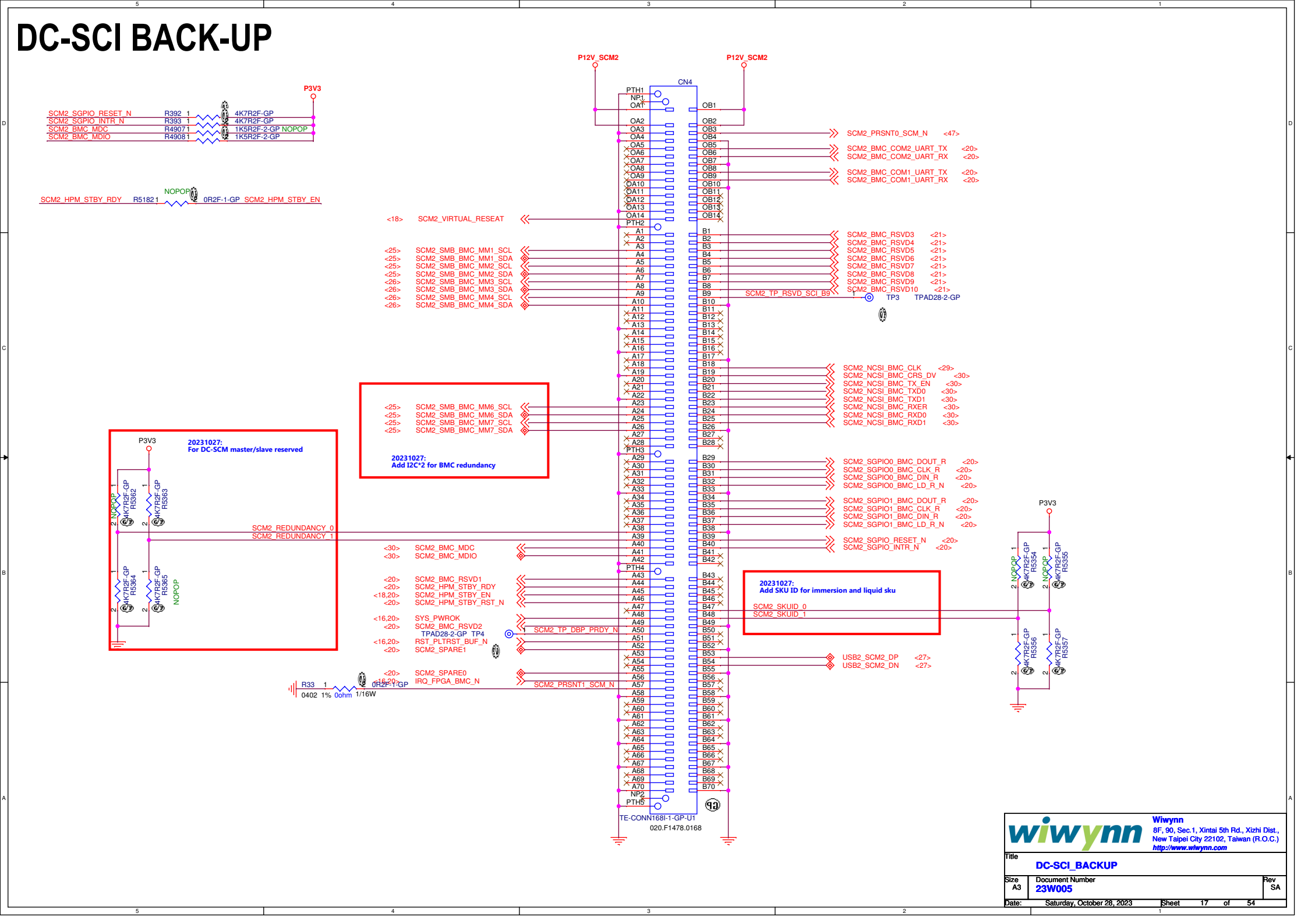
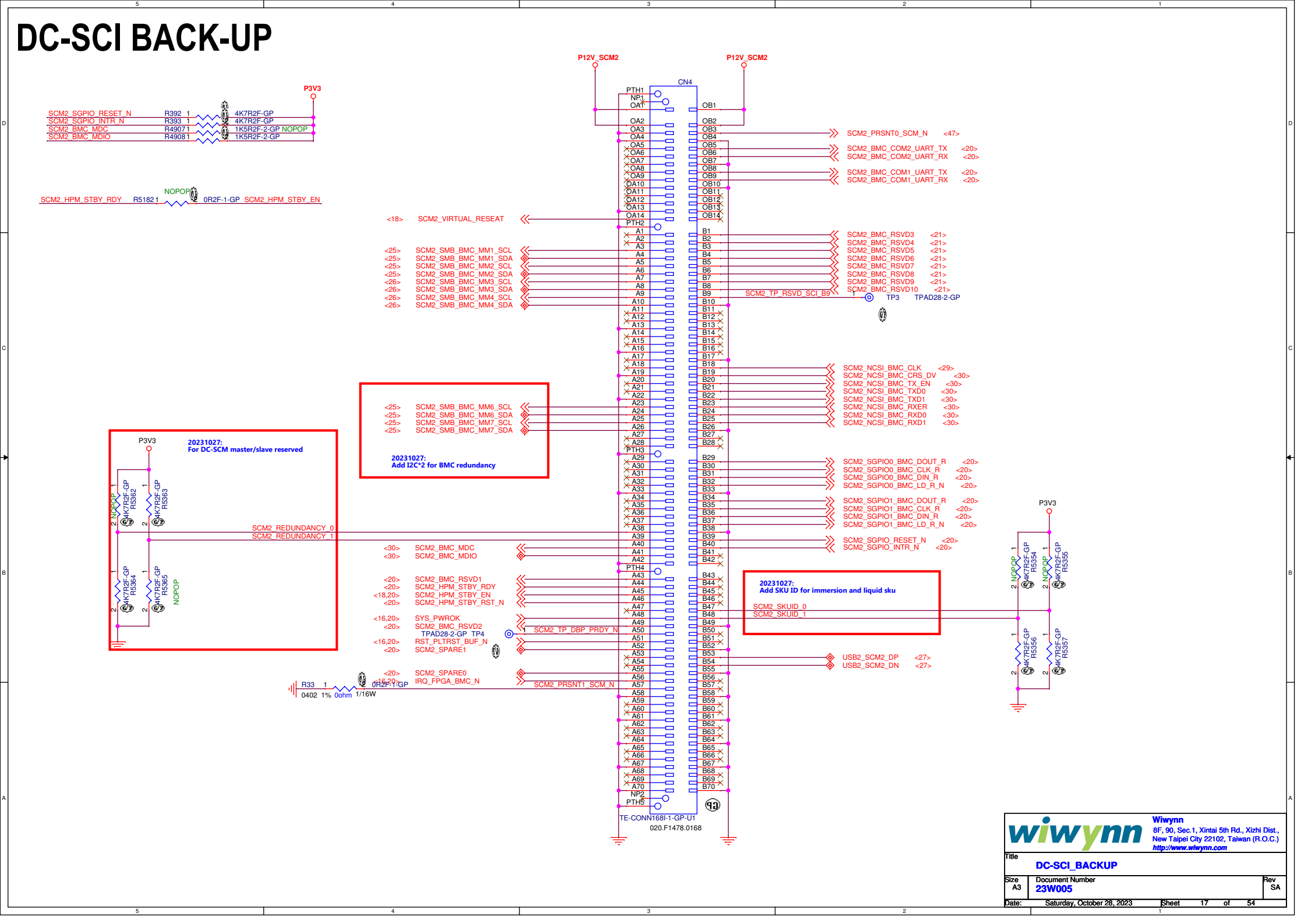


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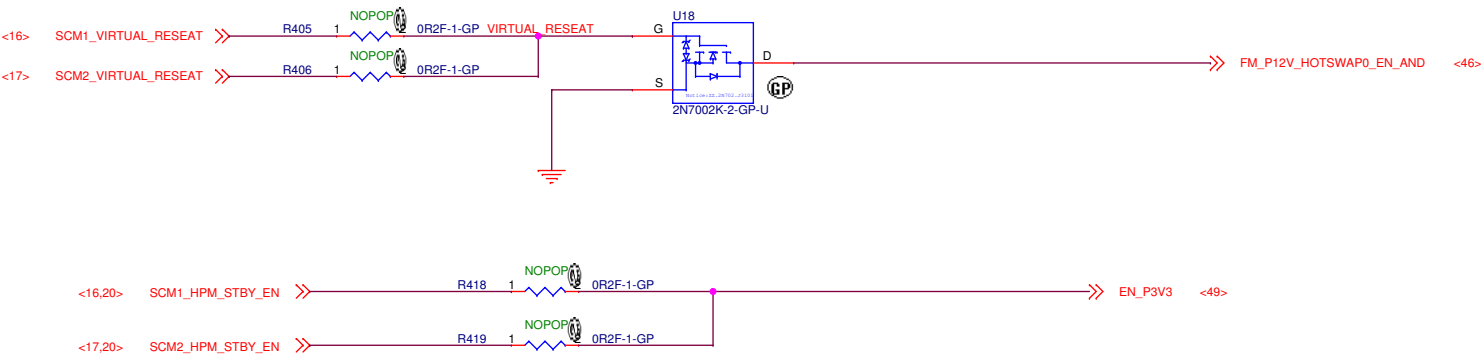
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DC-SCI HOST

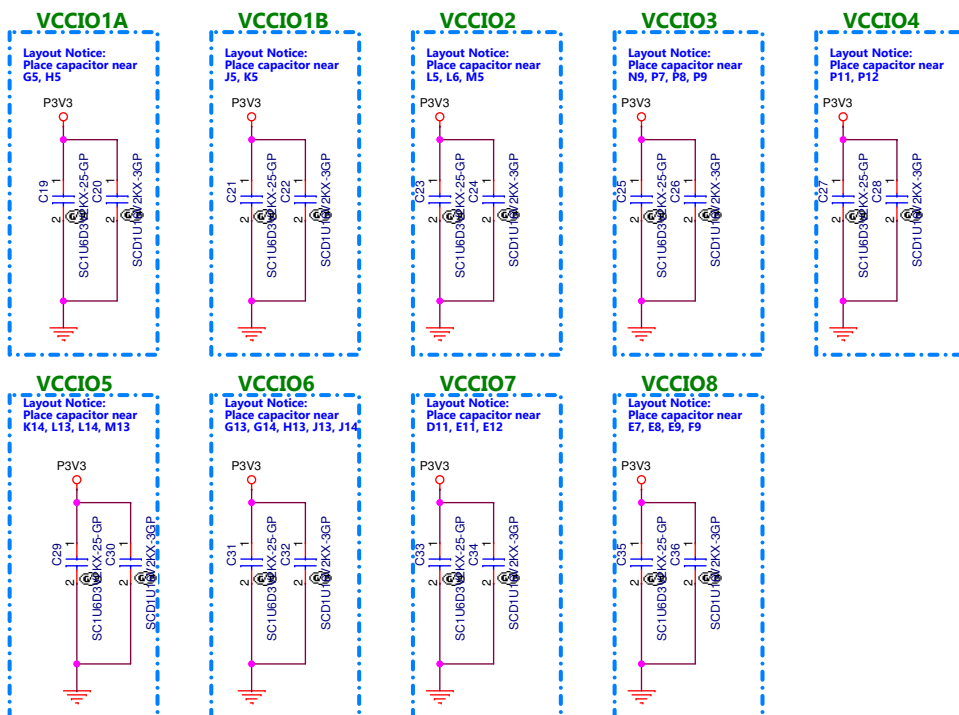
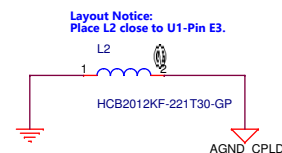
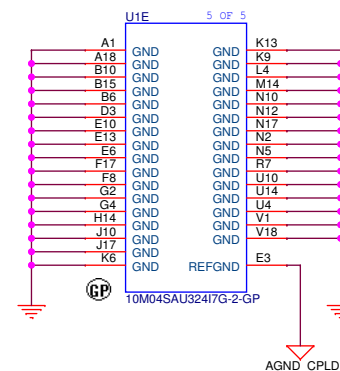
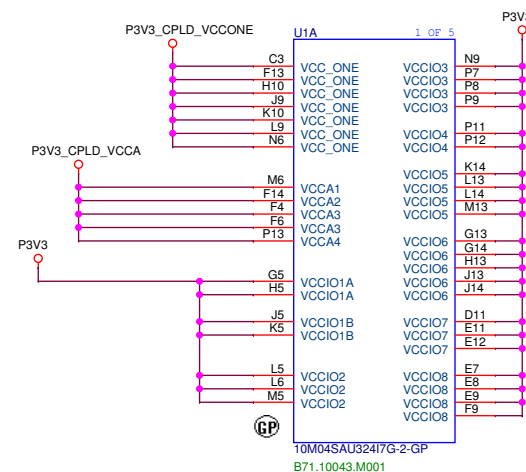
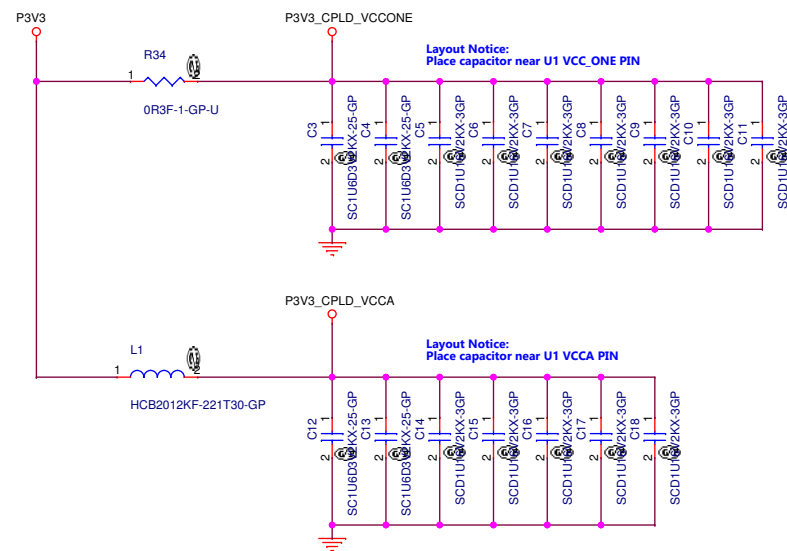


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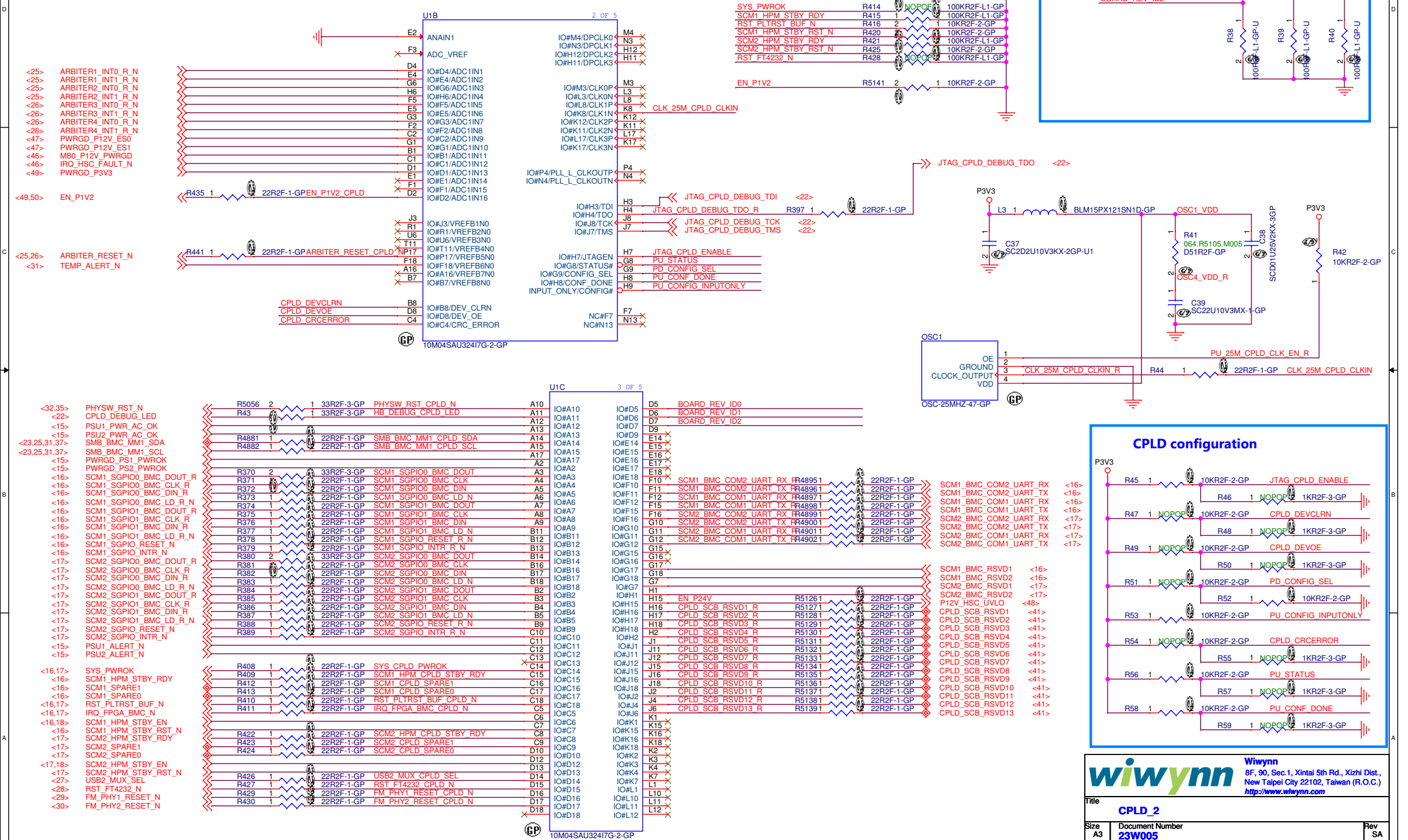
SIDE BAND



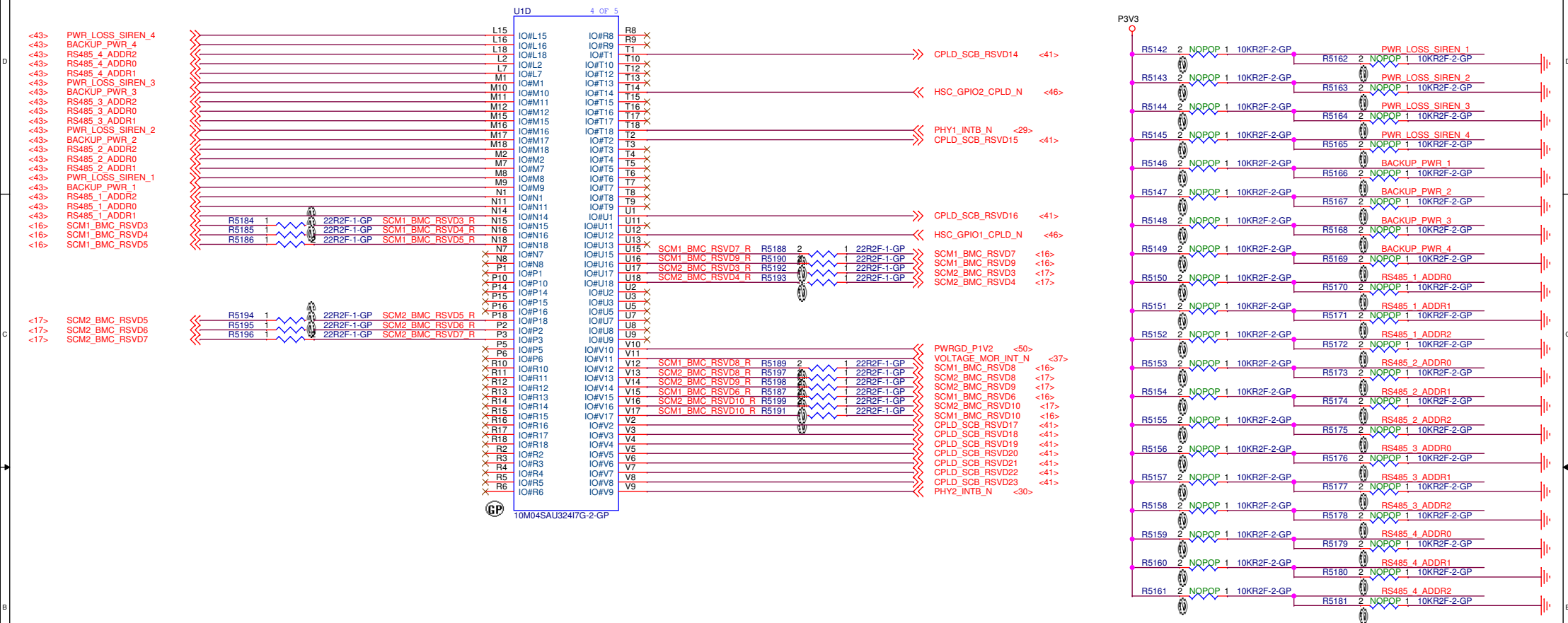
CPLD (1/4)



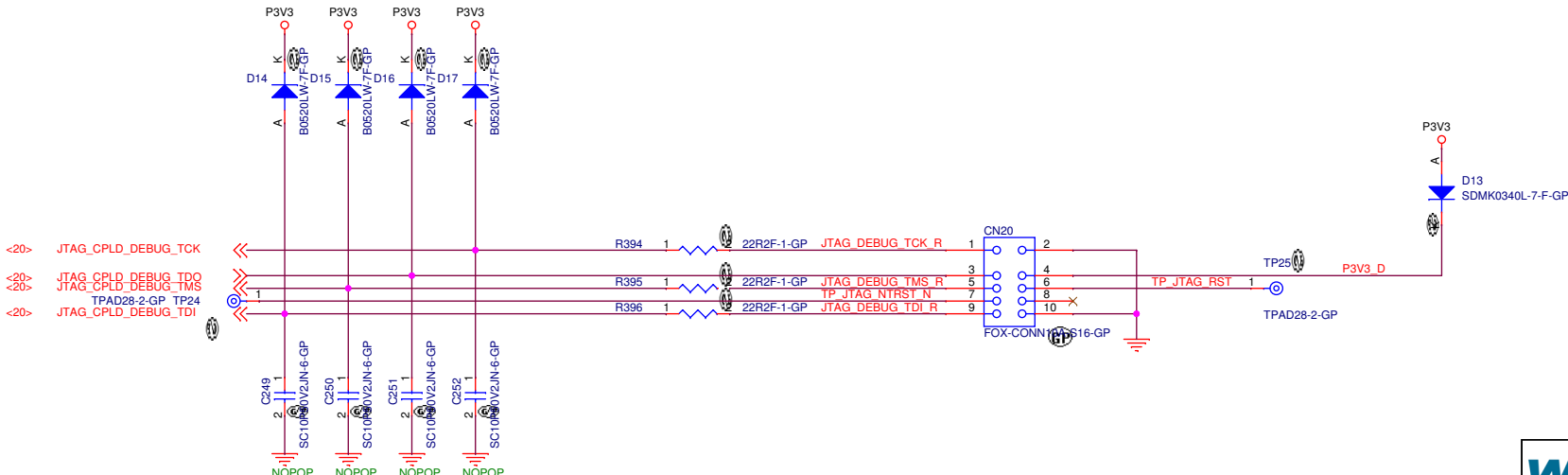
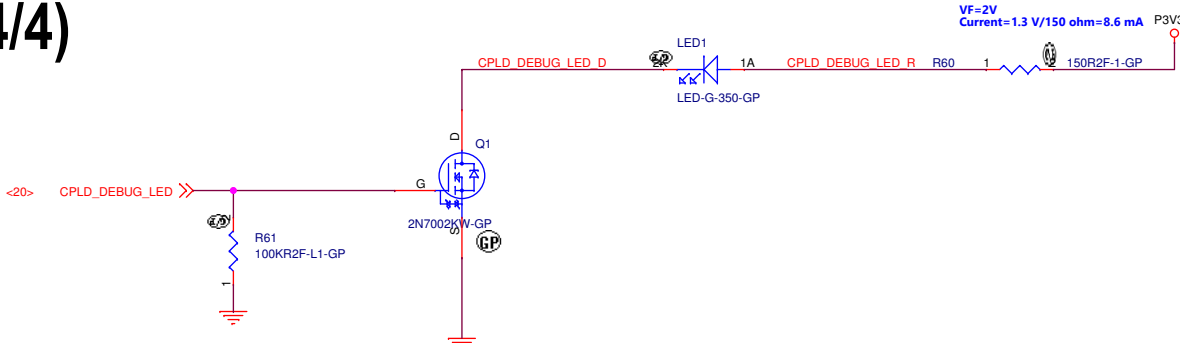
CPLD (2/4)



CPLD (3/4)



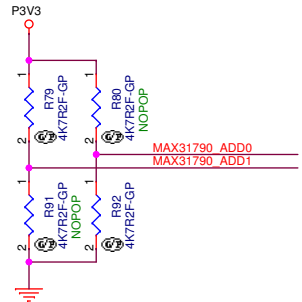
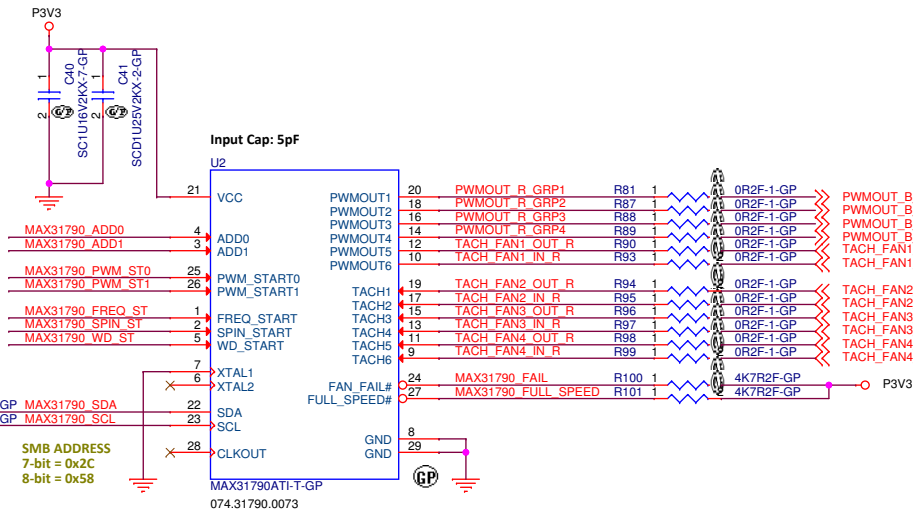
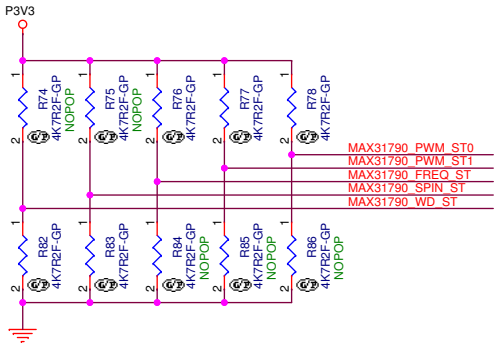
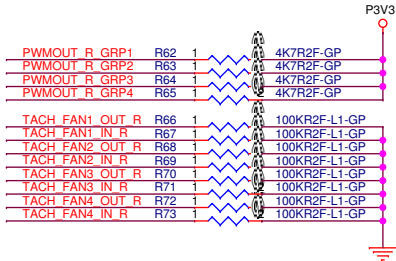
CPLD (4/4)



FAN CONTROLLER

PIN#	CONNECTION	MODE
FREQ_START	Vcc	PWM Output Frequency 25kHz
SPIN_START	GND	Spin-up Disable
WD_START	GND	Watchdog Disable

POR CONDITION		
PWM_START0	PWM_START1	PWM DUTY CYCLE (%)
GND	GND	0
GND	Unconnected	30
GND	V _{CC}	40
Unconnected	GND	50
Unconnected	V _{CC}	60
V _{CC}	GND	75
V _{CC}	V _{CC}	100



<20,25,31,37> SMB_BMC_MM1_SDA
<20,25,31,37> SMB_BMC_MM1_SCL

R102 1 0R2F-1-GP MAX31790 SDA
R103 1 0R2F-1-GP MAX31790 SCL
SMB ADDRESS
7-bit = 0x2C
8-bit = 0x58

ADD1 CONNECTION	ADD0 CONNECTION	SLAVE ADDRESS BYTE (HEX)
GND	GND	40
GND	SCL	42
GND	SDA	44
GND	VCC	46
SCL	GND	48
SCL	SCL	4A
SCL	SDA	4C
SCL	VCC	4E
SDA	GND	50
SDA	SCL	52
SDA	SDA	54
SDA	VCC	56
VCC	GND	58
VCC	SCL	5A
VCC	SDA	5C
VCC	VCC	5E

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Title			
FAN_CONNECTOR			
Size A3	Document Number 23W005		Rev SA
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I2C ARBITER (1/2)

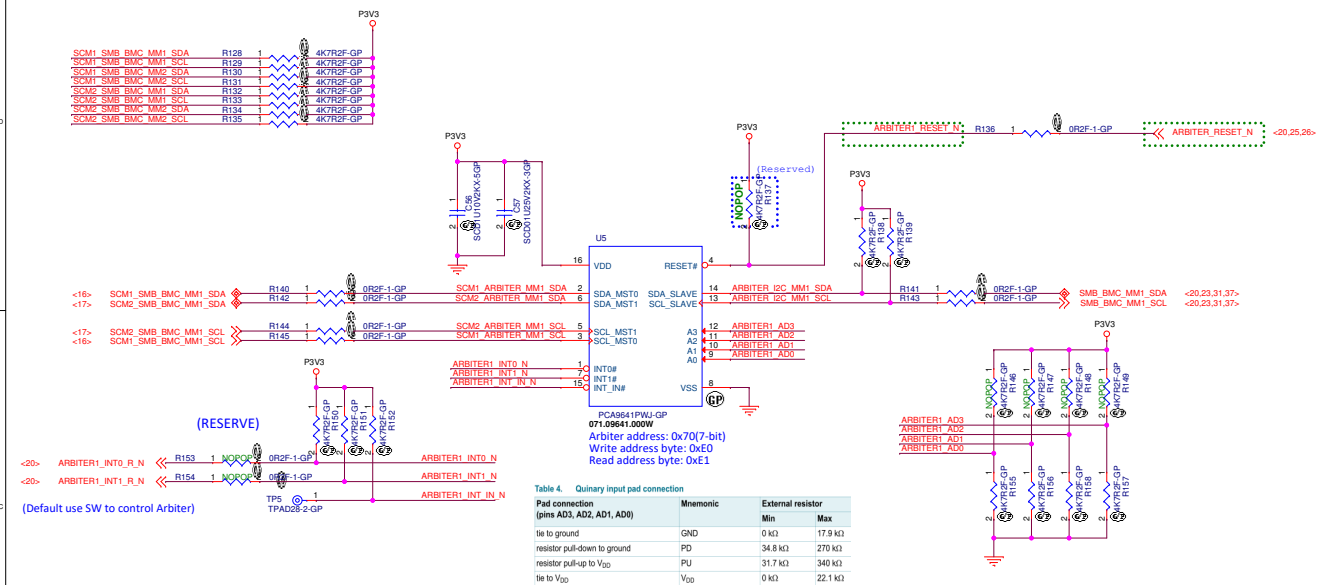


Table 4. Quinary input pad connection

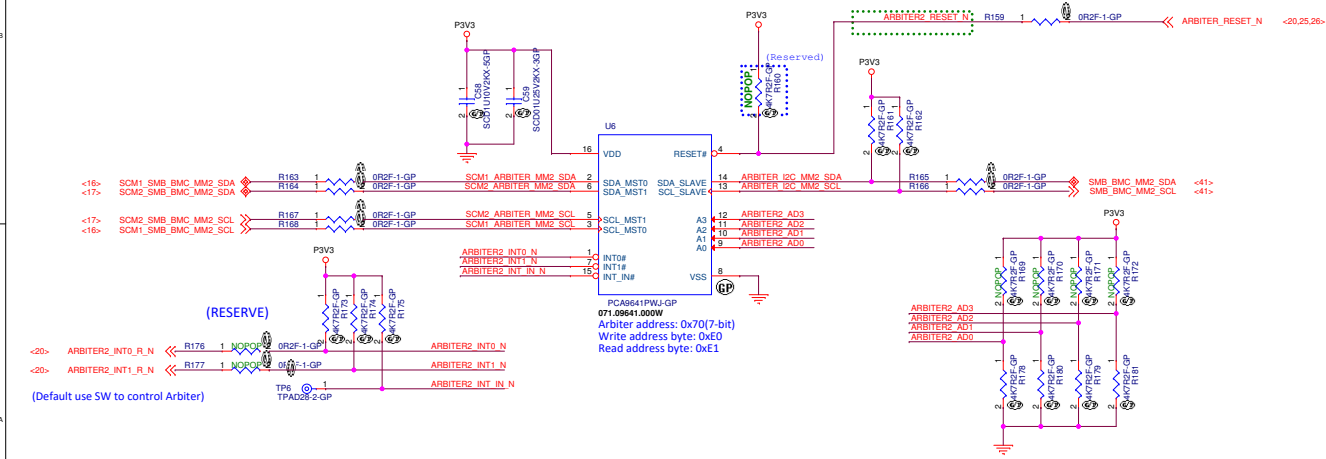
Pad connection (pins AD3, AD2, AD1, AD0)	Mnemonic	External resistor	
		Min	Max
tie to ground	GND	0 k Ω	17.9 k Ω
resistor pull-down to ground	PD	34.8 k Ω	270 k Ω
resistor pull-up to V_{DD}	PU	31.7 k Ω	340 k Ω
tie to V_{DD}	V_{DD}	0 k Ω	22.1 k Ω

7.2 Address maps

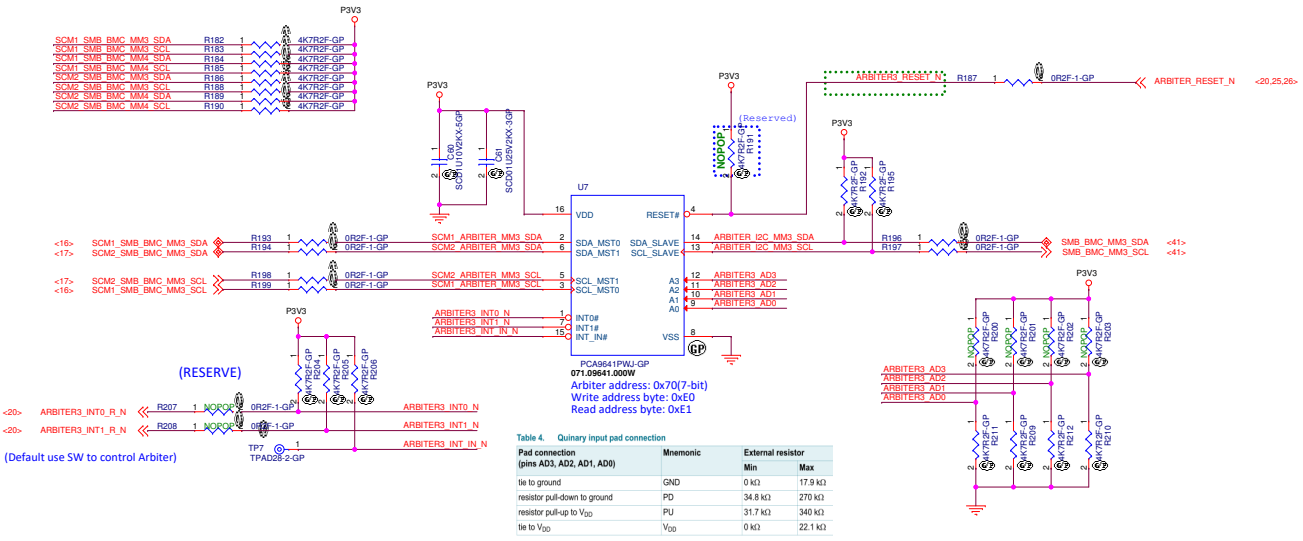
Table 5. Address maps

Do not use any other combination addresses to decode hardware addresses.

Pin connectivity				Address of PCA9641								Address byte value		7-bit hexadecimal address without R/W
AD3	AD2	AD1	AD0	A6	A5	A4	A3	A2	A1	A0	R/ $\overline{W}$	Write	Read	
V _{SS}	V _{SS}	V _{SS}	V _{SS}	1	1	1	0	0	0	0	-	E0h	E1h	70h
V _{SS}	V _{SS}	V _{SS}	V _{DD}	1	1	1	0	0	0	1	-	E2h	E3h	71h
V _{SS}	V _{SS}	V _{DD}	V _{SS}	1	1	1	0	0	1	0	-	E4h	E5h	72h
V _{SS}	V _{SS}	V _{DD}	V _{DD}	1	1	1	0	0	1	1	-	E6h	E7h	73h



I2C ARBITER (2/2)

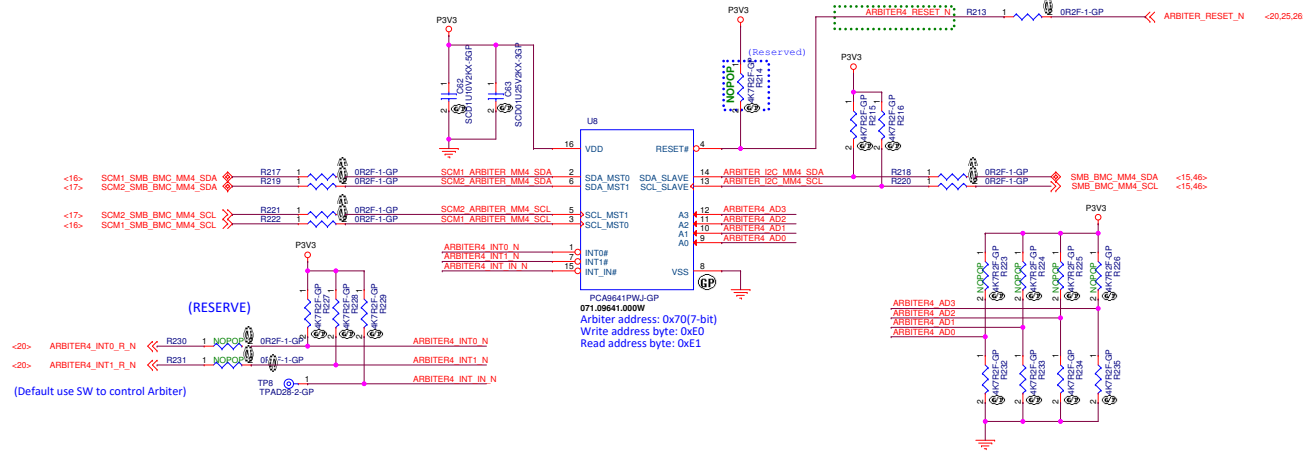


7.2 Address maps

Table 5. Address maps

Do not use any other combination addresses to decode hardware addresses.

Pin connectivity		Address of PCA9641										Address byte value		7-bit hexadecimal address without R/W	
AD3	AD2	AD1	AD0	A6	A5	A4	A3	A2	A1	A0	R/W	Write	Read		
V _{SS}	V _{SS}	V _{SS}	V _{SS}	1	1	1	0	0	0	0	-	E0h	E1h	70h	
V _{SS}	V _{SS}	V _{SS}	V _{DD}	1	1	1	0	0	0	1	-	E2h	E3h	71h	
V _{SS}	V _{SS}	V _{DD}	V _{SS}	1	1	1	0	0	1	0	-	E4h	E5h	72h	
V _{SS}	V _{SS}	V _{DD}	V _{DD}	1	1	1	0	0	1	1	-	E6h	E7h	73h	



USB SW

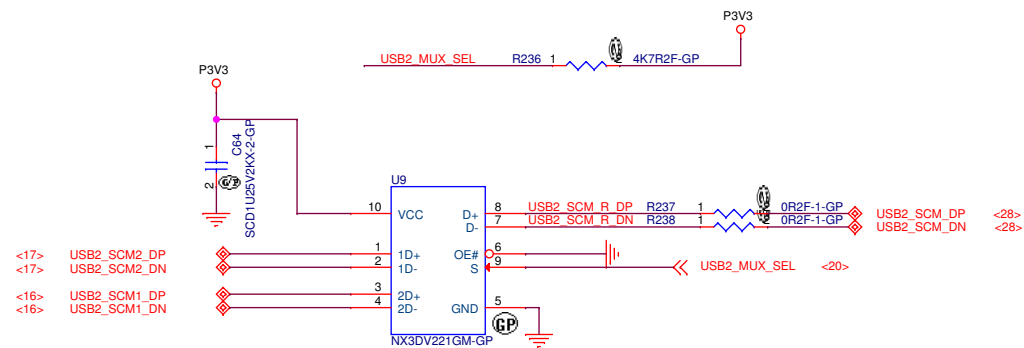


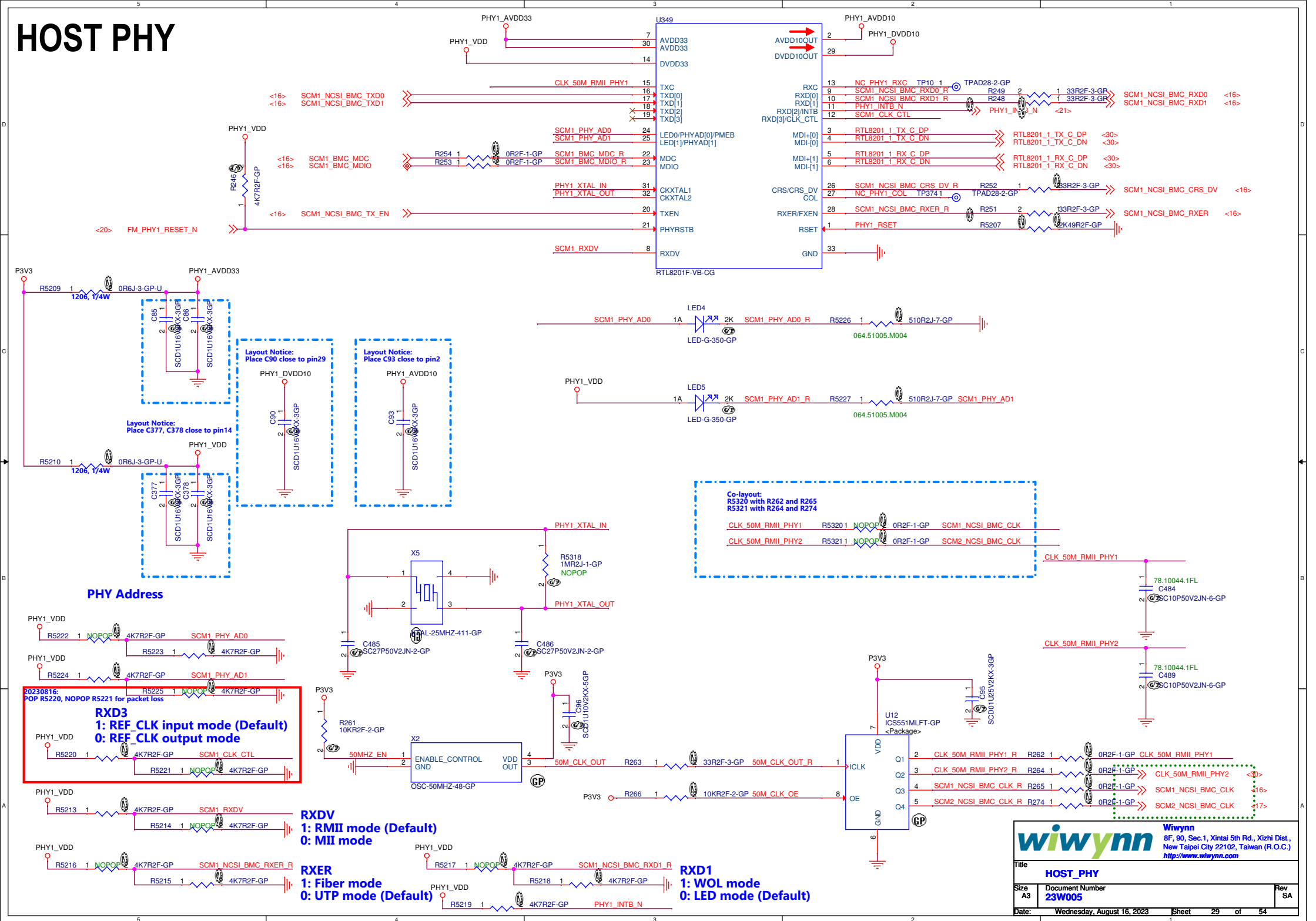
Table 4. Function table^[1]

Input		Channel
S	OE	
L	L	D+ = 1D+; D- = 1D-
H	L	D+ = 2D+; D- = 2D-
X	H	switches off

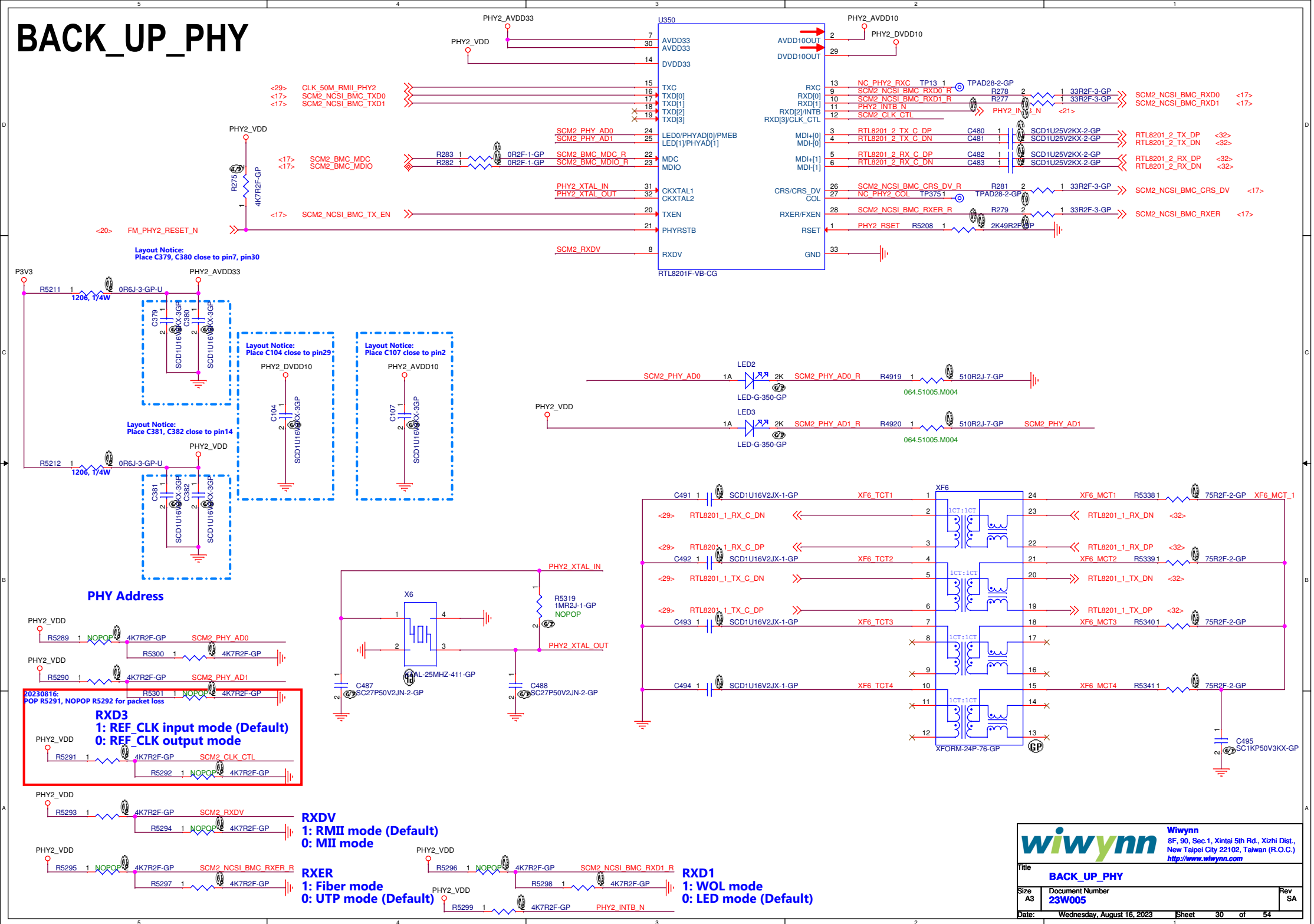
[1] H = HIGH voltage level; L = LOW voltage level; X = don't care.

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HOST PHY



BACK UP PHY



EEPROM

DESIGN NOTE
 SMBUS ADDRESS :
 0x51 (7-bit slave address)
 0xA2 (8-bit write address)
 0xA3 (8-bit read address)

Table 2. Device Select Code

Device Select Code	Device Type Identifier ¹				Chip Enable Address ²			RW
	b7	b6	b5	b4	b3	b2	b1	b0
	1	0	1	0	E2	E1	E0	RW

Note: 1. The most significant bit, b7, is sent first.
 2. E0, E1 and E2 are compared against the respective external pins on the memory device.

DESIGN NOTE
 SMBUS ADDRESS :
 0x48 (7-bit slave address)
 0x90 (8-bit write address)
 0x91 (8-bit read address)

Inlet Sensor

Address table for TPM75

A2	A1	A0	ADDR
0	0	0	90
0	0	1	92

A2	A1	A0	SLAVE ADDRESS
0	0	0	1001000
0	0	1	1001001
0	1	0	1001010
0	1	1	1001011
1	0	0	1001100
1	0	1	1001101
1	1	0	1001110
1	1	1	1001111

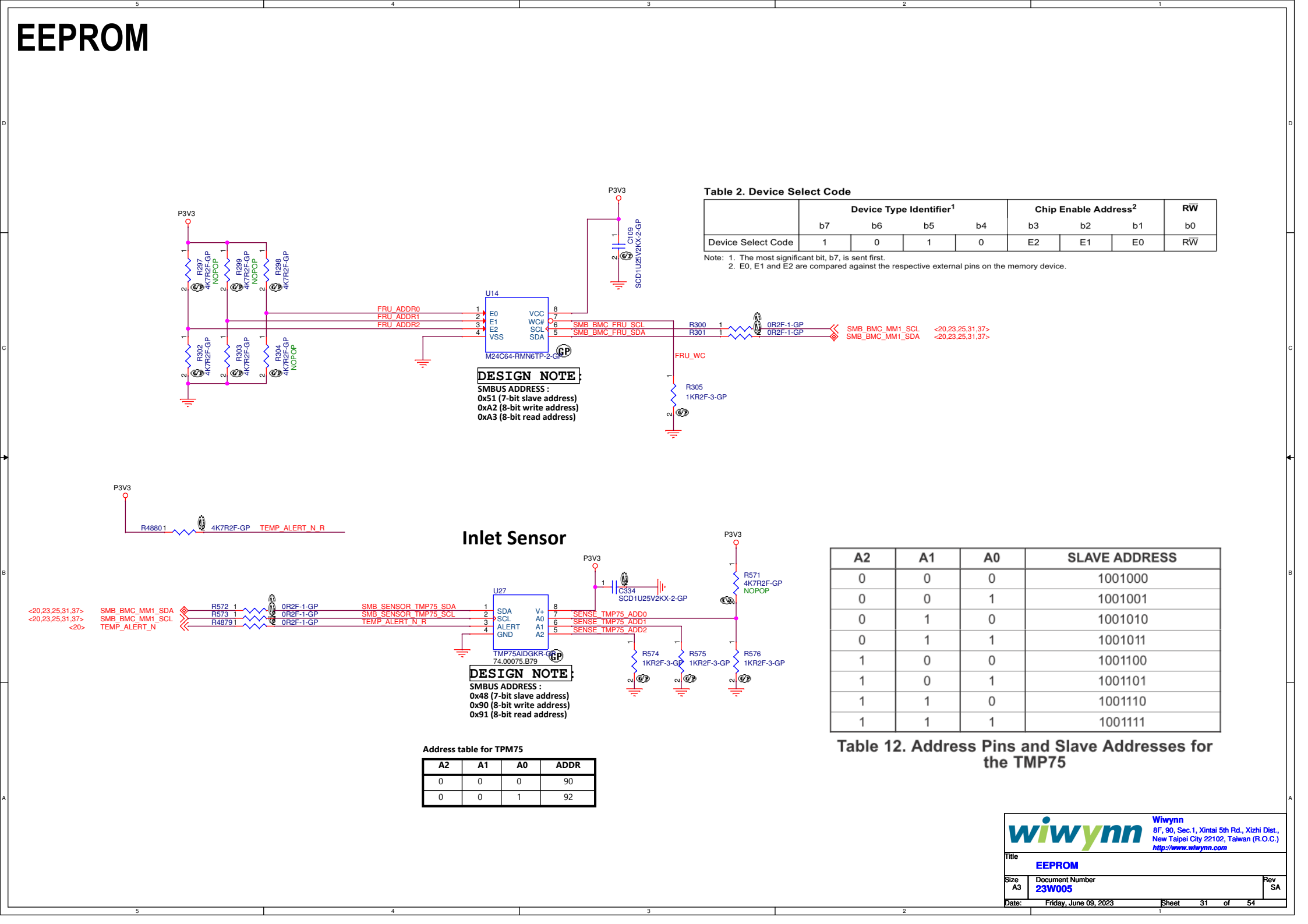
Table 12. Address Pins and Slave Addresses for the TMP75

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EEPROM

EEPROM

EEPROM

DESIGN NOTE
 SMBUS ADDRESS :
 0x51 (7-bit slave address)
 0xA2 (8-bit write address)
 0xA3 (8-bit read address)

Table 2. Device Select Code

Device Select Code	Device Type Identifier ¹				Chip Enable Address ²			RW
	b7	b6	b5	b4	b3	b2	b1	b0
	1	0	1	0	E2	E1	E0	RW

Note: 1. The most significant bit, b7, is sent first.
 2. E0, E1 and E2 are compared against the respective external pins on the memory device.

DESIGN NOTE
 SMBUS ADDRESS :
 0x48 (7-bit slave address)
 0x90 (8-bit write address)
 0x91 (8-bit read address)

Inlet Sensor

A2	A1	A0	SLAVE ADDRESS
0	0	0	1001000
0	0	1	1001001
0	1	0	1001010
0	1	1	1001011
1	0	0	1001100
1	0	1	1001101
1	1	0	1001110
1	1	1	1001111

Table 12. Address Pins and Slave Addresses for the TMP75

Address table for TPM75

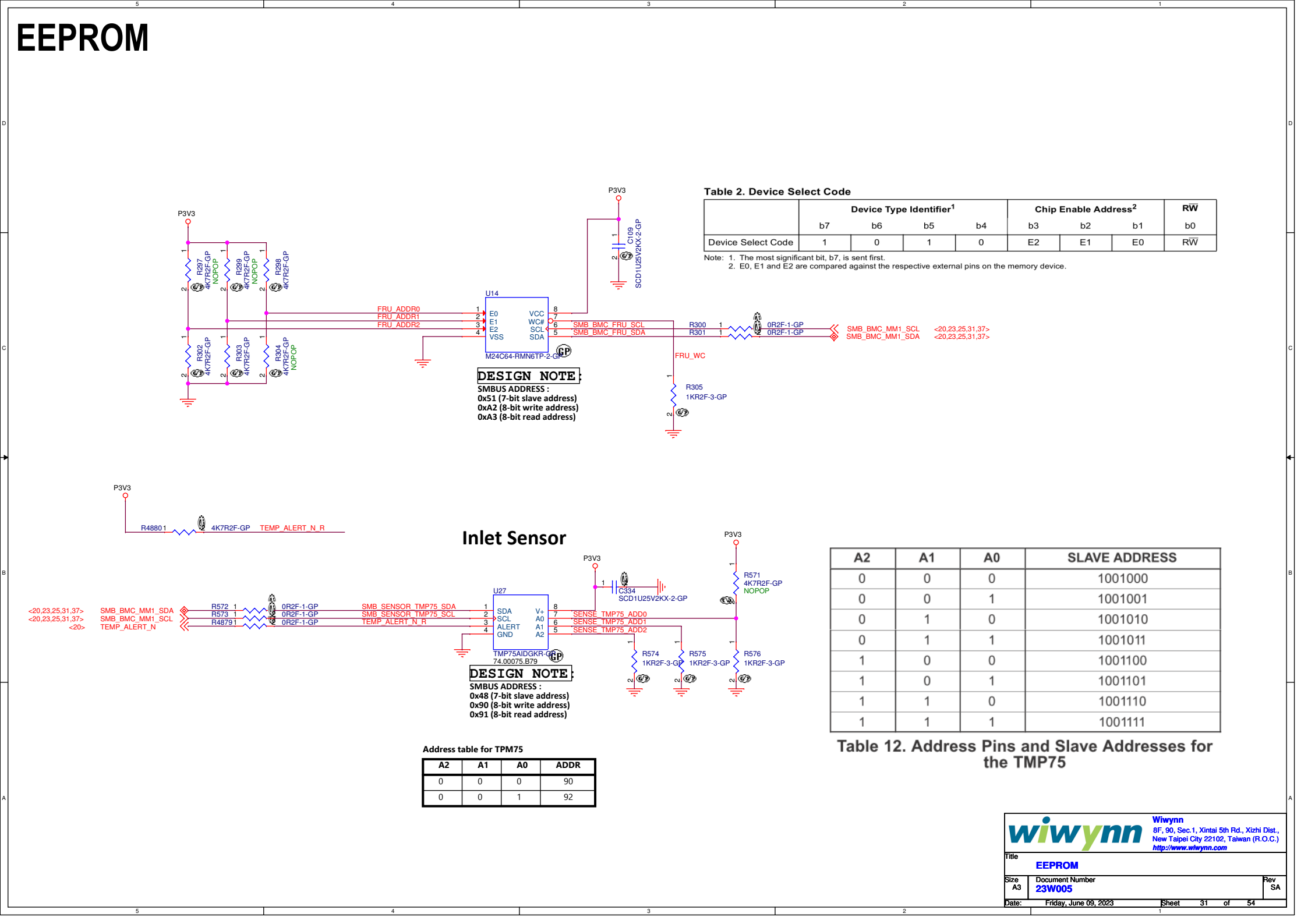
A2	A1	A0	ADDR
0	0	0	90
0	0	1	92

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EEPROM

EEPROM

Table 2. Device Select Code

Device Select Code	Device Type Identifier ¹				Chip Enable Address ²			RW
	b7	b6	b5	b4	b3	b2	b1	b0
	1	0	1	0	E2	E1	E0	RW

Note: 1. The most significant bit, b7, is sent first.
2. E0, E1 and E2 are compared against the respective external pins on the memory device.

DESIGN NOTE
SMBUS ADDRESS :
0x51 (7-bit slave address)
0xA2 (8-bit write address)
0xA3 (8-bit read address)

Inlet Sensor

DESIGN NOTE
SMBUS ADDRESS :
0x48 (7-bit slave address)
0x90 (8-bit write address)
0x91 (8-bit read address)

Address table for TPM75

A2	A1	A0	ADDR
0	0	0	90
0	0	1	92

A2	A1	A0	SLAVE ADDRESS
0	0	0	1001000
0	0	1	1001001
0	1	0	1001010
0	1	1	1001011
1	0	0	1001100
1	0	1	1001101
1	1	0	1001110
1	1	1	1001111

Table 12. Address Pins and Slave Addresses for the TMP75

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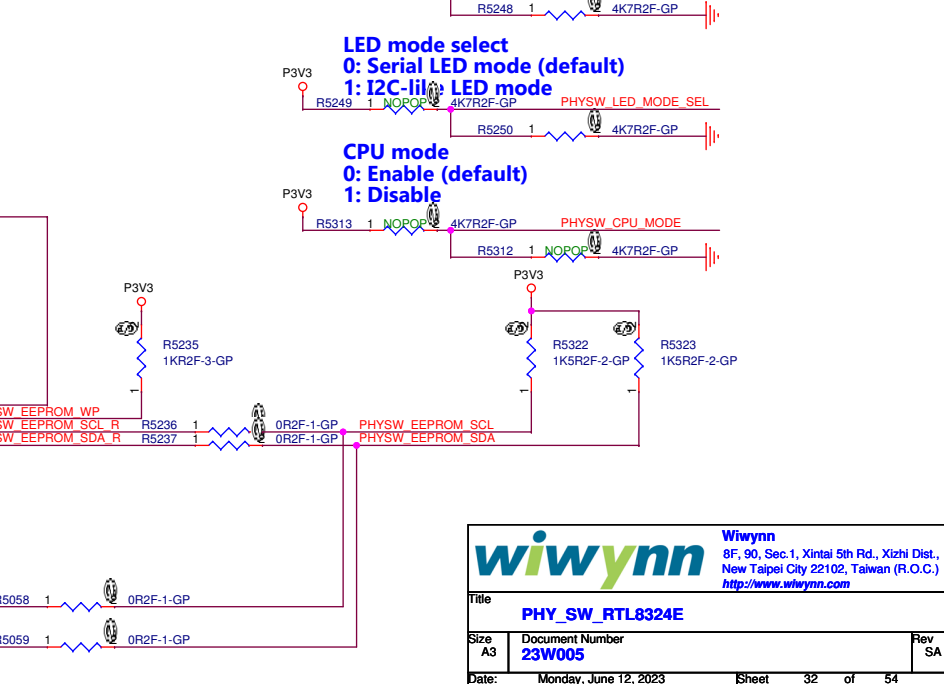
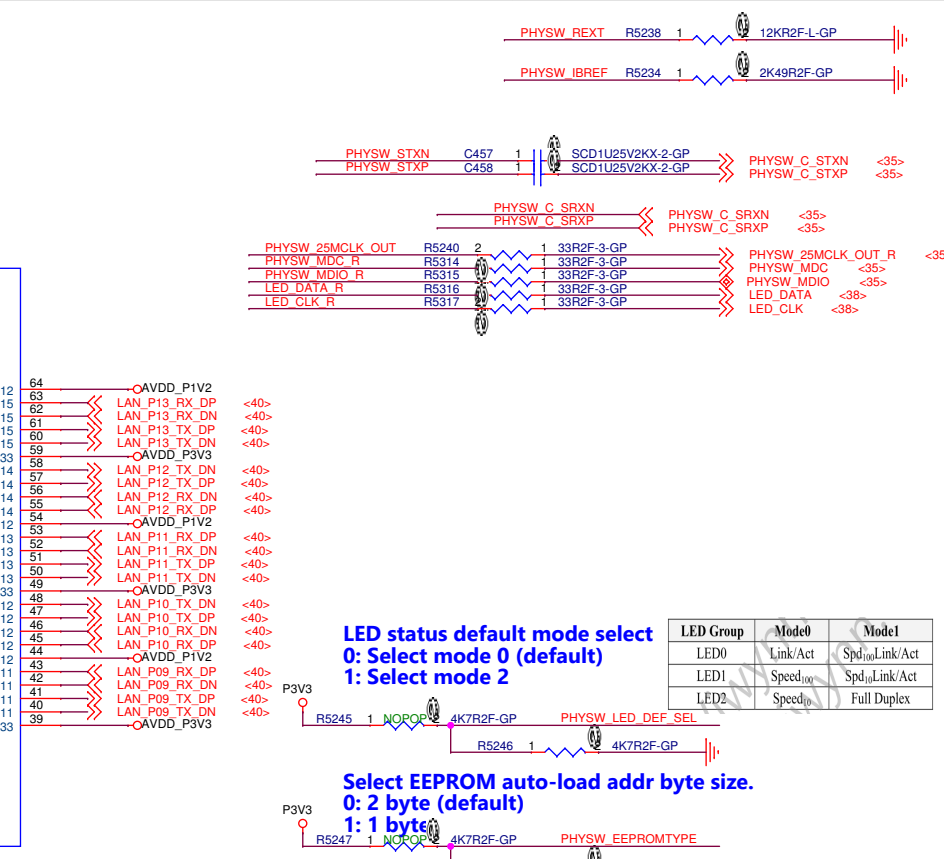
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EEPROM

EEPROM

Pin 1-20 of the P1 header:

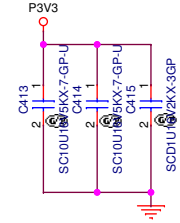
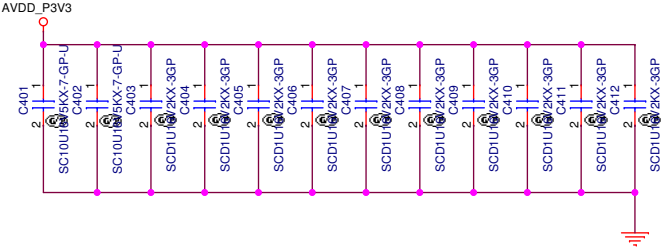
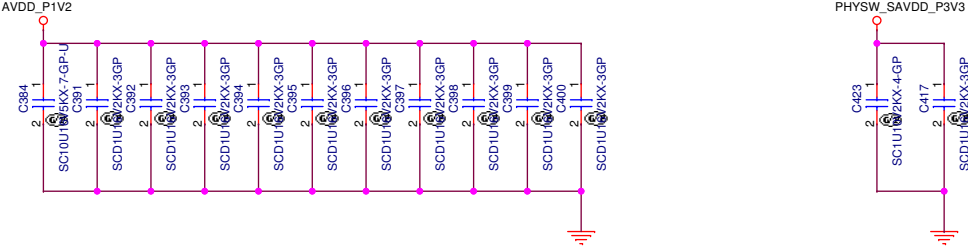
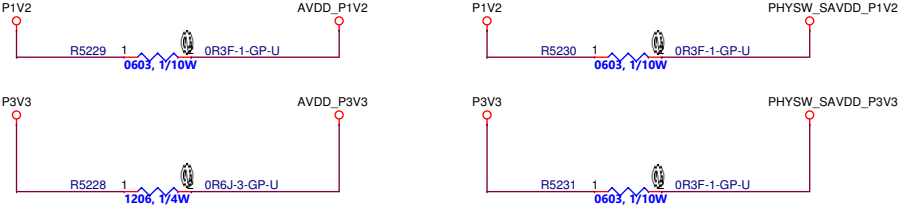
- Pin 1: P1V2
- Pin 2: PHYSW_RST_N
- Pin 3: PHYSW_FDR_N
- Pin 4: LED_CLK_R
- Pin 5: PHYSW_LED_R
- Pin 6: PHYSW_EEPROM_SCL
- Pin 7: PHYSW_EEPROM_SDA
- Pin 8: P3V3
- Pin 9: PHYSW_MDIO_R
- Pin 10: PHYSW_MDC_R
- Pin 11: PHY_SW_XTAL_25M_IN
- Pin 12: PHY_SW_XTAL_25M_OUT
- Pin 13: PHYSW_RESET
- Pin 14: PHYSW_STXN
- Pin 15: PHYSW_STXP
- Pin 16: PHYSW_25MCLK_OUT
- Pin 17: PHYSW_C_SRXP
- Pin 18: PHYSW_C_SRPXN
- Pin 19: PHYSW_DIS_FC
- Pin 20: PHYSW_EN_TST



LED Group	Mode0	Mode1
LED0	Link/Act	Spd ₁₀₀ Link/Act
LED1	Speed ₁₀₀	Spd ₁₀ Link/Act
LED2	Speed ₁₀	Full Duplex

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PHY_SW_RTL8324E			
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PHY_SW_RTL8324E_PWRCAP



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A3

Document Number
23W005

Rev
SA

Date:

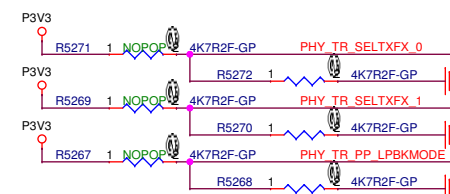
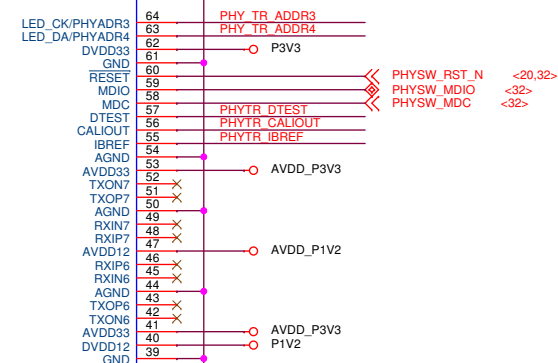
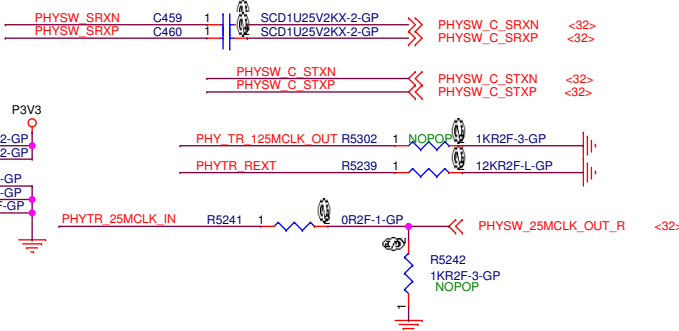
Monday, May 22, 2023

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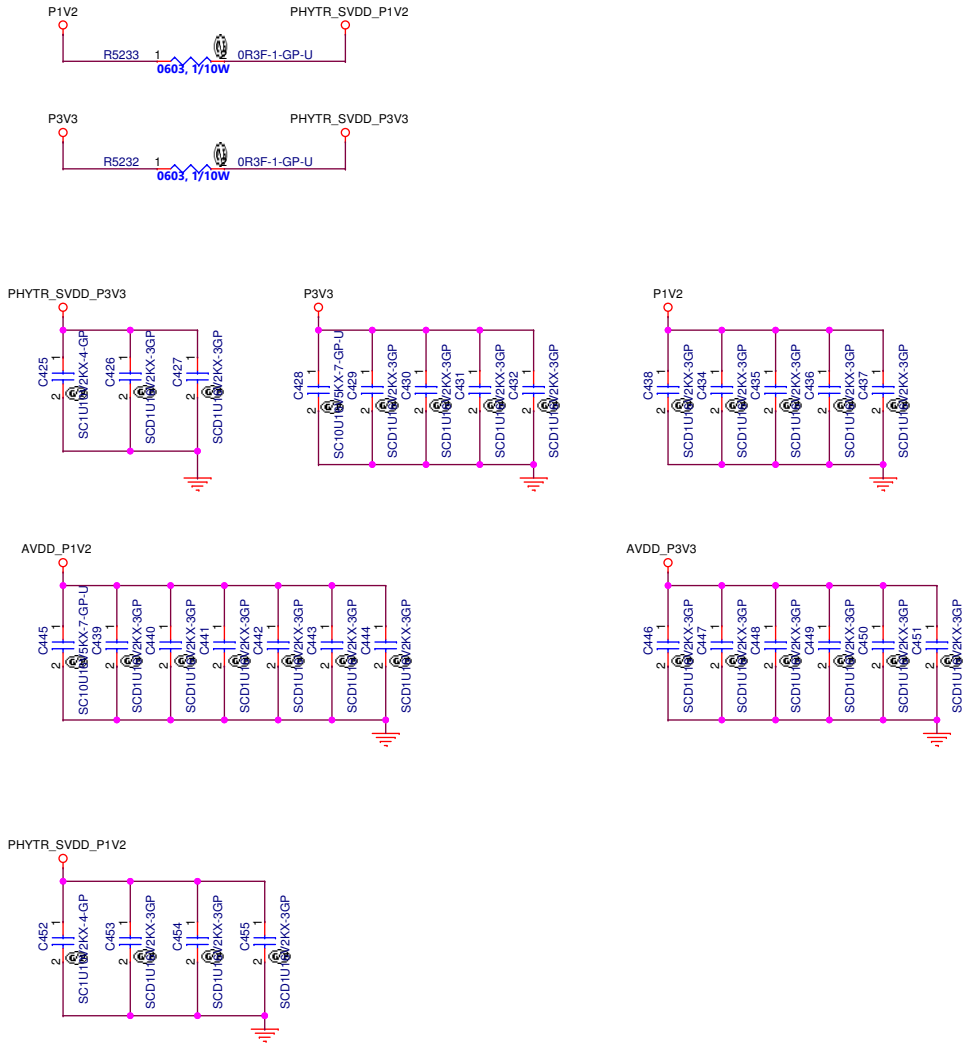
LED mode select
00: 3-bit serial stream mode (default)
01: 2-bit serial stream mode
10: 3-bit for Bi-color LED
11: 1-bit serial stream



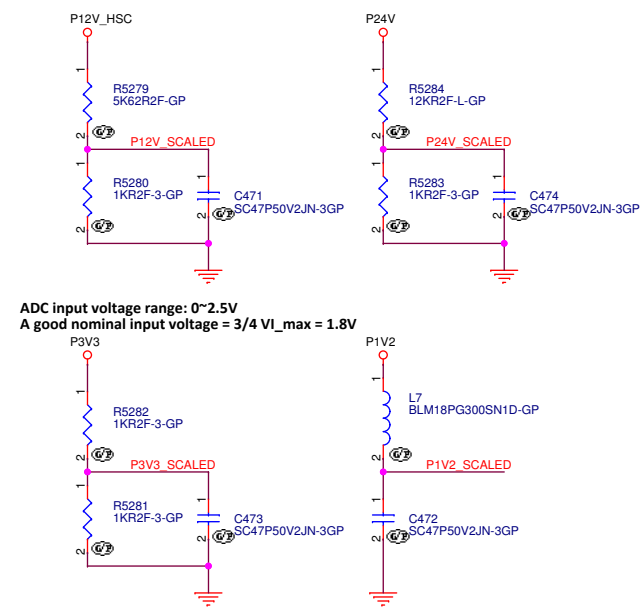
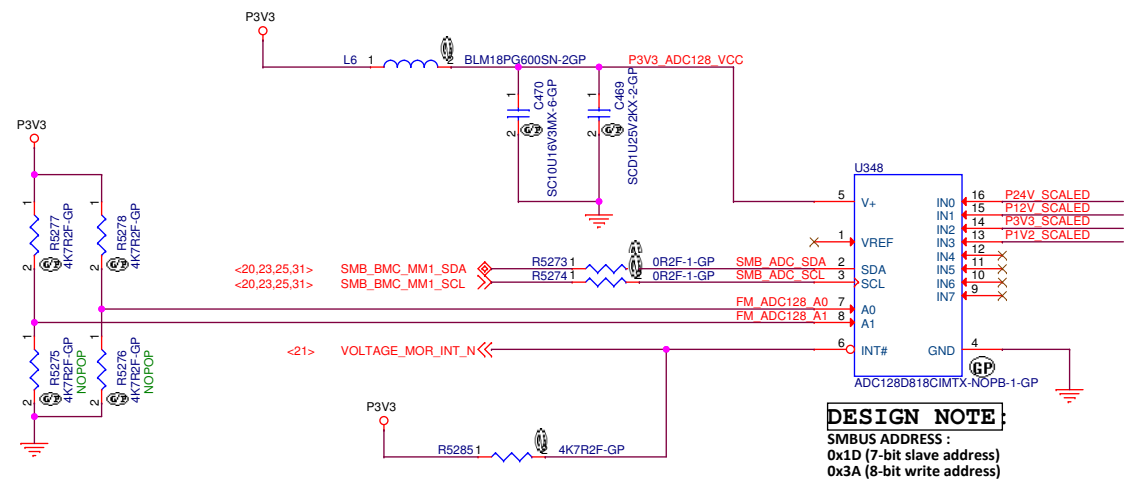
PHY_TR_XTAL 25M O



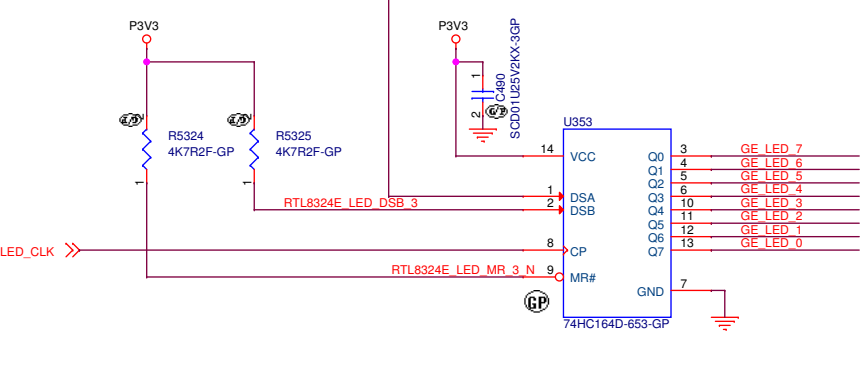
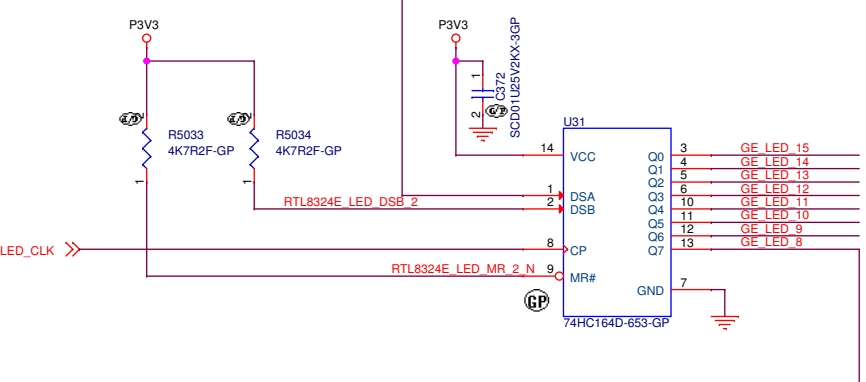
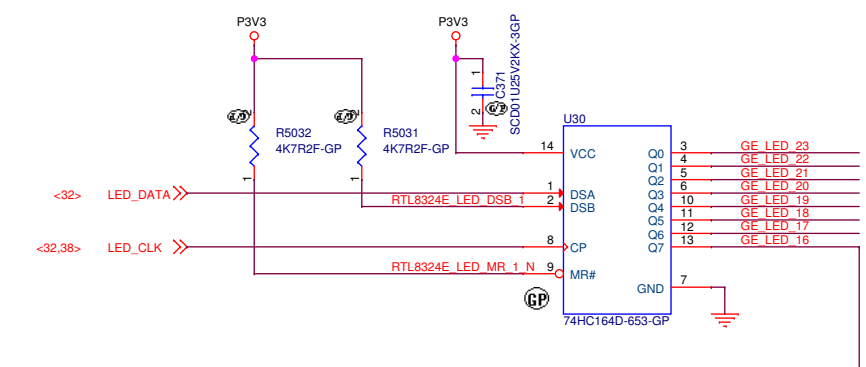
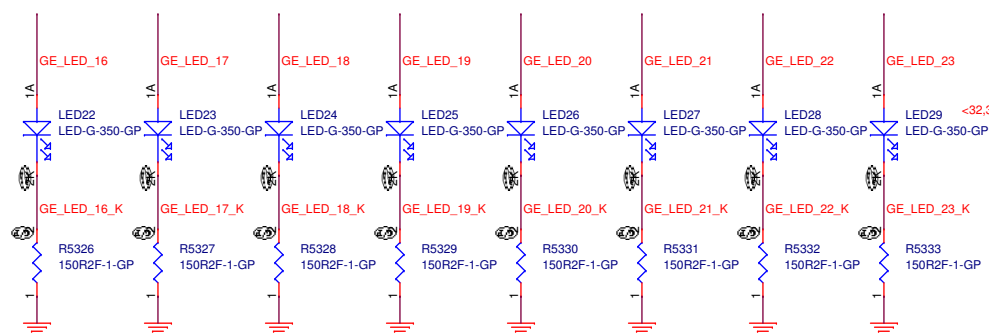
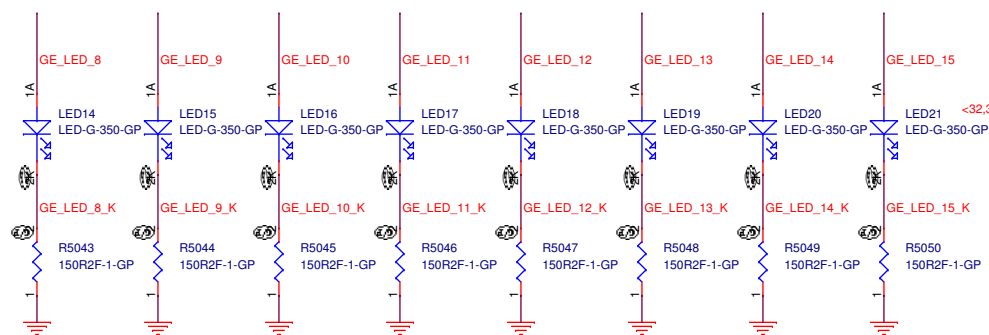
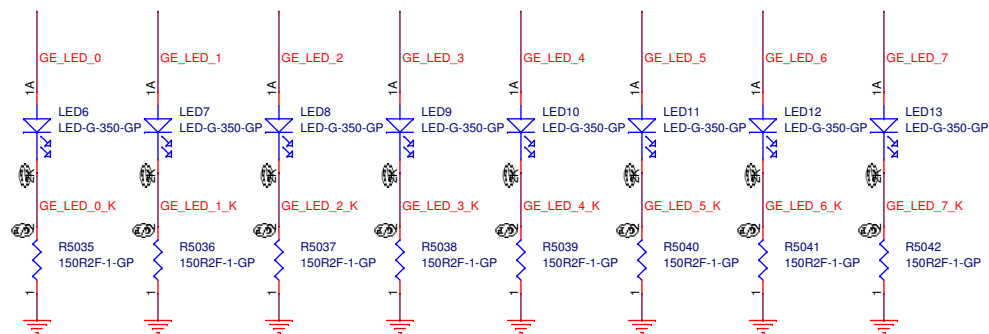
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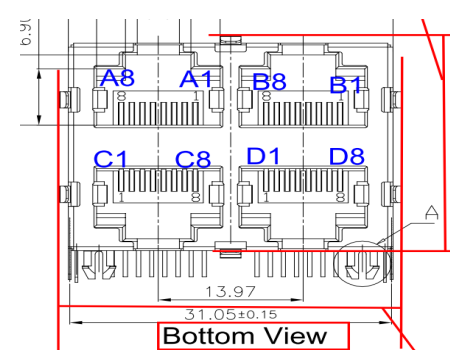
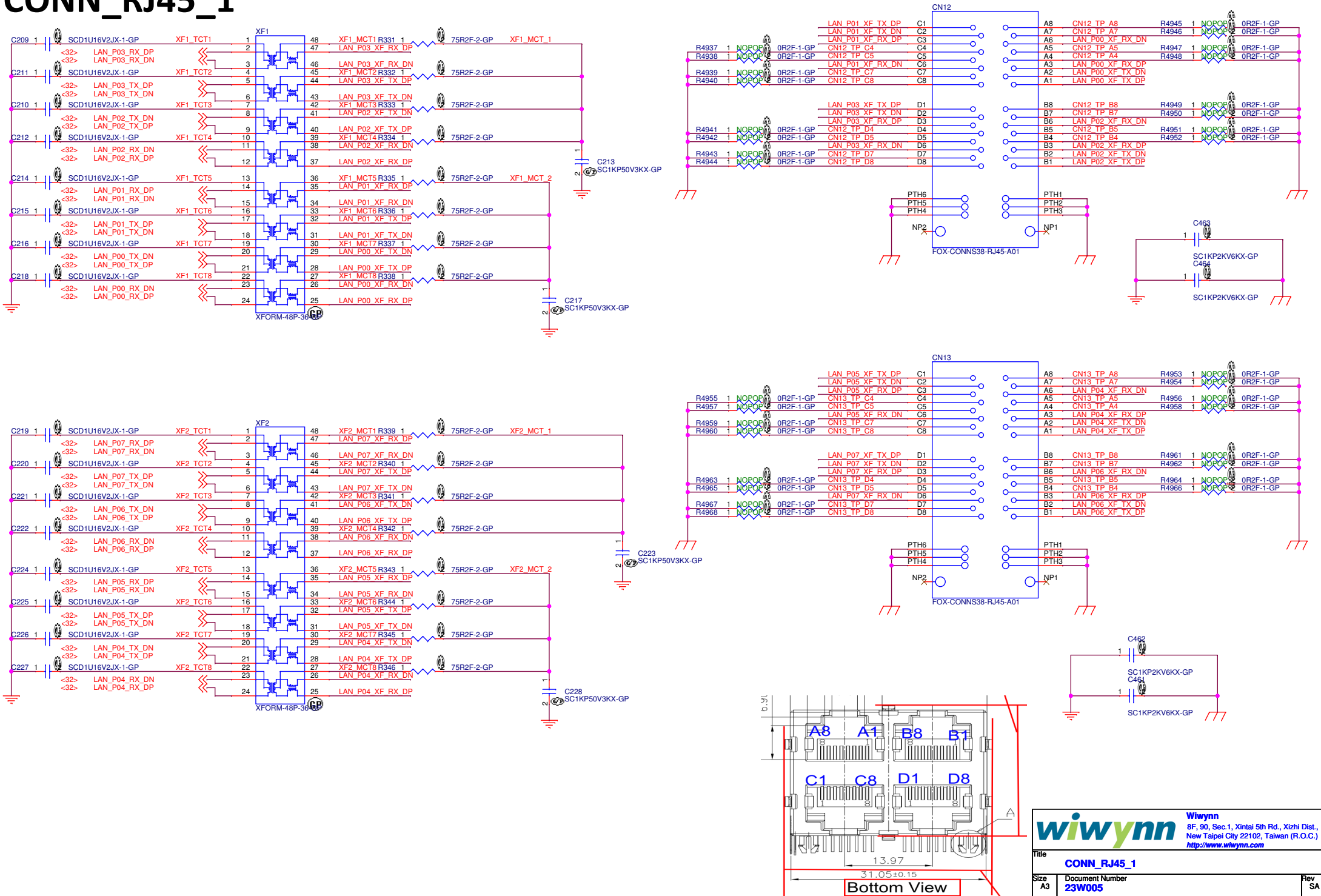
POWER MONITOR



PHY_SW_LED



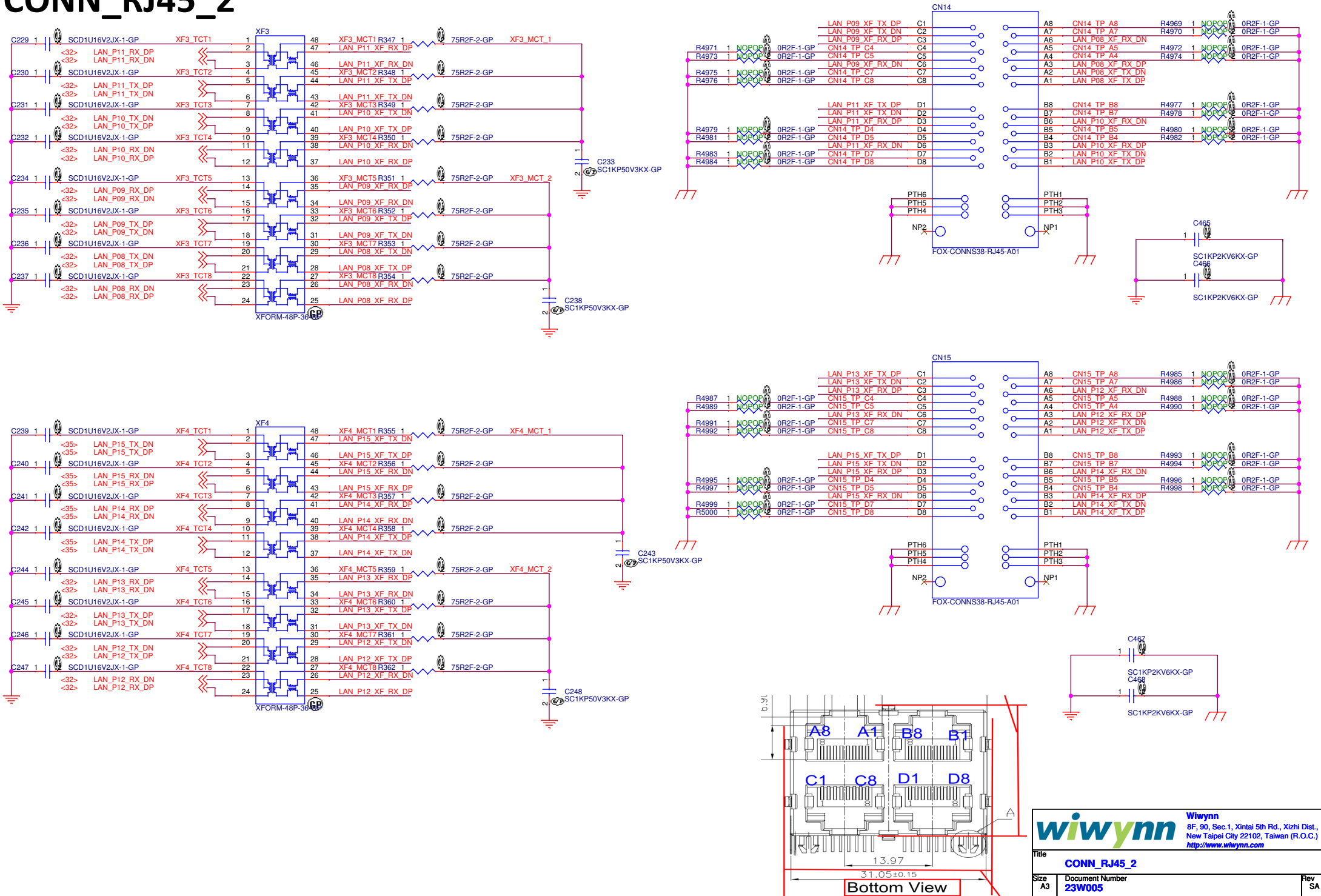
CONN_RJ45_1



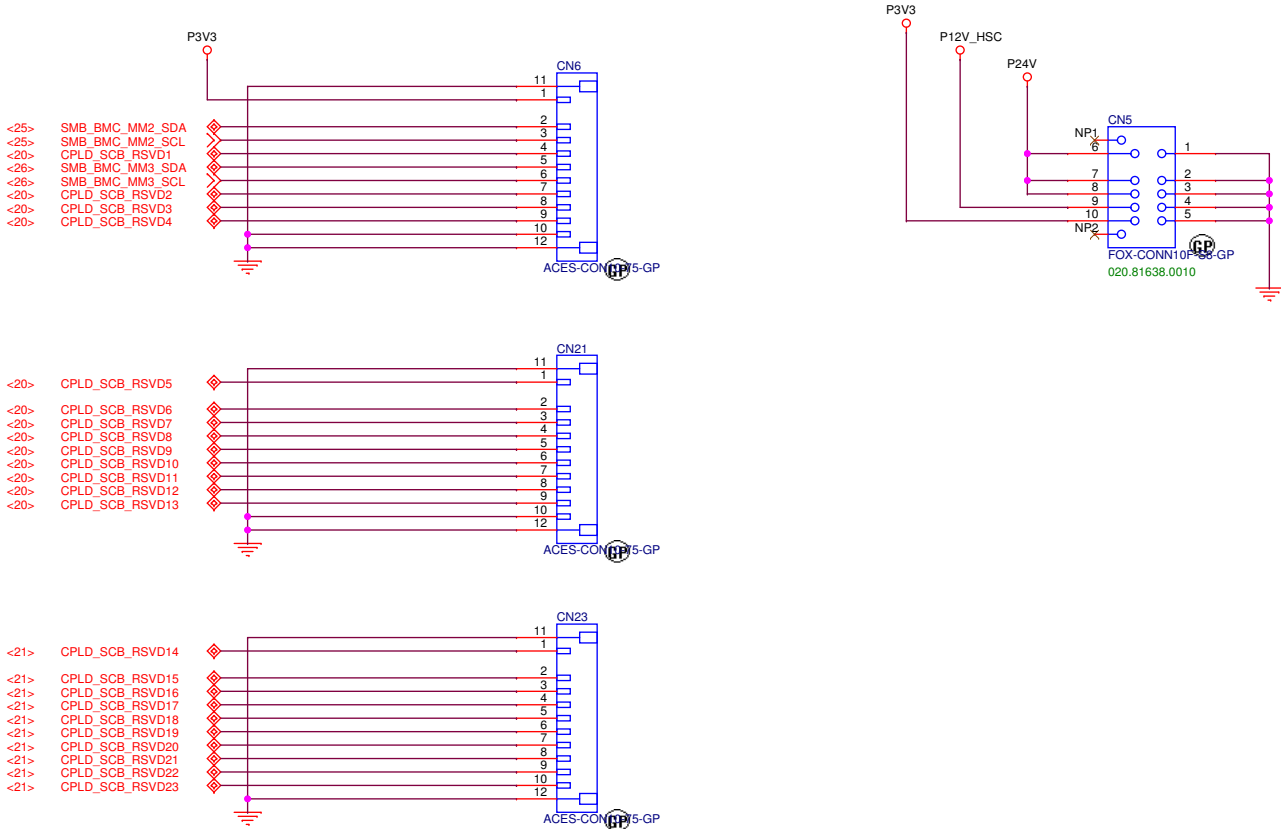
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New Taipei City 22102, Taiwan (R.O.C.)
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Title CONN_RJ45_1		
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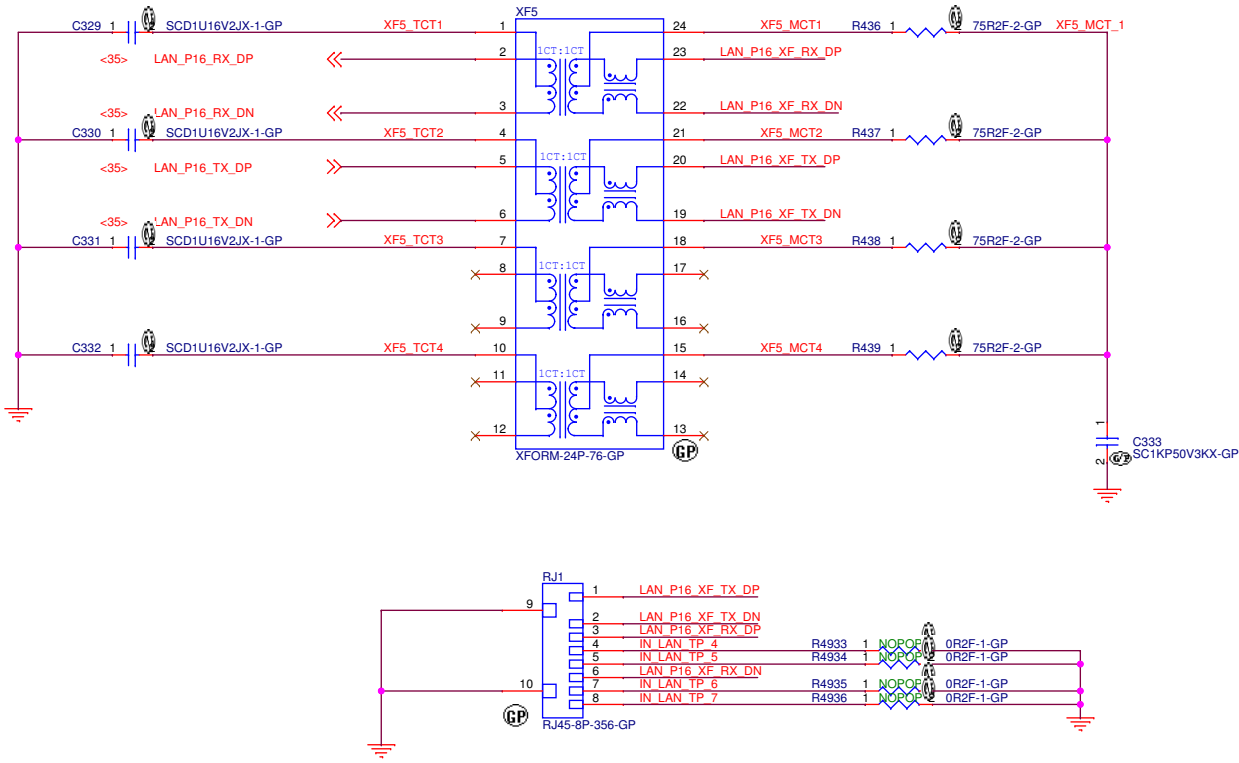
CONN RJ45 2



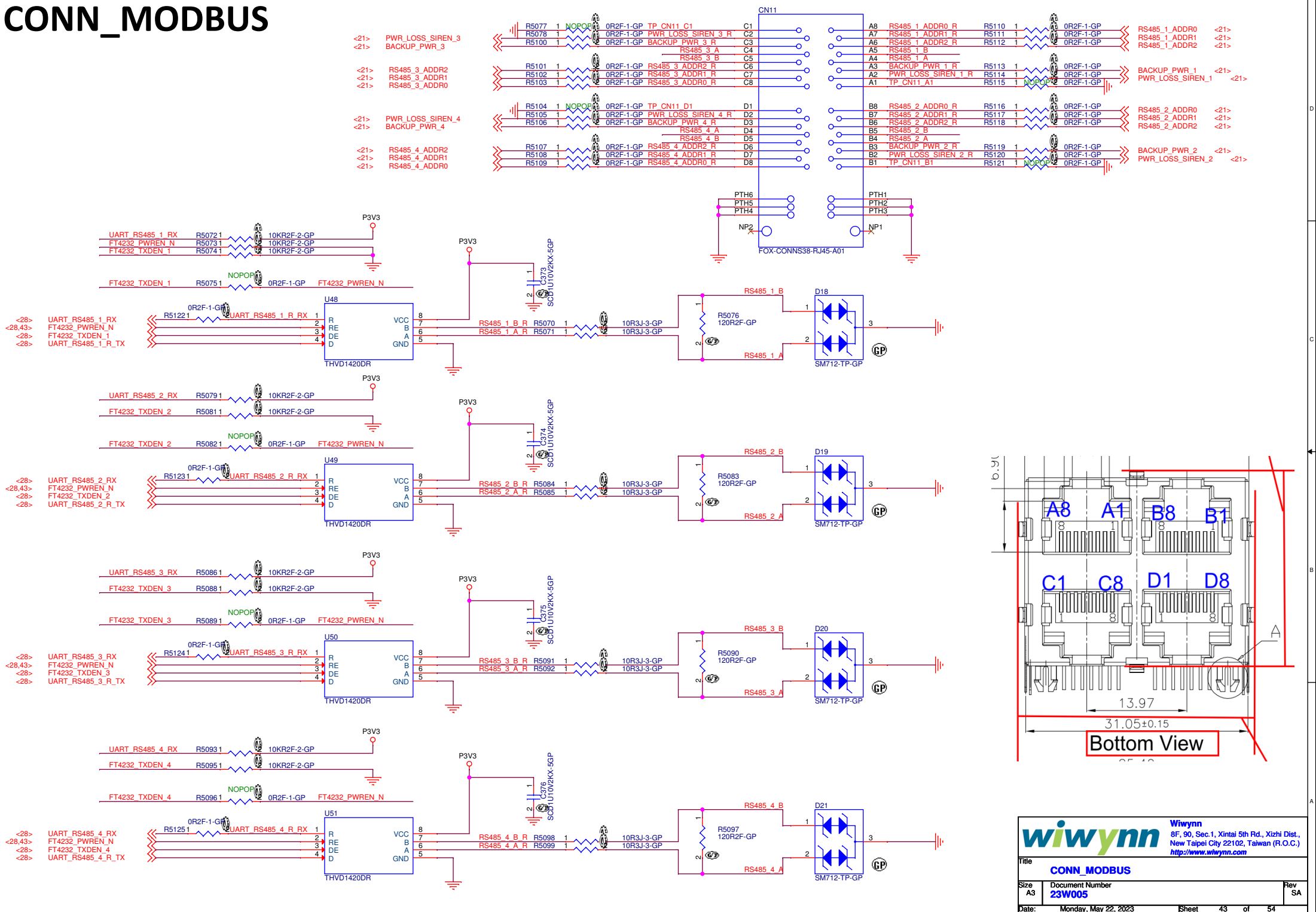
I2C_BLOCK_DIAGRAM



INTERNAL_RJ45



CONN_MODBUS





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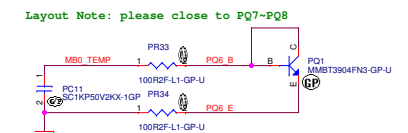
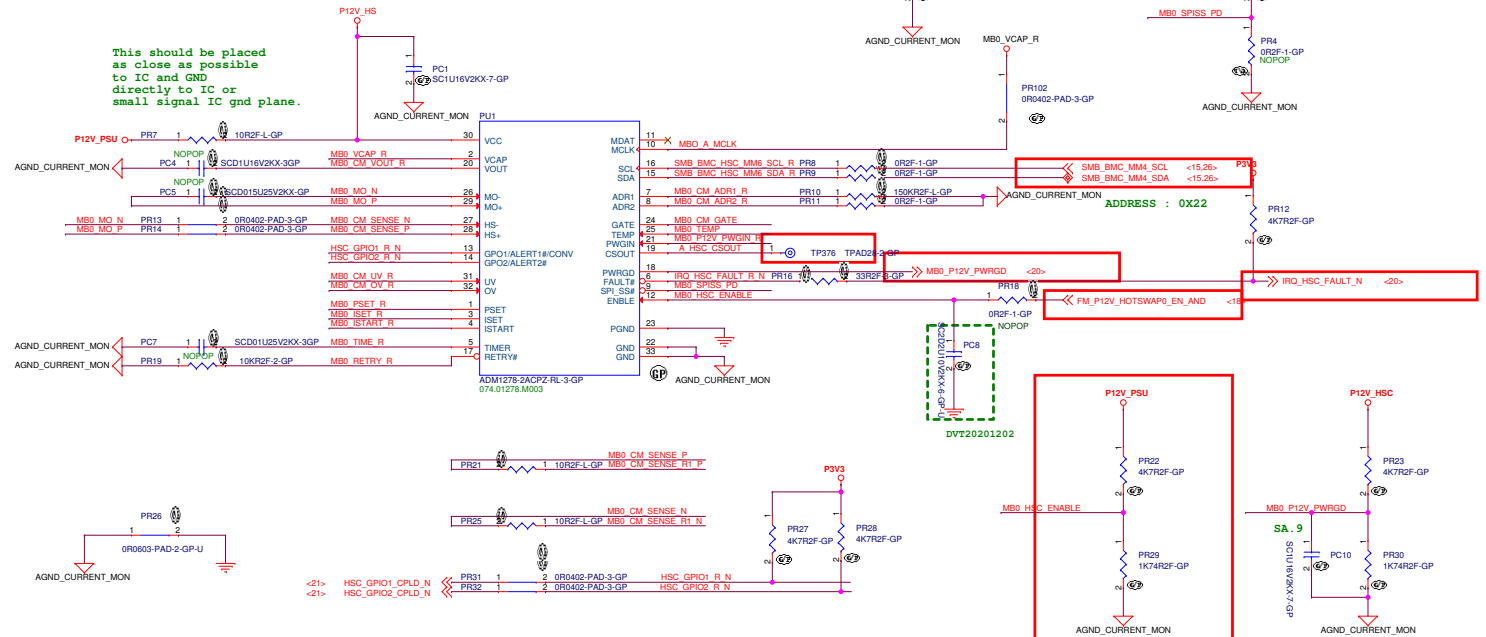
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Title		
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Size	Document Number	Rev
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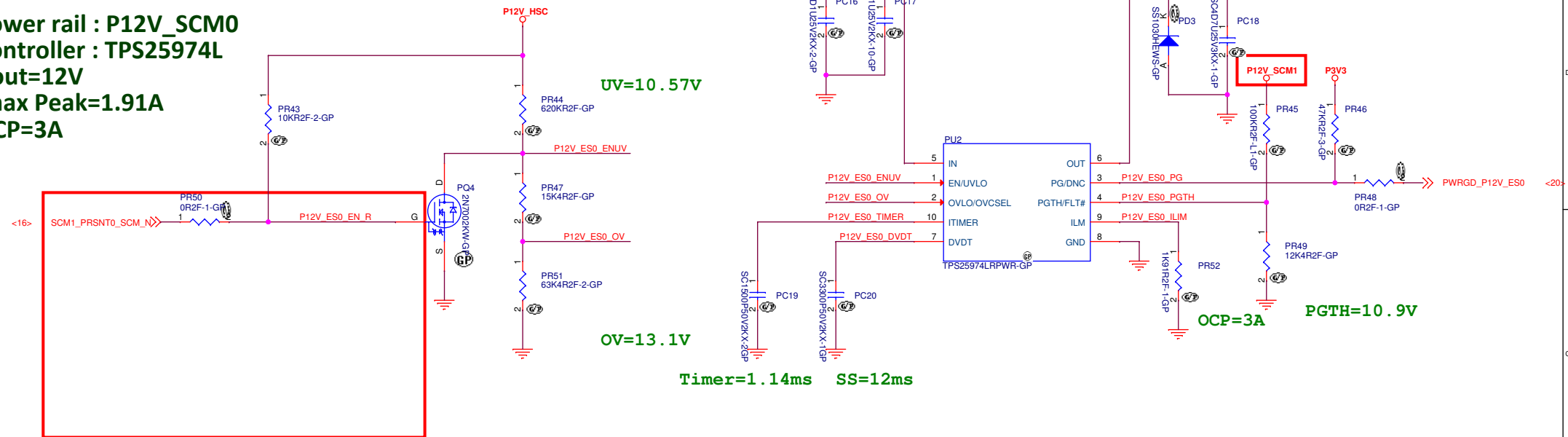
Imax: 46A
OCP : 60A
OC ALERT : 55A
UV ALERT : 11.5V

Imax: 46A
OCP : 60A
OC ALERT : 55A
UV ALERT : 11.5V



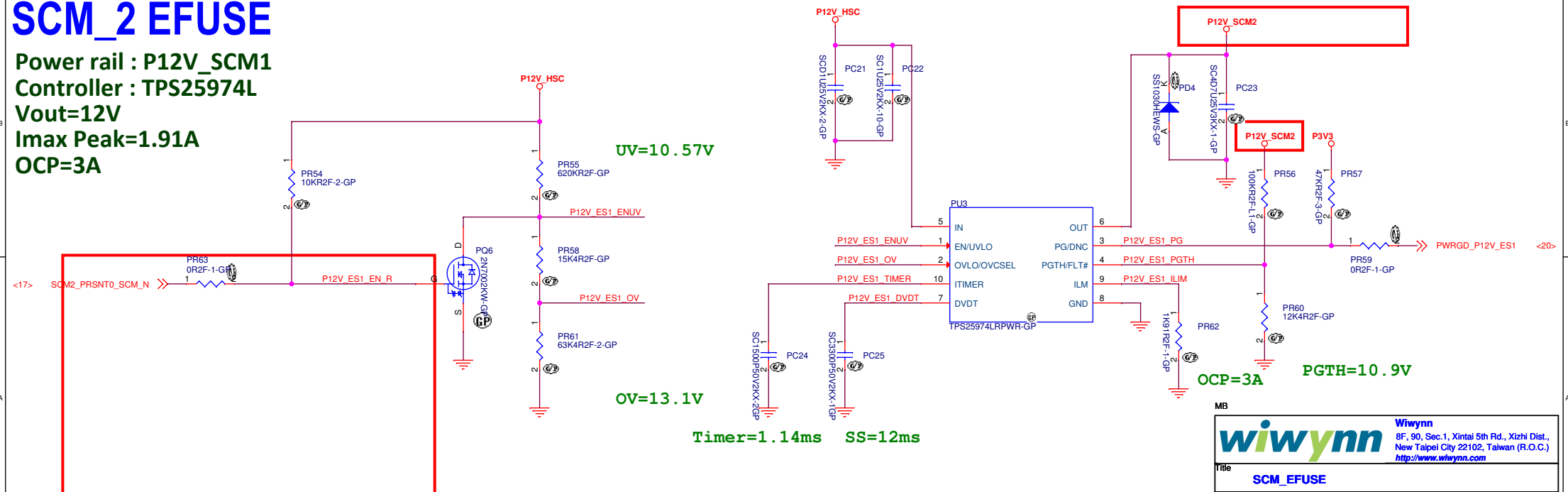
SCM_1 EFUSE

Power rail : P12V_SCM0
Controller : TPS25974L
Vout=12V
Imax Peak=1.91A
OCP=3A

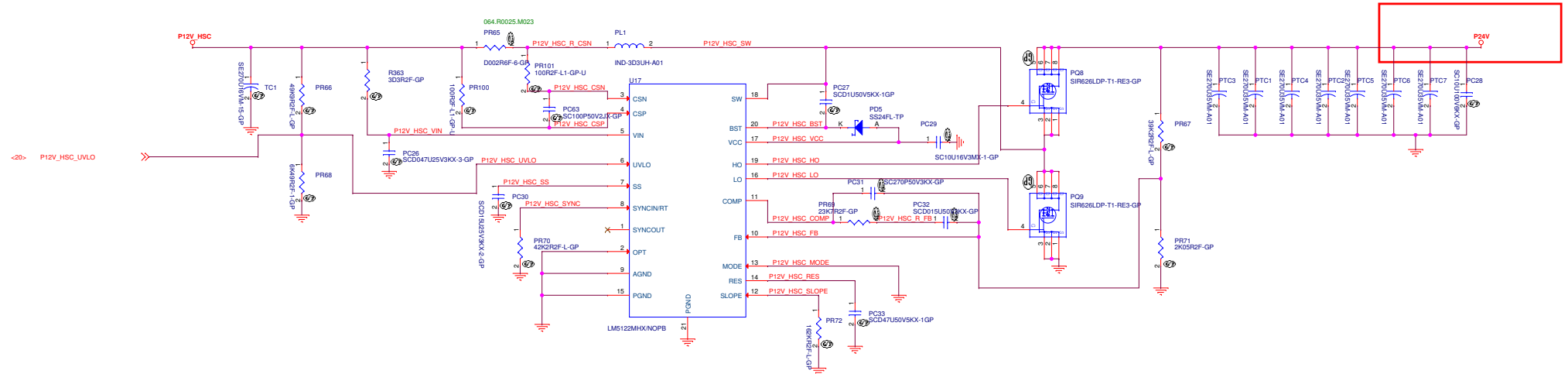


SCM_2 EFUSE

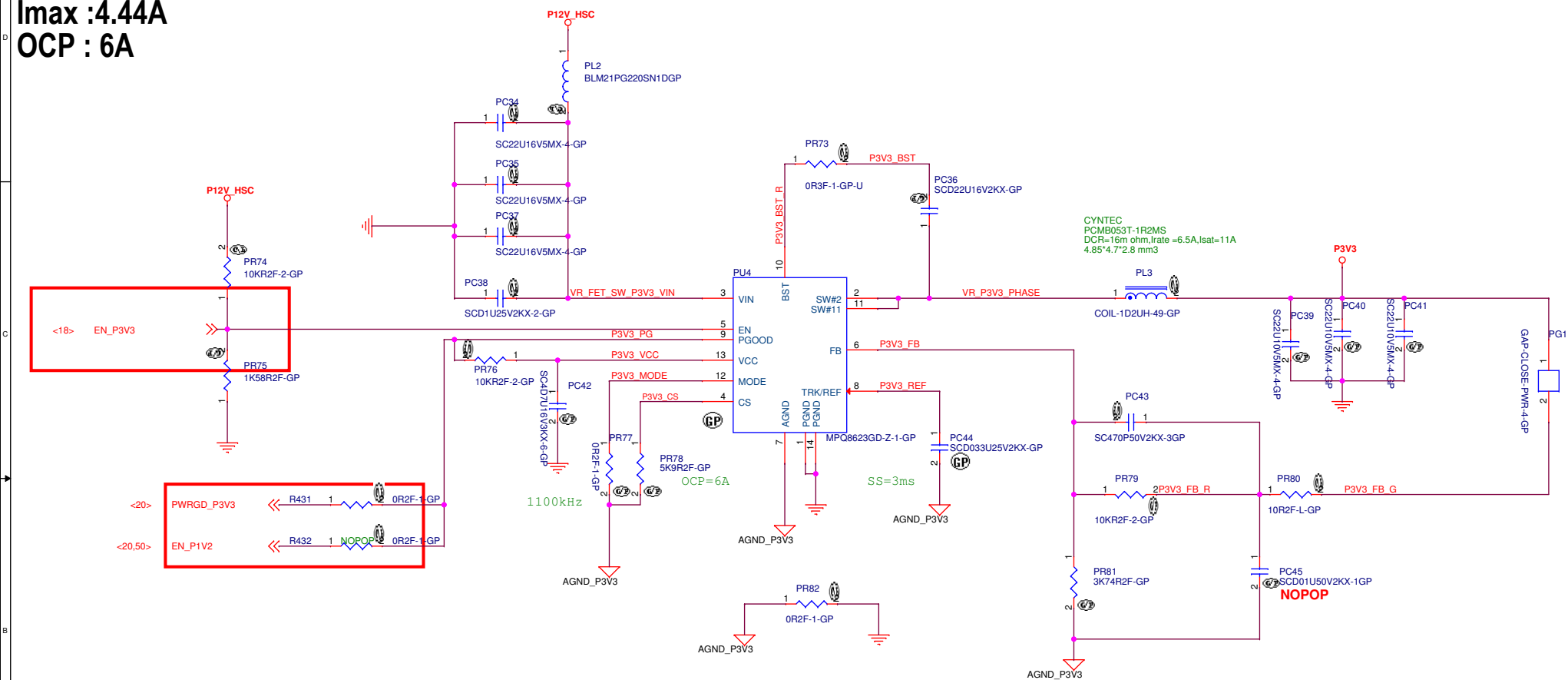
Power rail : P12V_SCM1
Controller : TPS25974L
Vout=12V
Imax Peak=1.91A
OCP=3A



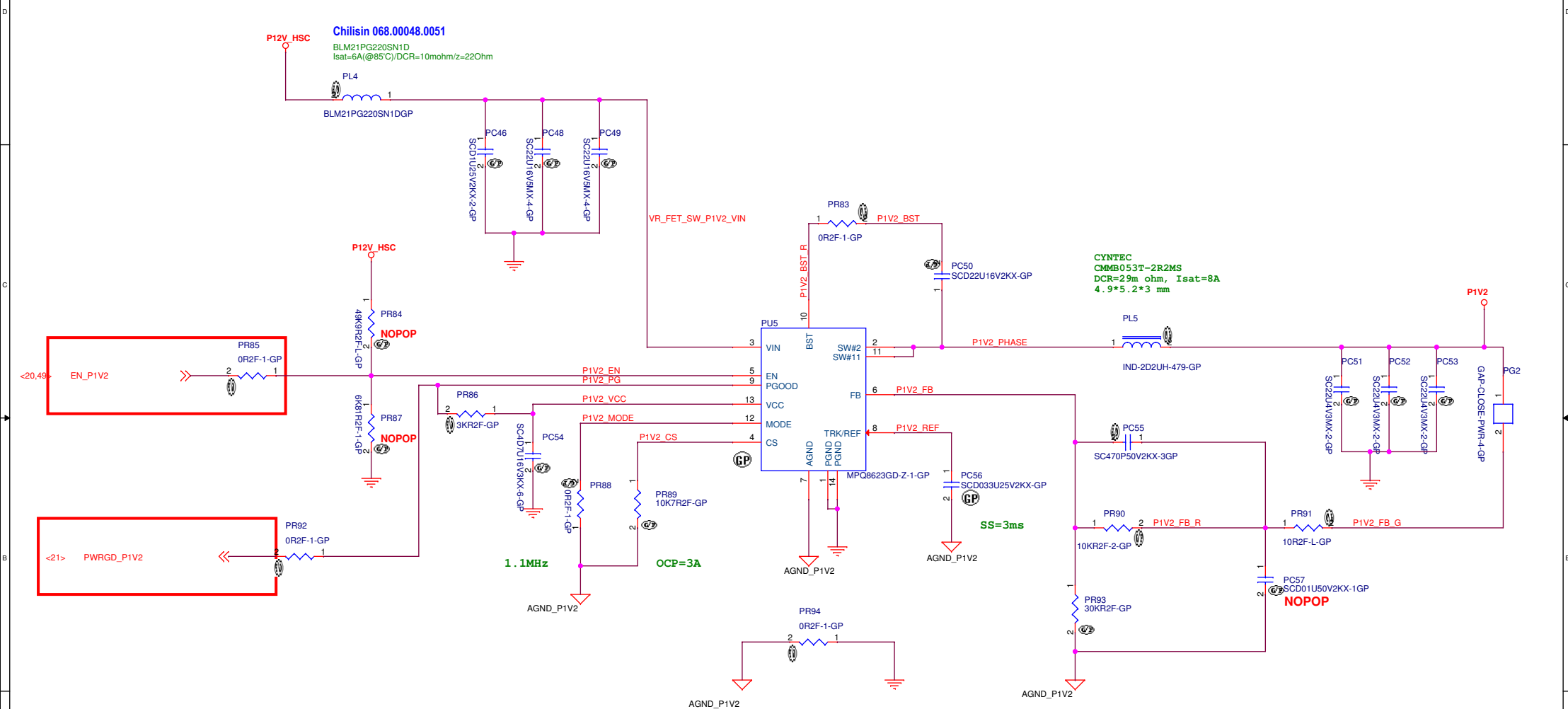
Power rail : P24V
Controller : LM5122MHX/NOPB
TDC :11.67A
Imax :11.67A
OCP : 18.75A



Power rail : P3V3
Controller : MPQ8623
TDC :4.44A
Imax :4.44A
OCP : 6A



Power rail : P1V2
Controller : MPQ8623
Vout : 1.2V
I_{max} : 1.5A
OCP : 3A



MB

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Title	P1V2 MPQ8623
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Monday, May 22, 2023

Rev

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Sheet

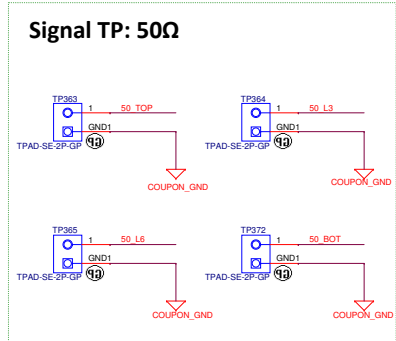
51

of

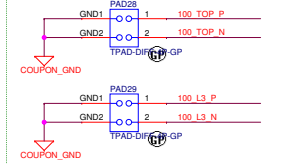
54

DELTA-L/IMPEDANCE COUPON

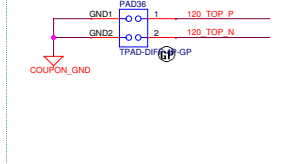
Impedance Coupon



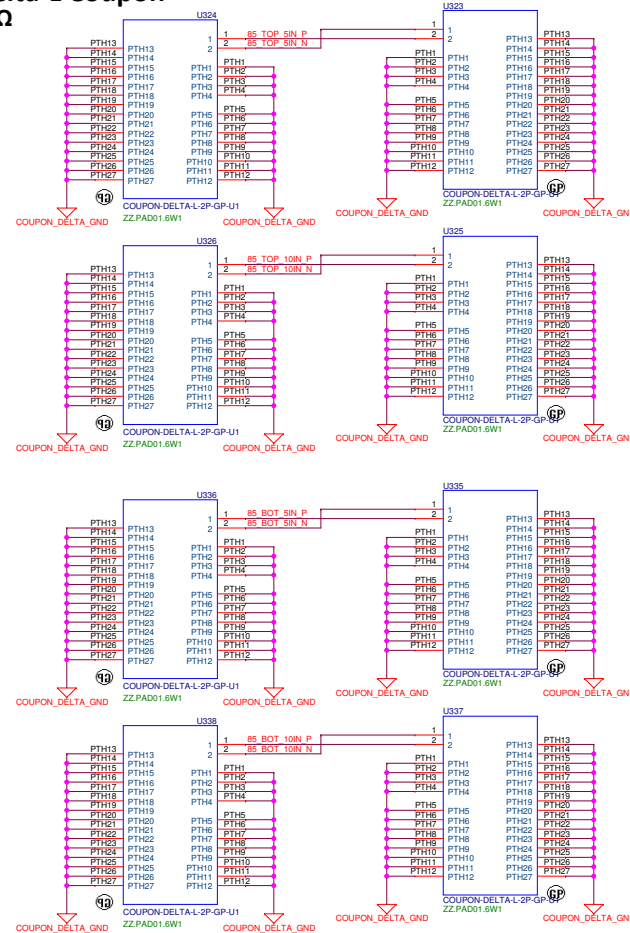
Differential Pair TP: 100Ω



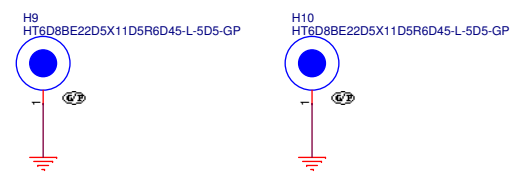
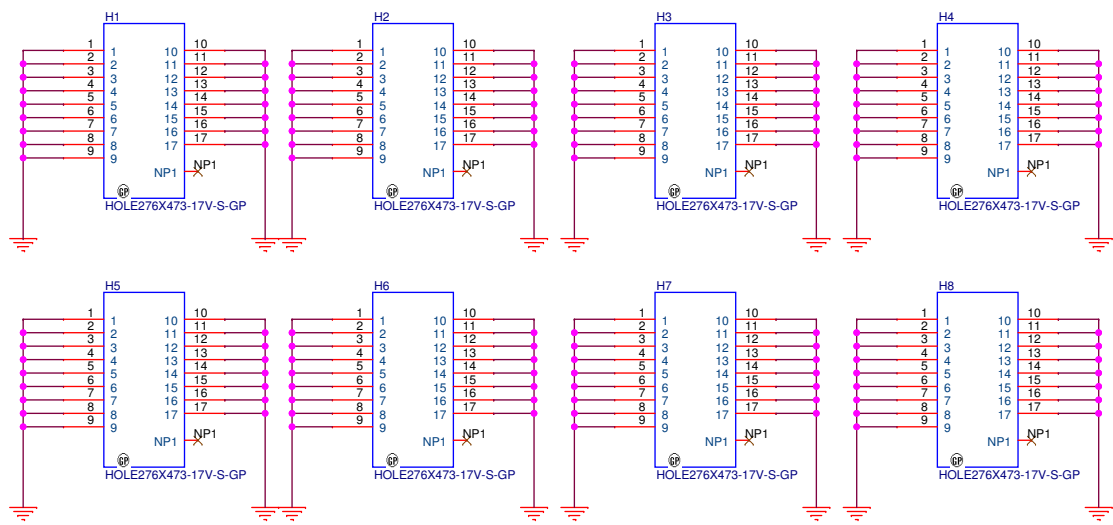
Differential Pair TP: 120Ω



Delta-L Coupon 85Ω



MECHANICAL



History

Version	Date	Item	Page	DESCRIPTION	BOM change	Layout change	Reason
SB	2023 08/16	1	29, 30	POP R5291, R5220 and NOPOP R5292, R5221 for packet loss issue.	✓		Wrong clock mode setting lead to packet loss issue.
SB	2023 08/16	2	29, 31	POP R5261, NOPOP R5262 for wrong address	✓		RTL8208D doesn't wrok due to wrong address setting.
1	2023 10/27	3	16, 17	Add 2*I2C between SCM0 and SCM1		✓	For BMC redundancy function by BMC team
1	2023 10/27	4	16, 17	Add sku ID for immersion and liquid sku		✓	Add sku ID for immersion and liquid sku
1	2023 10/27	5	16, 17	Add PU/PD option on CN3/CN4 A38/A39		✓	For DC-SCM master/slave reserved. Add sku ID for immersion and liquid sku